

C8000V IPsec Testbed

Test evidence — every command run on every device

Run started	07 September 2026, 19:21 UTC
Duration	19.4 minutes
Result	208 of 208 tests passed
Suites	22
Device commands recorded	1406
Topology	hub-and-spoke: R1 hub, R2 and R3 spokes, one Linux host per router
Encryption	IKEv2 + static VTI, ESP-AES-256 / SHA256, one tunnel per spoke
Routing	eBGP inside the tunnels; AS 65001 hub, AS 65002 / 65003 spokes

Suite results

Suite	Tests	Passed	Failed	Commands	Time
01 Platform	5	5	0	10	1s
02 Configuration	5	5	0	14	2s
03 Wan	6	6	0	19	1s
04 Dmvpn	13	13	0	37	25s
05 Capture State	12	12	0	79	16s
06 Wire Encryption	3	3	0	27	34s
07 Ipsec Negative	3	3	0	63	124s
08 Management Api	6	6	0	7	3s
09 Bgp	26	26	0	623	302s
10 Host To Host	10	10	0	47	79s
11 Throughput	3	3	0	10	81s
12 Nat	10	10	0	51	79s
13 Nat Pool	13	13	0	50	89s
14 Ntp	12	12	0	39	3s
15 Syslog	10	10	0	21	47s
16 Snmpv3	13	13	0	87	15s

17 Snmp Polling	14	14	0	74	17s
18 Mtu	8	8	0	17	21s
19 Oob	9	9	0	68	31s
20 Tacacs	12	12	0	35	25s
21 Vty Acl	9	9	0	22	1s
22 Banner	6	6	0	6	21s

Each test below lists the commands it issued, in order, with the device that ran them and the output it returned. Router commands are marked in teal, Linux host commands in amber. Long output is trimmed to the first 14 lines; complete captures are in results/.

01 Platform

PASS Both Routers Report Catalyst 8000V Software (0.2s, 2 commands)

R1 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 by mcpre
```

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documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines

R2 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 22-Jul-26 03:39 by mcpre
```

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GPL code under the terms of GPL Version 2.0. For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines

PASS Hostnames Match The Day0 Bootstrap Configs (0.2s, 2 commands)

R1 show running-config | include ^hostname

```
hostname c8000v-r1
```

R2 show running-config | include ^hostname

```
hostname c8000v-r2
```

PASS Management Interfaces Are Up With DHCP Addresses (0.1s, 2 commands)

R1 show ip interface brief GigabitEthernet1

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up

R2 show ip interface brief GigabitEthernet1

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up

PASS Crypto Feature License Is Active (0.1s, 2 commands)

Without a feature license the crypto CLI does not exist at all, so every IPsec test downstream depends on this.

R1 show version | include License Level

```
License Level: network-premier
Addon License Level: dna-premier
```

R2 show version | include License Level

```
License Level: network-premier
Addon License Level: dna-premier
```

PASS Management Plane Services Are Enabled (0.2s, 2 commands)

```
R1 show running-config | include ^netconf-yang|^restconf
netconf-yang
restconf

R2 show running-config | include ^netconf-yang|^restconf
netconf-yang
restconf
```

02 Configuration

PASS Create A Loopback Interface (0.1s, 6 commands)

```
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 interface Loopback100
  ${output} =
R1 description ROBOT-MANAGED
  ${output} =
R1 ip address 192.0.2.1 255.255.255.0
  ${output} =
R1 no shutdown
  ${output} =
R1 end
  ${output} =
```

PASS Loopback Appears In The Interface Table (0.0s, 1 commands)

```
R1 show ip interface brief Loopback100
Interface          IP-Address      OK? Method Status          Protocol
Loopback100        192.0.2.1       YES manual up              up
```

PASS Loopback Is Reachable From The Device Itself (2.1s, 2 commands)

```
R1 ping 192.0.2.1 repeat 2
Type escape sequence to abort.
Sending 2, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:
..!
Success rate is 50 percent (1/2), round-trip min/avg/max = 1/1/1 ms
R1 ping 192.0.2.1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

PASS Loopback Is Present In The Running Config (0.0s, 1 commands)

```
R1 show running-config interface Loopback100
Building configuration...

Current configuration : 92 bytes
!
interface Loopback100
  description ROBOT-MANAGED
  ip address 192.0.2.1 255.255.255.0
end
```

PASS Removing The Loopback Cleans It Up (0.1s, 4 commands)

```
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 no interface Loopback100
  ${output} =
R1 end
  ${output} =
R1 show ip interface brief Loopback100
^
% Invalid input detected at '^' marker.
```

03 Wan

PASS Every Router Is On The Shared WAN Segment (0.1s, 3 commands)

```
R1 show ip interface brief | include GigabitEthernet2
GigabitEthernet2    100.64.0.1    YES TFTP    up          up

R2 show ip interface brief | include GigabitEthernet2
GigabitEthernet2    100.64.0.2    YES TFTP    up          up

R3 show ip interface brief | include GigabitEthernet2
GigabitEthernet2    100.64.0.3    YES TFTP    up          up
```

PASS Every Router Can Reach Every Other Router On It (0.2s, 6 commands)

Including spoke to spoke, which is the precondition for a direct tunnel and the property the VTI lab deliberately did not have. Tested at the underlay, before any tunnel exists.

```
R2 ping 100.64.0.2 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R3 ping 100.64.0.3 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R1 ping 100.64.0.1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R3 ping 100.64.0.3 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R1 ping 100.64.0.1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R2 ping 100.64.0.2 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

PASS The Spokes Reach Each Other Directly, Not Through The Hub (0.0s, 1 commands)

One hop at the underlay. If this became two the spokes would still have connectivity, and DMVPN would still form tunnels -- but every "direct" spoke-to-spoke path would quietly be transiting the hub anyway.

```
R2 traceroute 100.64.0.3 probe 1 timeout 1 numeric
Type escape sequence to abort.
Tracing the route to 100.64.0.3
VRF info: (vrf in name/id, vrf out name/id)
 1 100.64.0.3 1 msec
```

PASS Each Router Learns The Others On The Segment (0.1s, 3 commands)

A shared segment, so every router should have resolved every other -- unlike a point-to-point link with exactly one neighbour.

```
R1 show ip arp GigabitEthernet2
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 100.64.0.1      -          5254.00c8.0102 ARPA   GigabitEthernet2
Internet 100.64.0.2      67         5254.00c8.0202 ARPA   GigabitEthernet2
Internet 100.64.0.3      66         5254.00c8.0302 ARPA   GigabitEthernet2
```

R2 show ip arp GigabitEthernet2

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.64.0.2	-	5254.00c8.0202	ARPA	GigabitEthernet2
Internet	100.64.0.1	67	5254.00c8.0102	ARPA	GigabitEthernet2
Internet	100.64.0.3	66	5254.00c8.0302	ARPA	GigabitEthernet2

R3 show ip arp GigabitEthernet2

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.64.0.3	-	5254.00c8.0302	ARPA	GigabitEthernet2
Internet	100.64.0.1	67	5254.00c8.0102	ARPA	GigabitEthernet2
Internet	100.64.0.2	66	5254.00c8.0202	ARPA	GigabitEthernet2

PASS The Underlay Carries No Site Prefixes (0.1s, 3 commands)

The WAN is a transport, not a routing domain. A site prefix appearing here would mean traffic could reach a LAN without ever entering the tunnel -- unencrypted, and invisible to every test that watches the overlay.

R1 show ip route | include GigabitEthernet2

```
C      100.64.0.0/24 is directly connected, GigabitEthernet2
L      100.64.0.1/32 is directly connected, GigabitEthernet2
```

R2 show ip route | include GigabitEthernet2

```
C      100.64.0.0/24 is directly connected, GigabitEthernet2
L      100.64.0.2/32 is directly connected, GigabitEthernet2
```

R3 show ip route | include GigabitEthernet2

```
C      100.64.0.0/24 is directly connected, GigabitEthernet2
L      100.64.0.3/32 is directly connected, GigabitEthernet2
```

PASS No Routing Protocol Runs On The WAN Segment (0.1s, 3 commands)

BGP peers over the tunnel, so the underlay should carry no adjacency at all. One here would be a second, unprotected path for the control plane.

R1 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.2	4	65002	21	24	28	0	0	00:02:16	3
172.20.0.3	4	65003	100	110	28	0	0	00:14:48	3

R2 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	24	21	46	0	0	00:02:16	6

R3 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	110	100	146	0	0	00:14:49	6

04 Dmvpn

PASS Every Router Runs One Multipoint Tunnel With No Destination (0.1s, 4 commands)

The structural difference from a static VTI. A configured destination would make this point-to-point again and there would be nothing for NHRP to resolve.

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R1 show ip interface brief | count Tunnel

Number of lines which match regexp = 1

PASS The Tunnel Is Protected By IPsec On Every Router (0.1s, 3 commands)

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
```


... 6 more lines

R2 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

PASS NHRP Shares A Network ID And Authentication Everywhere (0.1s, 3 commands)

Both must match across the cloud; a mismatch in either leaves registrations silently unanswered.

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
```

```

ip address 172.20.0.3 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

PASS The Hub Is The Next Hop Server And The Spokes Point At It (0.1s, 4 commands)

R2 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.2 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

R3 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.3 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
... 6 more lines

```

R1 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 399 bytes
!
interface Tunnel0
description DMVPN hub
ip address 172.20.0.1 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp redirect
ip tcp adjust-mss 1360
... 6 more lines

```

R1 show dmvpn

```

Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
T1 - Route Installed, T2 - Nexthop-override, B - BGP
C - CTS Capable, I2 - Temporary
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

```

```

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

```

```

# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
... 2 more lines

```

PASS This Is Phase 3, Not Phase 1 Or 2 (0.1s, 3 commands)

Redirect on the hub and shortcut on the spokes are what make the overlay reorganise itself. Without them the tests below about direct paths cannot pass, and the design is a different one.

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines
```

PASS Both Spokes Register With The Hub (0.0s, 1 commands)

Registration is dynamic: the hub is configured with no spoke addresses at all and learns them as they arrive.

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
          N - NATed, L - Local, X - No Socket
          T1 - Route Installed, T2 - Nexthop-override, B - BGP
          C - CTS Capable, I2 - Temporary
          # Ent --> Number of NHRP entries with same NBMA peer
          NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
          UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
... 2 more lines
```

PASS The Hub Maps Each Spoke's Tunnel Address To Its NBMA Address (0.0s, 1 commands)

R1 show ip nhrp

```
172.20.0.2/32 via 172.20.0.2
  Tunnel0 created 00:00:49, expire 00:04:46
  Type: dynamic, Flags: registered nhop bfd
  NBMA address: 100.64.0.2
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:49, expire 00:04:10
  Type: dynamic, Flags: registered nhop bfd
  NBMA address: 100.64.0.3
```

PASS Every DMVPN Peering Is Encrypted (0.0s, 2 commands)

One IPsec profile protects the multipoint tunnel, so each peering formed over it -- including ones the configuration never named -- is protected by construction.

R2 show crypto session brief

```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
      K - No IKE
ivrf = (none)
Peer      I/F      Username      Group/Phase1_id      Uptime      Status
100.64.0.3  Tu0              100.64.0.3      00:00:45 UA
100.64.0.1  Tu0              100.64.0.1      00:00:49 UA
```

R3 show crypto session brief

```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
      K - No IKE
ivrf = (none)
Peer      I/F      Username      Group/Phase1_id      Uptime      Status
100.64.0.2  Tu0              100.64.0.2      00:00:45 UA
100.64.0.1  Tu0              100.64.0.1      00:00:49 UA
```

PASS A Spoke Learns The Far Spoke As The BGP Next Hop (0.1s, 2 commands)

The routing half of Phase 3, and the reason "next-hop-unchanged" is configured on the hub. If the hub rewrote the next hop to itself, every spoke-to-spoke prefix would resolve to the hub, NHRP would never be asked to find the far spoke, and no shortcut could form.

R2 show bgp ipv4 unicast 10.20.3.1

```
BGP routing table entry for 10.20.3.1/32, version 43
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 23:19:46 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1

```
BGP routing table entry for 10.20.2.1/32, version 144
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 23:20:08 UTC
```

PASS Spoke To Spoke Traffic Works Before Any Shortcut Exists (12.1s, 4 commands)

The fallback path. With the NHRP cache cleared there is no direct entry, so the first packets go through the hub -- which is what makes the shortcut an optimisation rather than a prerequisite for connectivity.

R2 clear ip nhrp

```
${output} =
```

R3 clear ip nhrp

```
${output} =
```

```
R2 show ip nhrp 172.20.0.3
${output} =

R2 ping 10.20.3.1 source Loopback1 repeat 30
Type escape sequence to abort.
Sending 30, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Success rate is 96 percent (29/30), round-trip min/avg/max = 1/1/7 ms
```

PASS A Shortcut Forms On Demand (12.1s, 6 commands)

The defining behaviour. Cleared first so the entry cannot be left over from an earlier test: it has to be created by this traffic, here.

```
R2 clear ip nhrp
${output} =

R3 clear ip nhrp
${output} =

R2 show ip nhrp 172.20.0.3
${output} =

R2 ping 10.20.3.1 source Loopback1 repeat 20
Type escape sequence to abort.
Sending 20, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!!!!!!!!!!!!!!!!!!!!!!
Success rate is 95 percent (19/20), round-trip min/avg/max = 1/1/1 ms

R2 show ip nhrp 172.20.0.3
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:02, expire 00:04:58
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3

R3 show ip nhrp 172.20.0.2
172.20.0.2/32 via 172.20.0.2
  Tunnel0 created 00:00:02, expire 00:04:58
  Type: dynamic, Flags: router implicit nhop bfd
  NBMA address: 100.64.0.2
```

PASS Traffic Then Leaves The Hub Out Of The Path (0.1s, 2 commands)

The point of the shortcut, and the strongest statement this suite makes: with the direct path installed, a trace from one spoke to the other reaches it in a single hop instead of transiting the hub.

```
R2 show ip nhrp 172.20.0.3
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:02, expire 00:04:57
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3

R2 traceroute 10.20.3.1 source Loopback1 probe 1 timeout 1 numeric
Type escape sequence to abort.
Tracing the route to 10.20.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.20.0.3 2 msec
```

PASS The Shortcut Gets Its Own IPsec Session (0.0s, 2 commands)

A direct path between two routers that were never configured with each other's addresses, protected all the same, because the profile is bound to the tunnel rather than to a peer.

```
R2 show ip nhrp 172.20.0.3
172.20.0.3/32 via 172.20.0.3
  Tunnel0 created 00:00:02, expire 00:04:57
  Type: dynamic, Flags: router nhop bfd
  NBMA address: 100.64.0.3

R2 show crypto session brief
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
      K - No IKE
ivrf = (none)
Peer      I/F      Username      Group/Phase1_id      Uptime      Status
100.64.0.3  Tu0      100.64.0.3    100.64.0.3           00:00:02    UA
100.64.0.1  Tu0      100.64.0.1    100.64.0.1           00:01:14    UA
```

05 Capture State

PASS Capture Running Config From Every Router (13.1s, 3 commands)

R1 show running-config

```
Building configuration...

Current configuration : 21292 bytes
!
! Last configuration change at 23:22:01 UTC Mon Sep 7 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 624 more lines
```

R2 show running-config

```
Building configuration...

Current configuration : 19872 bytes
!
! Last configuration change at 23:21:14 UTC Mon Sep 7 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 582 more lines
```

R3 show running-config

```
Building configuration...

Current configuration : 19832 bytes
!
! Last configuration change at 23:21:14 UTC Mon Sep 7 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 579 more lines
```

PASS Capture Routing Table From Every Router (0.2s, 3 commands)

R1 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

R2 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

```

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines

R3 show ip route

```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

```

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines

PASS Capture BGP Table And Neighbours From Every Router (0.2s, 6 commands)

R1 show bgp ipv4 unicast

```

BGP table version is 28, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	172.20.0.2	0		0	65002 i
*> 10.20.3.1/32	172.20.0.3	0		0	65003 i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	172.20.0.2	0		0	65002 i

... 4 more lines

R1 show bgp ipv4 unicast summary

```

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 28, main routing table version 28
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 136/127 prefixes, 139/130 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:48:55.594 ago)

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
...	2	more	lines						

R2 show bgp ipv4 unicast

```

BGP table version is 46, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	172.20.0.1	0		0	65001 i
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	172.20.0.3	0		0	65001 65003 i
*> 10.30.1.0/24	172.20.0.1	0		0	65001 i
*> 10.30.2.0/24	0.0.0.0	0		32768	i

```

... 4 more lines

R2 show bgp ipv4 unicast summary
BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 46, main routing table version 46
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 160/151 prefixes, 162/153 paths, scan interval 60 secs
31 networks peaked at 22:34:42 Sep 7 2026 UTC (00:48:12.674 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines

```

```

R3 show bgp ipv4 unicast
BGP table version is 146, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
*>  10.20.1.1/32    172.20.0.1           0             0 65001 i
*>  10.20.2.1/32    172.20.0.2           0             0 65001 65002 i
*>  10.20.3.1/32    0.0.0.0              0            32768 i
*>  10.30.1.0/24    172.20.0.1           0             0 65001 i
*>  10.30.2.0/24    172.20.0.2           0             0 65001 65002 i
... 4 more lines

```

```

R3 show bgp ipv4 unicast summary
BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 146, main routing table version 146
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 72/63 prefixes, 77/68 paths, scan interval 60 secs
9 networks peaked at 22:15:22 Sep 7 2026 UTC (01:07:32.746 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines

```

PASS Capture Crypto State From Every Router (0.3s, 6 commands)

```

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
  current_peer 100.64.0.3 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 604, #pkts encrypt: 604, #pkts digest: 604
    #pkts decaps: 594, #pkts decrypt: 594, #pkts verify: 594
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines

```

```

R1 show crypto ikev2 sa
IPv4 Crypto IKEv2 SA

Tunnel-id Local              Remote              fvrf/ivrf          Status
1          100.64.0.1/500          100.64.0.2/500          none/none          READY
    Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
    Life/Active Time: 86400/100 sec
    CE id: 1029, Session-id: 26
    Local spi: 674CC240C9A38803      Remote spi: 6B960B9A3952A45C

```



```

Tunnel-id Local          Remote          fvrf/ivrf      Status
2      100.64.0.1/500    100.64.0.3/500 none/none      READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/100 sec
      CE id: 1030, Session-id: 27
... 3 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 1135, #pkts encrypt: 1135, #pkts digest: 1135
#pkts decaps: 1133, #pkts decrypt: 1133, #pkts verify: 1133
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

R2 show crypto ikev2 sa

```

IPv4 Crypto IKEv2 SA

Tunnel-id Local          Remote          fvrf/ivrf      Status
1      100.64.0.2/500    100.64.0.1/500 none/none      READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/101 sec
      CE id: 1029, Session-id: 24
      Local spi: 6B960B9A3952A45C      Remote spi: 674CC240C9A38803

IPv6 Crypto IKEv2 SA

```

R3 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.3

protected vrf: (none)
local  ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 9361, #pkts encrypt: 9361, #pkts digest: 9361
#pkts decaps: 9395, #pkts decrypt: 9395, #pkts verify: 9395
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

R3 show crypto ikev2 sa

```

IPv4 Crypto IKEv2 SA

Tunnel-id Local          Remote          fvrf/ivrf      Status
1      100.64.0.3/500    100.64.0.1/500 none/none      READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/101 sec
      CE id: 1026, Session-id: 26
      Local spi: 54006597FF55CAD2      Remote spi: C7ABC0CAC6503940

IPv6 Crypto IKEv2 SA

```

PASS Capture NAT State From Every Router (0.5s, 6 commands)

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.88	---	---
---	10.30.1.129	192.168.10.85	---	---
---	10.30.1.113	192.168.10.93	---	---
---	10.30.1.114	192.168.10.69	---	---
---	10.30.1.116	192.168.10.73	---	---
---	10.30.1.119	192.168.10.71	---	---
---	10.30.1.123	192.168.10.78	---	---
---	10.30.1.120	192.168.10.82	---	---
---	10.30.1.126	192.168.10.76	---	---
---	10.30.1.125	192.168.10.84	---	---
---	10.30.1.101	192.168.10.90	---	---
---	10.30.1.102	192.168.10.89	---	---

```

--- 10.30.1.111          192.168.10.70      ---
... 28 more lines

```

R1 show ip nat statistics

```

Total active translations: 40 (10 static, 30 dynamic; 0 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 457 Misses: 123
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 123
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
  pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.1.100 end 10.30.1.129
... 9 more lines

```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.89	---	---
---	10.30.2.129	192.168.20.84	---	---
---	10.30.2.106	192.168.20.68	---	---
---	10.30.2.105	192.168.20.88	---	---
---	10.30.2.111	192.168.20.92	---	---
---	10.30.2.108	192.168.20.80	---	---
---	10.30.2.101	192.168.20.64	---	---
---	10.30.2.102	192.168.20.65	---	---
---	10.30.2.126	192.168.20.77	---	---
---	10.30.2.125	192.168.20.86	---	---
---	10.30.2.123	192.168.20.79	---	---
---	10.30.2.120	192.168.20.74	---	---
---	10.30.2.116	192.168.20.82	---	---
---	10.30.2.117	192.168.20.83	---	---
---	10.30.2.118	192.168.20.85	---	---
---	10.30.2.119	192.168.20.87	---	---
---	10.30.2.121	192.168.20.81	---	---
---	10.30.2.122	192.168.20.78	---	---
---	10.30.2.124	192.168.20.90	---	---
---	10.30.2.127	192.168.20.81	---	---
---	10.30.2.121	192.168.20.80	---	---
---	10.30.2.122	192.168.20.77	---	---
---	10.30.2.118	192.168.20.86	---	---
---	10.30.2.117	192.168.20.75	---	---
---	10.30.2.115	192.168.20.82	---	---
---	10.30.2.114	192.168.20.81	---	---
---	10.30.2.113	192.168.20.80	---	---
---	10.30.2.112	192.168.20.79	---	---
---	10.30.2.111	192.168.20.92	---	---
---	10.30.2.110	192.168.20.88	---	---
---	10.30.2.109	192.168.20.84	---	---
---	10.30.2.108	192.168.20.80	---	---
---	10.30.2.107	192.168.20.68	---	---
---	10.30.2.106	192.168.20.68	---	---
---	10.30.2.105	192.168.20.88	---	---
---	10.30.2.104	192.168.20.87	---	---
---	10.30.2.103	192.168.20.86	---	---
---	10.30.2.102	192.168.20.65	---	---
---	10.30.2.101	192.168.20.64	---	---
---	10.30.2.100	192.168.20.83	---	---
---	10.30.2.099	192.168.20.82	---	---
---	10.30.2.098	192.168.20.81	---	---
---	10.30.2.097	192.168.20.80	---	---
---	10.30.2.096	192.168.20.79	---	---
---	10.30.2.095	192.168.20.78	---	---
---	10.30.2.094	192.168.20.77	---	---
---	10.30.2.093	192.168.20.76	---	---
---	10.30.2.092	192.168.20.75	---	---
---	10.30.2.091	192.168.20.74	---	---
---	10.30.2.090	192.168.20.73	---	---
---	10.30.2.089	192.168.20.72	---	---
---	10.30.2.088	192.168.20.71	---	---
---	10.30.2.087	192.168.20.70	---	---
---	10.30.2.086	192.168.20.69	---	---
---	10.30.2.085	192.168.20.68	---	---
---	10.30.2.084	192.168.20.67	---	---
---	10.30.2.083	192.168.20.66	---	---
---	10.30.2.082	192.168.20.65	---	---
---	10.30.2.081	192.168.20.64	---	---
---	10.30.2.080	192.168.20.63	---	---
---	10.30.2.079	192.168.20.62	---	---
---	10.30.2.078	192.168.20.61	---	---
---	10.30.2.077	192.168.20.60	---	---
---	10.30.2.076	192.168.20.59	---	---
---	10.30.2.075	192.168.20.58	---	---
---	10.30.2.074	192.168.20.57	---	---
---	10.30.2.073	192.168.20.56	---	---
---	10.30.2.072	192.168.20.55	---	---
---	10.30.2.071	192.168.20.54	---	---
---	10.30.2.070	192.168.20.53	---	---
---	10.30.2.069	192.168.20.52	---	---
---	10.30.2.068	192.168.20.51	---	---
---	10.30.2.067	192.168.20.50	---	---
---	10.30.2.066	192.168.20.49	---	---
---	10.30.2.065	192.168.20.48	---	---
---	10.30.2.064	192.168.20.47	---	---
---	10.30.2.063	192.168.20.46	---	---
---	10.30.2.062	192.168.20.45	---	---
---	10.30.2.061	192.168.20.44	---	---
---	10.30.2.060	192.168.20.43	---	---
---	10.30.2.059	192.168.20.42	---	---
---	10.30.2.058	192.168.20.41	---	---
---	10.30.2.057	192.168.20.40	---	---
---	10.30.2.056	192.168.20.39	---	---
---	10.30.2.055	192.168.20.38	---	---
---	10.30.2.054	192.168.20.37	---	---
---	10.30.2.053	192.168.20.36	---	---
---	10.30.2.052	192.168.20.35	---	---
---	10.30.2.051	192.168.20.34	---	---
---	10.30.2.050	192.168.20.33	---	---
---	10.30.2.049	192.168.20.32	---	---
---	10.30.2.048	192.168.20.31	---	---
---	10.30.2.047	192.168.20.30	---	---
---	10.30.2.046	192.168.20.29	---	---
---	10.30.2.045	192.168.20.28	---	---
---	10.30.2.044	192.168.20.27	---	---
---	10.30.2.043	192.168.20.26	---	---
---	10.30.2.042	192.168.20.25	---	---
---	10.30.2.041	192.168.20.24	---	---
---	10.30.2.040	192.168.20.23	---	---
---	10.30.2.039	192.168.20.22	---	---
---	10.30.2.038	192.168.20.21	---	---
---	10.30.2.037	192.168.20.20	---	---
---	10.30.2.036	192.168.20.19	---	---
---	10.30.2.035	192.168.20.18	---	---
---	10.30.2.034	192.168.20.17	---	---
---	10.30.2.033	192.168.20.16	---	---
---	10.30.2.032	192.168.20.15	---	---
---	10.30.2.031	192.168.20.14	---	---
---	10.30.2.030	192.168.20.13	---	---
---	10.30.2.029	192.168.20.12	---	---
---	10.30.2.028	192.168.20.11	---	---
---	10.30.2.027	192.168.20.10	---	---
---	10.30.2.026	192.168.20.09	---	---
---	10.30.2.025	192.168.20.08	---	---
---	10.30.2.024	192.168.20.07	---	---
---	10.30.2.023	192.168.20.06	---	---
---	10.30.2.022	192.168.20.05	---	---
---	10.30.2.021	192.168.20.04	---	---
---	10.30.2.020	192.168.20.03	---	---
---	10.30.2.019	192.168.20.02	---	---
---	10.30.2.018	192.168.20.01	---	---
---	10.30.2.017	192.168.20.00	---	---
---	10.30.2.016	192.168.20.00	---	---
---	10.30.2.015	192.168.20.00	---	---
---	10.30.2.014	192.168.20.00	---	---
---	10.30.2.013	192.168.20.00	---	---
---	10.30.2.012	192.168.20.00	---	---
---	10.30.2.011	192.168.20.00	---	---
---	10.30.2.010	192.168.20.00	---	---
---	10.30.2.009	192.168.20.00	---	---
---	10.30.2.008	192.168.20.00	---	---
---	10.30.2.007	192.168.20.00	---	---
---	10.30.2.006	192.168.20.00	---	---
---	10.30.2.005	192.168.20.00	---	---
---	10.30.2.004	192.168.20.00	---	---
---	10.30.2.003	192.168.20.00	---	---
---	10.30.2.002	192.168.20.00	---	---
---	10.30.2.001	192.168.20.00	---	---
---	10.30.2.000	192.168.20.00	---	---

R2 show ip nat statistics

```

Total active translations: 40 (10 static, 30 dynamic; 0 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 337 Misses: 83
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 83
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
  pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.2.100 end 10.30.2.129
... 9 more lines

```

R3 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.3.104	192.168.30.84	---	---
---	10.30.3.107	192.168.30.93	---	---
---	10.30.3.109	192.168.30.92	---	---
---	10.30.3.110	192.168.30.89	---	---
---	10.30.3.103	192.168.30.85	---	---
---	10.30.3.100	192.168.30.83	---	---
---	10.30.3.124	192.168.30.90	---	---
---	10.30.3.127	192.168.30.81	---	---
---	10.30.3.121	192.168.30.80	---	---
---	10.30.3.122	192.168.30.77	---	---
---	10.30.3.118	192.168.30.86	---	---
---	10.30.3.117	192.168.30.75	---	---
---	10.30.3.115	192.168.30.82	---	---
---	10.30.3.114	192.168.30.81	---	---
---	10.30.3.113	192.168.30.80	---	---
---	10.30.3.112	192.168.30.79	---	---
---	10.30.3.111	192.168.30.78	---	---
---	10.30.3.110	192.168.30.77	---	---
---	10.30.3.109	192.168.30.76	---	---
---	10.30.3.108	192.168.30.75	---	---
---	10.30.3.107	192.168.30.74	---	---
---	10.30.3.106	192.168.30.73	---	---
---	10.30.3.105	192.168.30.72	---	---
---	10.30.3.104	192.168.30.71	---	---
---	10.30.3.103	192.168.30.70	---	---
---	10.30.3.102	192.168.30.69	---	---
---	10.30.3.101	192.168.30.68	---	---
---	10.30.3.100	192.168.30.67	---	---
---	10.30.3.099	192.168.30.66	---	---
---	10.30.3.098	192.168.30.65	---	---
---	10.30.3.097	192.168.30.64	---	---
---	10.30.3.096	192.168.30.63	---	---
---	10.30.3.095	192.168.30.62	---	---
---	10.30.3.094	192.168.30.61	---	---
---	10.30.3.093	192.168.30.60	---	---
---	10.30.3.092	192.168.30.59	---	---
---	10.30.3.091	192.168.30.58	---	---
---	10.30.3.090	192.168.30.57	---	---
---	10.30.3.089	192.168.30.56	---	---
---	10.30.3.088	192.168.30.55	---	---
---	10.30.3.087	192.168.30.54	---	---
---	10.30.3.086	192.168.30.53	---	---
---	10.30.3.085	192.168.30.52	---	---
---	10.30.3.084	192.168.30.51	---	---
---	10.30.3.083	192.168.30.50	---	---
---	10.30.3.082	192.168.30.49	---	---
---	10.30.3.081	192.168.30.48	---	---
---	10.30.3.080	192.168.30.47	---	---
---	10.30.3.079	192.168.30.46	---	---
---	10.30.3.078	192.168.30.45	---	---
---	10.30.3.077	192.168.30.44	---	---
---	10.30.3.076	192.168.30.43	---	---
---	10.30.3.075	192.168.30.42	---	---
---	10.30.3.074	192.168.30.41	---	---
---	10.30.3.073	192.168.30.40	---	---
---	10.30.3.072	192.168.30.39	---	---
---	10.30.3.071	192.168.30.38	---	---
---	10.30.3.070	192.168.30.37	---	---
---	10.30.3.069	192.168.30.36	---	---
---	10.30.3.068	192.168.30.35	---	---
---	10.30.3.067	192.168.30.34	---	---
---	10.30.3.066	192.168.30.33	---	---
---	10.30.3.065	192.168.30.32	---	---
---	10.30.3.064	192.168.30.31	---	---
---	10.30.3.063	192.168.30.30	---	---
---	10.30.3.062	192.168.30.29	---	---
---	10.30.3.061	192.168.30.28	---	---
---	10.30.3.060	192.168.30.27	---	---
---	10.30.3.059	192.168.30.26	---	---
---	10.30.3.058	192.168.30.25	---	---
---	10.30.3.057	192.168.30.24	---	---
---	10.30.3.056	192.168.30.23	---	---
---	10.30.3.055	192.168.30.22	---	---
---	10.30.3.054	192.168		

```
pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.3.100 end 10.30.3.129
... 9 more lines
```

PASS Capture NTP State From Every Router (0.1s, 6 commands)

R1 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
ntp uptime is 410100 (1/100 of seconds), resolution is 4000
reference time is EE49C50C.33B64630 (23:19:40.202 UTC Mon Sep 7 2026)
clock offset is 1.1201 msec, root delay is 50.71 msec
root dispersion is 10.94 msec, peer dispersion is 5.58 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008787 s/s
system poll interval is 128, last update was 195 sec ago.
```

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    61   128  377  0.928  1.120  5.584
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 410100 (1/100 of seconds), resolution is 4000
reference time is EE49C517.28B439C8 (23:19:51.159 UTC Mon Sep 7 2026)
clock offset is -0.2035 msec, root delay is 49.85 msec
root dispersion is 69.46 msec, peer dispersion is 3.79 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007002 s/s
system poll interval is 64, last update was 184 sec ago.
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    24    64   17  0.062  -0.203  3.798
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0021 Hz, precision is 2**10
ntp uptime is 410100 (1/100 of seconds), resolution is 4000
reference time is EE49C439.2A7EFA50 (23:16:09.166 UTC Mon Sep 7 2026)
clock offset is 0.5161 msec, root delay is 47.91 msec
root dispersion is 13.66 msec, peer dispersion is 4.91 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008467 s/s
system poll interval is 128, last update was 406 sec ago.
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3     2   128  377  0.058  0.516  4.914
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

PASS Capture Logging State From Every Router (0.2s, 3 commands)

R1 show logging | include Trap logging|Logging to|Syslog logging

```
Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
filtering disabled)
  Trap logging: level informational, 1046 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
000142: *Sep 7 22:14:37.628 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 0 CLI
Request Triggered
000144: *Sep 7 22:14:37.632 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - CLI initiated
```

R2 show logging | include Trap logging|Logging to|Syslog logging

```
Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
filtering disabled)
  Trap logging: level informational, 1336 message lines logged
    Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
000718: .Sep 7 22:47:06.797 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
000721: .Sep 7 22:47:20.338 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
001203: .Sep 7 23:17:12.492 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
001206: .Sep 7 23:17:30.372 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - reconnection
```

R3 show logging | include Trap logging|Logging to|Syslog logging

```
Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
filtering disabled)
```

```

Trap logging: level informational, 873 message lines logged
Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
000142: *Sep  7 22:14:37.619 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 0 CLI
Request Triggered
000143: *Sep  7 22:14:37.620 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
(MGMT) started - CLI initiated

```

PASS Capture SNMP State From Every Router (0.2s, 9 commands)

R1 show snmp user

```

User name: labmon
Engine ID: 800000090300525400C80101
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP

```

R1 show snmp host

```

Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT   type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT   type: trap
user: labmon      security model: v3 priv

```

R1 show snmp engineID

```

Local SNMP engineID: 800000090300525400C80101
Remote Engine ID      IP-addr      Port

```

R2 show snmp user

```

User name: labmon
Engine ID: 800000090300525400C80201
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP

```

R2 show snmp host

```

Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT   type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT   type: trap
user: labmon      security model: v3 priv

```

R2 show snmp engineID

```

Local SNMP engineID: 800000090300525400C80201
Remote Engine ID      IP-addr      Port

```

R3 show snmp user

```

User name: labmon
Engine ID: 800000090300525400C80301
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP

```

R3 show snmp host

```

Notification host: 192.168.99.10      udp-port: 1162  VRFName: MGMT   type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10      udp-port: 162   VRFName: MGMT   type: trap
user: labmon      security model: v3 priv

```

R3 show snmp engineID

```

Local SNMP engineID: 800000090300525400C80301
Remote Engine ID      IP-addr      Port

```

PASS Capture The NMS View Of Every Router (0.5s, 16 commands)

The same system group, but seen over SNMPv3 from the NMS rather than read off the console. Captured next to the CLI output so the two can be compared in the evidence PDF.

NMS ip -br addr; ip route

```

lo                UNKNOWN      127.0.0.1/8 ::1/128
ens3              UP              10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4              UP              192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5              UP              192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100

```

```

10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10

NMS ip -br addr; ip route
lo UNKNOWN 127.0.0.1/8 ::1/128
ens3 UP 10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4 UP 192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5 UP 192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10

NMS snmpget --version 2>&1 | head -1
NET-SNMP version: 5.9.4.pre2

NMS snmpget --version 2>&1 | head -1
NET-SNMP version: 5.9.4.pre2

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (415000) 1:09:10.00
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (415000) 1:09:10.00
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-

```

```

UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (415086) 1:09:10.86
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (415086) 1:09:10.86
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (414990) 1:09:09.90
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (414990) 1:09:09.90
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2

```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

PASS Capture Collector State From The NMS (0.3s, 12 commands)

What the collectors actually hold: the tail of each router's syslog file and of snmptrapd's decrypted log. This is the management-plane evidence that used to come off h1.

NMS systemctl is-active rsyslog snmptrapd lab-trapcap

```
active
active
active
```

NMS systemctl is-active rsyslog snmptrapd lab-trapcap

```
active
active
active
```

NMS sudo ls -l /var/log/lab

```
total 2660
-rw-r--r-- 1 syslog syslog 128123 Sep  7 23:22 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 159924 Sep  7 23:22 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 105045 Sep  7 23:22 192.168.99.3.log
-rw-r--r-- 1 root  root   881546 Sep  7 23:22 snmptrapd.log
-rw-r--r-- 1 root  root  1416130 Sep  7 23:22 traps-raw.log
```

NMS sudo ls -l /var/log/lab

```
total 2660
-rw-r--r-- 1 syslog syslog 128123 Sep  7 23:22 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 159924 Sep  7 23:22 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 105045 Sep  7 23:22 192.168.99.3.log
-rw-r--r-- 1 root  root   881546 Sep  7 23:22 snmptrapd.log
-rw-r--r-- 1 root  root  1416130 Sep  7 23:22 traps-raw.log
```

NMS sudo tail -40 /var/log/lab/192.168.10.1.log

tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.10.1.log

tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.20.1.log

tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.20.1.log

tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.30.1.log

tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/192.168.30.1.log

tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory

NMS sudo tail -40 /var/log/lab/snmptrapd.log

```
2026-09-07 23:22:26 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:39.57 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.29 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.29 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.29.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.29.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.29.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.29.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.29.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.29.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.29.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.29.1 0
2026-09-07 23:22:26 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:40.66 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.29 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.29 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.29.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.29.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.29.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.29.1 1
```



```

iso.3.6.1.4.1.9.9.171.1.3.3.1.10.29.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.29.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.29.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.29.1 0
2026-09-07 23:22:27 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:40.41 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708049 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708049 0
iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:27 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.50 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708035 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708035 0
iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.71 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.29 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.29 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.29 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.29 4
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.29 213 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.28 213
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.28 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.28 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.28 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.28 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.28 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.28 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.28 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.28 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.28 0
... 26 more lines

```

NMS sudo tail -40 /var/log/lab/snmptrapd.log

```

2026-09-07 23:22:26 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:39.57 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.29 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.29 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.29.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.29.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.29.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.29.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.29.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.29.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.29.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.29.1 0
2026-09-07 23:22:26 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:40.66 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.29 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.29 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.29.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.29.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.29.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.29.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.29.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.29.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.29.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.29.1 0
2026-09-07 23:22:27 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:40.41 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708049 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708049 0
iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:27 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.50 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708035 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708035 0
iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.71 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.29 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.29 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.29 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.29 4
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.29 213 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.28 213
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.28 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.28 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.28 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.28 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.28 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.28 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.28 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.28 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.28 0
... 26 more lines

```

PASS Capture Config And Routing Table From Every Host (0.1s, 6 commands)

H1 ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever

```



```

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 82334sec preferred_lft 71534sec
   inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
       valid_lft 86212sec preferred_lft 14212sec
   inet6 fe80::5054:ff:feaa:101/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines

```

H1 ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1

```

H2 ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 82337sec preferred_lft 71537sec
   inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
       valid_lft 86083sec preferred_lft 14083sec
   inet6 fe80::5054:ff:feaa:201/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines

```

H2 ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1

```

H3 ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 82336sec preferred_lft 71536sec
   inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
       valid_lft 86292sec preferred_lft 14292sec
   inet6 fe80::5054:ff:feaa:301/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines

```

H3 ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10

```

PASS Every Router Routing Table Holds All Three LANs (0.2s, 3 commands)

A content check on what was just captured: in a working hub-and-spoke every router reaches every LAN, the spokes reaching each other's through the hub.

R1 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, 0 - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
 n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 H - NHRP, G - NHRP registered, g - NHRP registration summary
 o - ODR, P - periodic downloaded static route, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR
 & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
 ... 24 more lines

R2 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, 0 - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
 n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 H - NHRP, G - NHRP registered, g - NHRP registration summary
 o - ODR, P - periodic downloaded static route, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR
 & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
 ... 24 more lines

R3 show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, 0 - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
 n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 H - NHRP, G - NHRP registered, g - NHRP registration summary
 o - ODR, P - periodic downloaded static route, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR
 & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
 ... 24 more lines

06 Wire Encryption

PASS Protected Traffic Appears On The Link As ESP (15.7s, 9 commands)

```
R1 no monitor capture ROBOTCAP
Capture does not exist

R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
${output} =

R1 monitor capture ROBOTCAP match ipv4 any any
${output} =

R1 monitor capture ROBOTCAP buffer size 5
${output} =

R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP

R1 ping 172.16.2.1 source Loopback0 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
.....
Success rate is 0 percent (0/5)

R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP

R1 show monitor capture ROBOTCAP buffer brief
-----
#   size   timestamp      source          destination      dscp   protocol
-----
0  138     0.000000    100.64.0.1      -> 100.64.0.2      48 CS6   ESP
1  138     0.000992    100.64.0.2      -> 100.64.0.1      48 CS6   ESP
2  138     0.123986    100.64.0.1      -> 100.64.0.3      48 CS6   ESP
3  138     0.123986    100.64.0.3      -> 100.64.0.1      48 CS6   ESP
4  138     0.137993    100.64.0.1      -> 100.64.0.3      48 CS6   ESP
5  138     0.333982    100.64.0.2      -> 100.64.0.1      48 CS6   ESP
6  138     0.369036    100.64.0.3      -> 100.64.0.1      48 CS6   ESP
7  138     0.369036    100.64.0.1      -> 100.64.0.3      48 CS6   ESP
8  138     0.412033    100.64.0.2      -> 100.64.0.1      48 CS6   ESP
9  138     0.412033    100.64.0.1      -> 100.64.0.2      48 CS6   ESP
10 138     0.421035    100.64.0.1      -> 100.64.0.2      48 CS6   ESP
... 275 more lines

R1 no monitor capture ROBOTCAP
${output} =
```

PASS Protected Endpoint Addresses Never Appear In Cleartext On The Link (15.7s, 9 commands)

The strongest statement this lab can make about encryption: traffic between the loopbacks crosses the wire without either loopback address being visible, and with no cleartext ICMP.

```
R1 no monitor capture ROBOTCAP
Capture does not exist

R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
${output} =

R1 monitor capture ROBOTCAP match ipv4 any any
${output} =

R1 monitor capture ROBOTCAP buffer size 5
${output} =

R1 monitor capture ROBOTCAP start
Started capture point : ROBOTCAP

R1 ping 172.16.2.1 source Loopback0 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
.....
Success rate is 0 percent (0/5)

R1 monitor capture ROBOTCAP stop
Stopped capture point : ROBOTCAP

R1 show monitor capture ROBOTCAP buffer brief
-----
#   size   timestamp      source          destination      dscp   protocol
-----
0  138     0.000000    100.64.0.3      -> 100.64.0.1      48 CS6   ESP
```

```

1 138 0.000000 100.64.0.1 -> 100.64.0.3 48 CS6 ESP
2 138 0.016998 100.64.0.1 -> 100.64.0.2 48 CS6 ESP
3 138 0.026000 100.64.0.1 -> 100.64.0.2 48 CS6 ESP
4 138 0.026992 100.64.0.2 -> 100.64.0.1 48 CS6 ESP
5 138 0.159996 100.64.0.1 -> 100.64.0.3 48 CS6 ESP
6 138 0.160988 100.64.0.3 -> 100.64.0.1 48 CS6 ESP
7 138 0.279985 100.64.0.2 -> 100.64.0.1 48 CS6 ESP
8 138 0.279985 100.64.0.1 -> 100.64.0.2 48 CS6 ESP
9 138 0.331984 100.64.0.3 -> 100.64.0.1 48 CS6 ESP
10 138 0.391978 100.64.0.3 -> 100.64.0.1 48 CS6 ESP
... 269 more lines

```

R1 no monitor capture ROBOTCAP

```

${output} =

```

PASS Unprotected Link Traffic Is Visible As Cleartext ICMP (2.5s, 9 commands)

Positive control for the capture itself. Without this, the absence asserted above could just mean the capture was broken.

R1 no monitor capture ROBOTCAP

```

Capture does not exist

```

R1 monitor capture ROBOTCAP interface GigabitEthernet2 both

```

${output} =

```

R1 monitor capture ROBOTCAP match ipv4 any any

```

${output} =

```

R1 monitor capture ROBOTCAP buffer size 5

```

${output} =

```

R1 monitor capture ROBOTCAP start

```

Started capture point : ROBOTCAP

```

R1 ping 100.64.0.2 repeat 5

```

Type escape sequence to abort.

```

```

Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:

```

```

!!!!

```

```

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

```

R1 monitor capture ROBOTCAP stop

```

Stopped capture point : ROBOTCAP

```

R1 show monitor capture ROBOTCAP buffer brief

```

-----
#   size  timestamp      source          destination    dscp  protocol
-----
0  138    0.000000    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
1  138    0.000000    100.64.0.1      -> 100.64.0.3    48 CS6  ESP
2  114    0.020995    100.64.0.1      -> 100.64.0.2    0 BE    ICMP
3  114    0.020995    100.64.0.2      -> 100.64.0.1    0 BE    ICMP
4  114    0.021987    100.64.0.1      -> 100.64.0.2    0 BE    ICMP
5  114    0.022994    100.64.0.2      -> 100.64.0.1    0 BE    ICMP
6  114    0.023986    100.64.0.1      -> 100.64.0.2    0 BE    ICMP
7  114    0.023986    100.64.0.2      -> 100.64.0.1    0 BE    ICMP
8  114    0.023986    100.64.0.1      -> 100.64.0.2    0 BE    ICMP
9  114    0.024993    100.64.0.2      -> 100.64.0.1    0 BE    ICMP
10 114    0.024993    100.64.0.1      -> 100.64.0.2    0 BE    ICMP
... 54 more lines

```

R1 no monitor capture ROBOTCAP

```

${output} =

```

07 Ipsec Negative

PASS IKEv2 Refuses To Establish With A Mismatched Pre-Shared Key (13.3s, 13 commands)

Proves authentication is actually enforced. Without this, a tunnel that accepted any peer would still pass suite 04.

```
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 crypto ikev2 keyring LAB-KR
  ${output} =
R2 peer ANY
  ${output} =
R2 pre-shared-key WrongKeyOnPurpose999
  ${output} =
R2 end
  ${output} =
R1 clear crypto sa
  ${output} =
R1 clear crypto ikev2 sa
  ${output} =
R2 clear crypto sa
  ${output} =
R2 clear crypto ikev2 sa
  ${output} =
R3 clear crypto sa
  ${output} =
R3 clear crypto ikev2 sa
  ${output} =
R1 ping 172.16.2.1 source Loopback0 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
  Packet sent with a source address of 172.16.1.1
  .....
  Success rate is 0 percent (0/5)
R2 show crypto ikev2 sa
  ${output} =
```

PASS Tunnel Recovers Once The Correct Key Is Restored (14.3s, 13 commands)

Also guarantees the previous test cannot leave the lab broken.

```
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 crypto ikev2 keyring LAB-KR
  ${output} =
R2 peer ANY
  ${output} =
R2 pre-shared-key LabPreSharedKey123
  ${output} =
R2 end
  ${output} =
R1 clear crypto sa
  ${output} =
R1 clear crypto ikev2 sa
  ${output} =
R2 clear crypto sa
  ${output} =
R2 clear crypto ikev2 sa
  ${output} =
R3 clear crypto sa
  ${output} =
R3 clear crypto ikev2 sa
  ${output} =
R1 ping 10.20.2.1 source Loopback1 repeat 5
  Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
...!!
Success rate is 40 percent (2/5), round-trip min/avg/max = 1/1/1 ms
```

R1 ping 10.20.2.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

PASS IPsec Rekeys And Traffic Still Flows Afterwards (96.1s, 37 commands)

Drops the SA lifetime to the 120s platform minimum, waits for the outbound SPI to rotate, then re-checks connectivity. Rekey is where real tunnels break.

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 crypto ipsec security-association lifetime seconds 120

Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec

R1 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 crypto ipsec security-association lifetime seconds 120

Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec

R2 end

#{output} =

R3 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3 crypto ipsec security-association lifetime seconds 120

Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec

R3 end

#{output} =

R1 clear crypto sa

#{output} =

R1 clear crypto ikev2 sa

#{output} =

R2 clear crypto sa

#{output} =

R2 clear crypto ikev2 sa

#{output} =

R3 clear crypto sa

#{output} =

R3 clear crypto ikev2 sa

#{output} =

R1 ping 10.20.2.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
...!!
Success rate is 40 percent (2/5), round-trip min/avg/max = 1/1/1 ms
```

R1 ping 10.20.2.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 68, #pkts encrypt: 68, #pkts digest: 68
    #pkts decaps: 69, #pkts decrypt: 69, #pkts verify: 69
    #pkts compressed: 0, #pkts decompressed: 0
```

```
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 68, #pkts encrypt: 68, #pkts digest: 68
#pkts decaps: 69, #pkts decrypt: 69, #pkts verify: 69
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 133, #pkts encrypt: 133, #pkts digest: 133
#pkts decaps: 131, #pkts decrypt: 131, #pkts verify: 131
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 192, #pkts encrypt: 192, #pkts digest: 192
#pkts decaps: 191, #pkts decrypt: 191, #pkts verify: 191
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 251, #pkts encrypt: 251, #pkts digest: 251
#pkts decaps: 249, #pkts decrypt: 249, #pkts verify: 249
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 309, #pkts encrypt: 309, #pkts digest: 309
#pkts decaps: 309, #pkts decrypt: 309, #pkts verify: 309
```

```

#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 369, #pkts encrypt: 369, #pkts digest: 369
#pkts decaps: 373, #pkts decrypt: 373, #pkts verify: 373
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 436, #pkts encrypt: 436, #pkts digest: 436
#pkts decaps: 434, #pkts decrypt: 434, #pkts verify: 434
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 495, #pkts encrypt: 495, #pkts digest: 495
#pkts decaps: 494, #pkts decrypt: 494, #pkts verify: 494
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 550, #pkts encrypt: 550, #pkts digest: 550
#pkts decaps: 548, #pkts decrypt: 548, #pkts verify: 548
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

R1 ping 10.20.2.1 source Loopback1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms

R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R1 no crypto ipsec security-association lifetime seconds 120

```



```
    ${output} =  
R1 end  
    ${output} =  
R2 configure terminal  
    Enter configuration commands, one per line. End with CNTL/Z.  
R2 no crypto ipsec security-association lifetime seconds 120  
    ${output} =  
R2 end  
    ${output} =  
R3 configure terminal  
    Enter configuration commands, one per line. End with CNTL/Z.  
R3 no crypto ipsec security-association lifetime seconds 120  
    ${output} =  
R3 end  
    ${output} =
```

08 Management Api

PASS RESTCONF Reports The Same Hostname As The CLI (0.4s, 3 commands)

```
R1 show running-config | include ^hostname
hostname c8000v-r1
R2 show running-config | include ^hostname
hostname c8000v-r2
R3 show running-config | include ^hostname
hostname c8000v-r3
```

PASS RESTCONF Interface State Agrees With The CLI (0.1s, 1 commands)

```
R1 show ip interface brief GigabitEthernet2
Interface      IP-Address      OK? Method Status          Protocol
GigabitEthernet2 100.64.0.1      YES TFTP  up              up
```

PASS Config Pushed Over RESTCONF Appears In The CLI (0.3s, 1 commands)

Write path plus cross-check: RESTCONF and the CLI must be two views of one datastore, not two datastores.

```
R1 show running-config interface Loopback200
Building configuration...

Current configuration : 68 bytes
!
interface Loopback200
 ip address 198.51.100.1 255.255.255.0
end
```

PASS Config Deleted Over RESTCONF Disappears From The CLI (0.8s, 1 commands)

Runs after the PUT test, which creates the interface; a 404 here would mean the create silently failed.

```
R1 show ip interface brief Loopback200
^
% Invalid input detected at '^' marker.
```

PASS NETCONF Reports The Same Hostname As The CLI (0.8s, 1 commands)

```
R1 show running-config | include ^hostname
hostname c8000v-r1
```

PASS NETCONF Advertises The Expected Base Capabilities (0.5s, 0 commands)

IOS-XE does not list per-module URIs such as Cisco-IOS-XE-native in the hello; modules are exposed via yang-library. That the native model is usable is proven by the hostname test above, which queries the native namespace directly.

No device commands (assertion used earlier output).

09 Bgp

PASS BGP Sessions Are Established With The Expected Peer AS (0.1s, 4 commands)

The hub holds one session per spoke; each spoke holds exactly one.

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 40, main routing table version 40
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 139/130 prefixes, 145/136 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:51:43.657 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 40, main routing table version 40
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 139/130 prefixes, 145/136 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:51:43.692 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R2 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 70, main routing table version 70
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6168 total bytes of memory
BGP activity 172/163 prefixes, 174/165 paths, scan interval 60 secs
31 networks peaked at 22:34:42 Sep 7 2026 UTC (00:51:00.728 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

R3 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 152, main routing table version 152
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5952 total bytes of memory
BGP activity 75/66 prefixes, 80/71 paths, scan interval 60 secs
9 networks peaked at 22:15:22 Sep 7 2026 UTC (01:10:20.761 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

PASS Each Spoke Learns The Other Spoke Prefix Through The Hub (0.1s, 2 commands)

The defining property of this topology: spokes never peer with each other, so R3's prefix can only reach R2 via the hub, and its AS path must show both hops.

R2 show bgp ipv4 unicast 10.20.3.1

```
BGP routing table entry for 10.20.3.1/32, version 67
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 23:24:08 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1

```
BGP routing table entry for 10.20.2.1/32, version 150
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 7 2026 23:24:34 UTC
```

PASS Every Peer Is Configured With MD5 Authentication (0.4s, 3 commands)

The negative test below proves a wrong password is rejected; this proves no peer was left without one in the first place.

R1 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.2 remote-as 65002
neighbor 172.20.0.2 password LabBgpSecret
neighbor 172.20.0.3 remote-as 65003
neighbor 172.20.0.3 password LabBgpSecret
```

R2 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 password LabBgpSecret
```

R3 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 password LabBgpSecret
```

PASS A BFD Session Is Up For Every BGP Peer (0.1s, 3 commands)

One BFD session per peer, on the tunnel the peer address lives on -- the hub therefore holds two.

R1 show bfd neighbors

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.2	4097/4102	Up	Up	Tu0
172.20.0.3	4098/4097	Up	Up	Tu0

R2 show bfd neighbors

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.1	4102/4097	Up	Up	Tu0

R3 show bfd neighbors

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.1	4097/4098	Up	Up	Tu0

PASS BGP Uses BFD For Fall-over On Every Peer (0.4s, 6 commands)

A BFD session that no protocol consumes would detect failures and change nothing, so this checks BGP is the registered client rather than just that BFD is running.

R1 show running-config | include ^ neighbor .* remote-as|fall-over bfd

```
neighbor 172.20.0.2 remote-as 65002
neighbor 172.20.0.2 fall-over bfd
neighbor 172.20.0.3 remote-as 65003
neighbor 172.20.0.3 fall-over bfd
```

R1 show bfd neighbors client bgp

IPv4 Sessions				
NeighAddr	LD/RD	RH/RS	State	Int
172.20.0.2	4097/4102	Up	Up	Tu0

```

172.20.0.3          4098/4097      Up        Up        Tu0
R2 show running-config | include ^ neighbor .* remote-as|fall-over bfd
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 fall-over bfd
R2 show bfd neighbors client bgp
IPv4 Sessions
NeighAddr          LD/RD          RH/RS      State      Int
172.20.0.1         4102/4097      Up         Up         Tu0
R3 show running-config | include ^ neighbor .* remote-as|fall-over bfd
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 fall-over bfd
R3 show bfd neighbors client bgp
IPv4 Sessions
NeighAddr          LD/RD          RH/RS      State      Int
172.20.0.1         4097/4098      Up         Up         Tu0

```

PASS BFD Tears The Session Down Long Before The Hold Time (42.8s, 32 commands)

The test that makes BFD more than configuration. An ACL on the hub's tunnel drops only BFD control packets (UDP 3784 and 3785) and leaves BGP's TCP session untouched, so ordinary BGP liveness is unaffected. Without fall-over the session would survive until the 30s hold time expires; with it, BFD declares the peer down in about a second and BGP follows.

```

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 40, main routing table version 40
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 139/130 prefixes, 145/136 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:51:44.666 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 ip access-list extended BLOCK-BFD
${output} =
R1 deny udp any any eq 3784
${output} =
R1 deny udp any any eq 3785
${output} =
R1 permit ip any any
${output} =
R1 end
${output} =
R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Tunnel0
${output} =
R1 ip access-group BLOCK-BFD in
${output} =
R1 end
${output} =
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      17      19      40      0      0 00:01:35      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      17      20      40      0      0 00:01:36      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      0        0        1      0      0 00:00:00 Idle
R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Tunnel0
${output} =

```

```

R1 no ip access-group BLOCK-BFD in
  ${output} =
R1 end
  ${output} =
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 no ip access-list extended BLOCK-BFD
  ${output} =
R1 end
  ${output} =
R1 show bgp ipv4 unicast summary
  BGP router identifier 172.16.1.1, local AS number 65001
  BGP table version is 46, main routing table version 46
  3 network entries using 744 bytes of memory
  3 path entries using 408 bytes of memory
  1/1 BGP path/bestpath attribute entries using 296 bytes of memory
  0 BGP route-map cache entries using 0 bytes of memory
  0 BGP filter-list cache entries using 0 bytes of memory
  BGP using 1448 total bytes of memory
  BGP activity 139/136 prefixes, 145/142 paths, scan interval 60 secs
  22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:51:47.175 ago)

  Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
  172.20.0.2    4      65002      0       0        1    0    0 00:00:01 Idle
  172.20.0.3    4      65003      0       0        1    0    0 00:00:00 Idle

R1 show bgp ipv4 unicast summary
  BGP router identifier 172.16.1.1, local AS number 65001
  BGP table version is 49, main routing table version 49
  6 network entries using 1488 bytes of memory
  6 path entries using 816 bytes of memory
  4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
  1 BGP AS-PATH entries using 24 bytes of memory
  3 BGP community entries using 72 bytes of memory
  3 BGP route-map cache entries using 216 bytes of memory
  0 BGP filter-list cache entries using 0 bytes of memory
  BGP using 3800 total bytes of memory
  BGP activity 142/136 prefixes, 148/142 paths, scan interval 60 secs
  22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:51:52.214 ago)

  Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
  ... 2 more lines

R1 show bgp ipv4 unicast summary
  BGP router identifier 172.16.1.1, local AS number 65001
  BGP table version is 52, main routing table version 52
  9 network entries using 2232 bytes of memory
  9 path entries using 1224 bytes of memory
  7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
  2 BGP AS-PATH entries using 48 bytes of memory
  6 BGP community entries using 144 bytes of memory
  6 BGP route-map cache entries using 432 bytes of memory
  0 BGP filter-list cache entries using 0 bytes of memory
  BGP using 6152 total bytes of memory
  BGP activity 145/136 prefixes, 151/142 paths, scan interval 60 secs
  22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:51:57.255 ago)

  Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
  ... 2 more lines

R1 show ip route 10.20.2.1
  Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:05 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:00:05 ago
    opaque_ptr 0x747B4EAD51F8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none

R3 show ip route 10.20.2.1
  % Subnet not in table

R3 show ip route 10.20.2.1
  % Subnet not in table

R3 show ip route 10.20.2.1
  % Subnet not in table

```

```

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
% Subnet not in table

R3 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:05 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:05 ago
      opaque_ptr 0x75671E6AB998
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none

```

PASS Every Peer Has An Outbound Route-map Over Per-class Prefix-lists (0.6s, 13 commands)

One prefix-list per prefix class, and a route-map clause per class so each can carry its own community. The clauses are the whitelist: the route-map's implicit deny is what stops anything unnamed being advertised.

```

R2 show running-config | include route-map RM-OUT-R2 out
neighbor 172.20.0.2 route-map RM-OUT-R2 out

R3 show running-config | include route-map RM-OUT-R3 out
neighbor 172.20.0.3 route-map RM-OUT-R3 out

R1 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.1.1/32

R1 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.1.0/24

R1 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.10.0/24

R1 show running-config | include route-map RM-OUT-R1 out
neighbor 172.20.0.1 route-map RM-OUT-R1 out

R2 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.2.1/32

R2 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.2.0/24

R2 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.20.0/24

R1 show running-config | include route-map RM-OUT-R1 out
neighbor 172.20.0.1 route-map RM-OUT-R1 out

R3 show ip prefix-list PL-OWN-L0
ip prefix-list PL-OWN-L0: 1 entries
seq 5 permit 10.20.3.1/32

R3 show ip prefix-list PL-OWN-NAT
ip prefix-list PL-OWN-NAT: 1 entries
seq 5 permit 10.30.3.0/24

R3 show ip prefix-list PL-OWN-LAN
ip prefix-list PL-OWN-LAN: 1 entries
seq 5 permit 192.168.30.0/24

```

PASS Every Peer Has An Inbound Policy Matching Community And AS Path (0.6s, 11 commands)

*Both matches sit in one permit clause, so they are ANDed and the clause's implicit deny is the drop: a prefix with an unknown community *or* an unexpected AS path falls through.*

```

R2 show running-config | include route-map RM-IN-R2 in
neighbor 172.20.0.2 route-map RM-IN-R2 in

R2 show route-map RM-IN-R2

```

```

route-map RM-IN-R2, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 10
    community (community-list filter): CL-VALID
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
R3 show running-config | include route-map RM-IN-R3 in
neighbor 172.20.0.3 route-map RM-IN-R3 in
R3 show route-map RM-IN-R3
route-map RM-IN-R3, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 20
    community (community-list filter): CL-VALID
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
R1 show ip community-list CL-VALID
Named Community standard list CL-VALID
  permit 65001:1
  permit 65001:2
  permit 65001:3
  permit 65002:1
  permit 65002:2
  permit 65002:3
  permit 65003:1
  permit 65003:2
  permit 65003:3
R1 show running-config | include route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-IN-R1 in
R1 show route-map RM-IN-R1
route-map RM-IN-R1, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 10
    community (community-list filter): CL-VALID
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
R2 show ip community-list CL-VALID
Named Community standard list CL-VALID
  permit 65001:1
  permit 65001:2
  permit 65001:3
  permit 65002:1
  permit 65002:2
  permit 65002:3
  permit 65003:1
  permit 65003:2
  permit 65003:3
R1 show running-config | include route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-IN-R1 in
R1 show route-map RM-IN-R1
route-map RM-IN-R1, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 10
    community (community-list filter): CL-VALID
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
R3 show ip community-list CL-VALID
Named Community standard list CL-VALID
  permit 65001:1
  permit 65001:2
  permit 65001:3
  permit 65002:1
  permit 65002:2
  permit 65002:3
  permit 65003:1
  permit 65003:2
  permit 65003:3

```

PASS Every Received Prefix Carries The Community Of Its Origin (0.6s, 18 commands)

A prefix is tagged by the router that owns it, so the community identifies both the origin AS and the kind of prefix. Checked on the receiver, which is the only place it proves the tag survived the wire.

```

R1 show bgp ipv4 unicast 10.20.2.1/32
BGP routing table entry for 10.20.2.1/32, version 52

```



```

Paths: (1 available, best #1, table default)
Advertised to update-groups:
 33
Refresh Epoch 1
65002
 172.20.0.2 from 172.20.0.2 (172.16.2.1)
  Origin IGP, metric 0, localpref 100, valid, external, best
  Community: 65002:1
  rx pathid: 0, tx pathid: 0x0
  Updated on Sep 7 2026 23:25:51 UTC

R1 show bgp ipv4 unicast 10.20.3.1/32
BGP routing table entry for 10.20.3.1/32, version 49
Paths: (1 available, best #1, table default)
Advertised to update-groups:
 34
Refresh Epoch 1
65003
 172.20.0.3 from 172.20.0.3 (172.16.3.1)
  Origin IGP, metric 0, localpref 100, valid, external, best
  Community: 65003:1
  rx pathid: 0, tx pathid: 0x0
  Updated on Sep 7 2026 23:25:50 UTC

R1 show bgp ipv4 unicast 10.30.2.0/24
BGP routing table entry for 10.30.2.0/24, version 51
Paths: (1 available, best #1, table default)
Advertised to update-groups:
 33
Refresh Epoch 1
65002
 172.20.0.2 from 172.20.0.2 (172.16.2.1)
  Origin IGP, metric 0, localpref 100, valid, external, best
  Community: 65002:2
  rx pathid: 0, tx pathid: 0x0
  Updated on Sep 7 2026 23:25:51 UTC

R1 show bgp ipv4 unicast 10.30.3.0/24
BGP routing table entry for 10.30.3.0/24, version 48
Paths: (1 available, best #1, table default)
Advertised to update-groups:
 34
Refresh Epoch 1
65003
 172.20.0.3 from 172.20.0.3 (172.16.3.1)
  Origin IGP, metric 0, localpref 100, valid, external, best
  Community: 65003:2
  rx pathid: 0, tx pathid: 0x0
  Updated on Sep 7 2026 23:25:50 UTC

R1 show bgp ipv4 unicast 192.168.20.0/24
BGP routing table entry for 192.168.20.0/24, version 50
Paths: (1 available, best #1, table default)
Advertised to update-groups:
 33
Refresh Epoch 1
65002
 172.20.0.2 from 172.20.0.2 (172.16.2.1)
  Origin IGP, metric 0, localpref 100, valid, external, best
  Community: 65002:3
  rx pathid: 0, tx pathid: 0x0
  Updated on Sep 7 2026 23:25:51 UTC

R1 show bgp ipv4 unicast 192.168.30.0/24
BGP routing table entry for 192.168.30.0/24, version 47
Paths: (1 available, best #1, table default)
Advertised to update-groups:
 34
Refresh Epoch 1
65003
 172.20.0.3 from 172.20.0.3 (172.16.3.1)
  Origin IGP, metric 0, localpref 100, valid, external, best
  Community: 65003:3
  rx pathid: 0, tx pathid: 0x0
  Updated on Sep 7 2026 23:25:50 UTC

R2 show bgp ipv4 unicast 10.20.1.1/32
BGP routing table entry for 10.20.1.1/32, version 79
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
 172.20.0.1 from 172.20.0.1 (172.16.1.1)
  Origin IGP, metric 0, localpref 100, valid, external, best

```

Community: 65001:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:51 UTC

R2 show bgp ipv4 unicast 10.20.3.1/32

BGP routing table entry for 10.20.3.1/32, version 80
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65003
172.20.0.3 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:51 UTC

R2 show bgp ipv4 unicast 10.30.1.0/24

BGP routing table entry for 10.30.1.0/24, version 78
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:51 UTC

R2 show bgp ipv4 unicast 10.30.3.0/24

BGP routing table entry for 10.30.3.0/24, version 81
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65003
172.20.0.3 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:51 UTC

R2 show bgp ipv4 unicast 192.168.10.0/24

BGP routing table entry for 192.168.10.0/24, version 77
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:51 UTC

R2 show bgp ipv4 unicast 192.168.30.0/24

BGP routing table entry for 192.168.30.0/24, version 82
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65003
172.20.0.3 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:51 UTC

R3 show bgp ipv4 unicast 10.20.1.1/32

BGP routing table entry for 10.20.1.1/32, version 161
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:50 UTC

R3 show bgp ipv4 unicast 10.20.2.1/32

BGP routing table entry for 10.20.2.1/32, version 162
Paths: (1 available, best #1, table default)
Flag: 0x8100
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best

```
Community: 65002:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:26:21 UTC
```

R3 show bgp ipv4 unicast 10.30.1.0/24

```
BGP routing table entry for 10.30.1.0/24, version 160
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:50 UTC
```

R3 show bgp ipv4 unicast 10.30.2.0/24

```
BGP routing table entry for 10.30.2.0/24, version 163
Paths: (1 available, best #1, table default)
Flag: 0x8100
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:2
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:26:21 UTC
```

R3 show bgp ipv4 unicast 192.168.10.0/24

```
BGP routing table entry for 192.168.10.0/24, version 159
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001
172.20.0.1 from 172.20.0.1 (172.16.1.1)
Origin IGP, metric 0, localpref 100, valid, external, best
Community: 65001:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:50 UTC
```

R3 show bgp ipv4 unicast 192.168.20.0/24

```
BGP routing table entry for 192.168.20.0/24, version 164
Paths: (1 available, best #1, table default)
Flag: 0x8100
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:26:21 UTC
```

PASS Re-advertised Prefixes Keep Their Origin Community Through The Hub (0.1s, 4 commands)

A spoke-to-spoke prefix crosses the hub, which re-advertises it. If the hub re-tagged it the community would say 65001 and the origin would be lost, so this asserts the far spoke's own community survives the hop.

R2 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 80
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65003
172.20.0.3 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:25:51 UTC
```

R2 show bgp ipv4 unicast 192.168.30.0/24

```
BGP routing table entry for 192.168.30.0/24, version 82
Paths: (1 available, best #1, table default)
Not advertised to any peer
Refresh Epoch 1
65001 65003
172.20.0.3 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65003:3
rx pathid: 0, tx pathid: 0x0
```

Updated on Sep 7 2026 23:25:51 UTC

R3 show bgp ipv4 unicast 10.20.2.1/32

BGP routing table entry for 10.20.2.1/32, version 162
Paths: (1 available, best #1, table default)
Flag: 0x8100
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:1
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:26:21 UTC

R3 show bgp ipv4 unicast 192.168.20.0/24

BGP routing table entry for 192.168.20.0/24, version 164
Paths: (1 available, best #1, table default)
Flag: 0x8100
Not advertised to any peer
Refresh Epoch 1
65001 65002
172.20.0.2 from 172.20.0.1 (172.16.1.1)
Origin IGP, localpref 100, valid, external, best
Community: 65002:3
rx pathid: 0, tx pathid: 0x0
Updated on Sep 7 2026 23:26:21 UTC

PASS A Prefix With An Unknown Community Is Rejected Inbound (26.0s, 40 commands)

*The prefix is explicitly permitted *outbound* by a temporary clause, so the outbound whitelist cannot be what stops it -- only the inbound community match can. The test asserts R2 really did advertise it before asserting R1 refused it, otherwise it would pass whenever the advertisement failed.*

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback98

#{output} =

R2 ip address 10.88.88.88 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 route-map RM-OUT-R1 permit 5

#{output} =

R2 match ip address prefix-list PL-TEST

#{output} =

R2 set community 65500:9

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 address-family ipv4 unicast

#{output} =

R2 network 10.88.88.88 mask 255.255.255.255

#{output} =

R2 end

#{output} =

R2 clear ip bgp * soft out

#{output} =

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 82, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

```

        x best-external, a additional-path, c RIB-compressed,
        t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 10.88.88.88/32	0.0.0.0	0		32768	i
*> 192.168.20.0	0.0.0.0	0		32768	i

... 1 more lines

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```

BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 10.88.88.88/32	0.0.0.0	0		32768	i
*> 192.168.20.0	0.0.0.0	0		32768	i

... 1 more lines

R1 show bgp ipv4 unicast 10.88.88.88

% Network not in table

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 address-family ipv4 unicast

#{output} =

R2 no network 10.88.88.88 mask 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no route-map RM-OUT-R1 permit 5

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip prefix-list PL-TEST

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no interface Loopback98

#{output} =

R2 end

#{output} =

PASS Every Peer Has A Maximum-prefix Limit With Headroom (0.5s, 6 commands)

Filtering controls what a peer may send; this controls how much. The limit must sit above normal load or it would trip in steady state -- a hub holds 3 prefixes per spoke, a spoke 6 -- and far below anything that could exhaust memory.

R1 show running-config | include ^ neighbor .* remote-as|maximum-prefix

```

neighbor 172.20.0.2 remote-as 65002
neighbor 172.20.0.3 remote-as 65003
neighbor 172.20.0.2 maximum-prefix 15 80
neighbor 172.20.0.3 maximum-prefix 15 80

```

R1 show bgp ipv4 unicast summary | begin Neighbor

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.2	4	65002	18	16	52	0	0	00:01:02	3
172.20.0.3	4	65003	13	16	52	0	0	00:01:03	3

```

R2 show running-config | include ^ neighbor .* remote-as|maximum-prefix
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 maximum-prefix 15 80

R2 show bgp ipv4 unicast summary | begin Neighbor
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1    4      65001    16     18     83     0   0 00:01:02        6

R3 show running-config | include ^ neighbor .* remote-as|maximum-prefix
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 maximum-prefix 15 80

R3 show bgp ipv4 unicast summary | begin Neighbor
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1    4      65001    16     13    164     0   0 00:01:03        6

```

PASS A Peer That Floods Prefixes Is Torn Down Rather Than Absorbed (37.0s, 344 commands)

R2 originates 20 extra prefixes, tagged with a valid community and carrying a valid AS path, and explicitly permitted outbound -- so neither the whitelist, the community match nor the AS-path filter can be what stops them. Only the prefix count is left. The hub must drop the session instead of accepting an unbounded table.

```

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip prefix-list PL-FL00D seq 5 permit 10.99.0.0/16 le 32
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 route-map RM-OUT-R1 permit 7
${output} =

R2 match ip address prefix-list PL-FL00D
${output} =

R2 set community 65002:1
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.0.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.1.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.2.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.3.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.4.0 255.255.255.0 Null0
${output} =

```

```

R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.5.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.6.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.7.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.8.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.9.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.10.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.11.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.12.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.13.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.14.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.15.0 255.255.255.0 Null0

```



```

    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.16.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.17.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.18.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 ip route 10.99.19.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.0.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.1.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.2.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.3.0 mask 255.255.255.0
    ${output} =

```

```

R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.4.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.5.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.6.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.7.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.8.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.9.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002

```

```

    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.10.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.11.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.12.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.13.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.14.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.15.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 network 10.99.16.0 mask 255.255.255.0
    ${output} =

```

```

R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.17.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.18.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.19.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      18      17      52      0      0 00:01:06      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      19      17      52      0      0 00:01:11      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      19      18      52      0      0 00:01:16      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      20      18      52      0      0 00:01:21      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      20      19      52      0      0 00:01:26      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      21      19      52      0      0 00:01:31      3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      0      0      1      0      0 00:00:04 Idle (PfxCt)
R1 show bgp ipv4 unicast summary | include 172.20.0.2
172.20.0.2      4      65002      0      0      1      0      0 00:00:04 Idle (PfxCt)
R1 show logging | include MAXPFEXCEED
000408: Sep  7 22:33:59.581 UTC: %BGP-3-MAXPFEXCEED: Number of prefixes received from 172.20.0.2 1/1
(afi/safi): 16 exceeds limit 15
000748: Sep  7 23:03:35.410 UTC: %BGP-3-MAXPFEXCEED: Number of prefixes received from 172.20.0.2 1/1
(afi/safi): 16 exceeds limit 15
001152: Sep  7 23:27:24.049 UTC: %BGP-3-MAXPFEXCEED: Number of prefixes received from 172.20.0.2 1/1
(afi/safi): 16 exceeds limit 15
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.0.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal

```

```

Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.0.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.1.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.1.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.2.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.2.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.3.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.3.0 255.255.255.0 Null0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 no network 10.99.4.0 mask 255.255.255.0
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

```

```

R2 no ip route 10.99.4.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.5.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.5.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.6.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.6.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.7.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.7.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.8.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.8.0 255.255.255.0 Null0

```

```

    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.9.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.9.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.10.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.10.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.11.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.11.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.12.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.12.0 255.255.255.0 Null0
    ${output} =

```

```

R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.13.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.13.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.14.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.14.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.15.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.15.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.16.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no ip route 10.99.16.0 255.255.255.0 Null0
  ${output} =
R2 end

```



```

    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.17.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.17.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.18.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.18.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
    ${output} =
R2 address-family ipv4 unicast
    ${output} =
R2 no network 10.99.19.0 mask 255.255.255.0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.19.0 255.255.255.0 Null0
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 no route-map RM-OUT-R1 permit 7
    ${output} =
R2 end
    ${output} =
R2 configure terminal
    Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip prefix-list PL-FLOOD
    ${output} =
R2 end
    ${output} =
R1 clear ip bgp 172.20.0.2
    ${output} =

```

PASS The Session Recovers Once The Flood Stops (25.2s, 8 commands)

A maximum-prefix teardown is deliberately sticky -- no "restart" is configured -- so recovery takes an operator clearing it. This asserts the lab returns to a converged state rather than leaving the next suite a broken topology.

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 79, main routing table version 79
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
8/4 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
7 BGP route-map cache entries using 504 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5392 total bytes of memory
BGP activity 158/152 prefixes, 164/158 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:53:32.901 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 82, main routing table version 82
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
8/7 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
7 BGP route-map cache entries using 504 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6544 total bytes of memory
BGP activity 161/152 prefixes, 167/158 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:53:37.936 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:00 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:00 ago
      opaque_ptr 0x747B4EAD51F8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:03 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:03 ago
      opaque_ptr 0x75671E6AB998
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

PASS A Prefix With An Invalid AS Path Is Rejected Inbound (10.8s, 38 commands)

Same prefix, this time with a valid community but an AS path this topology cannot produce. Only the AS-path match differs, so a rejection here isolates that filter.

```

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback98
${output} =
R2 ip address 10.88.88.88 255.255.255.255
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 route-map RM-OUT-R1 permit 5
${output} =
R2 match ip address prefix-list PL-TEST
${output} =
R2 set community 65002:1
${output} =
R2 set as-path prepend 65099
${output} =
R2 end
${output} =
R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2 router bgp 65002
${output} =
R2 address-family ipv4 unicast
${output} =
R2 network 10.88.88.88 mask 255.255.255.255
${output} =
R2 end
${output} =
R2 clear ip bgp * soft out
${output} =
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 136, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32      0.0.0.0              0         32768 i
*>  10.30.2.0/24      0.0.0.0              0         32768 i
*>  192.168.20.0      0.0.0.0              0         32768 i

Total number of prefixes 3
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 137, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32      0.0.0.0              0         32768 i
*>  10.30.2.0/24      0.0.0.0              0         32768 i
*>  192.168.20.0      0.0.0.0              0         32768 i

Total number of prefixes 3
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 137, local router ID is 172.16.2.1

```

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	10.88.88.88/32	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

... 1 more lines

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

BGP table version is 137, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.2.1/32	0.0.0.0	0		32768	i
*>	10.30.2.0/24	0.0.0.0	0		32768	i
*>	10.88.88.88/32	0.0.0.0	0		32768	i
*>	192.168.20.0	0.0.0.0	0		32768	i

... 1 more lines

R1 show bgp ipv4 unicast 10.88.88.88

% Network not in table

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 address-family ipv4 unicast

#{output} =

R2 no network 10.88.88.88 mask 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no route-map RM-OUT-R1 permit 5

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip prefix-list PL-TEST

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 no interface Loopback98

#{output} =

R2 end

#{output} =

PASS Each Peer Is Advertised Exactly Its Whitelisted Prefixes (0.3s, 4 commands)

Not "at least" -- exactly. A whitelist that permits more than intended still passes a containment check, so the advertised set is compared for equality against what the peer should get: a spoke sends only its own prefixes, the hub sends its own plus the far spoke's.

R2 show bgp ipv4 unicast neighbors 172.20.0.2 advertised-routes

BGP table version is 82, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

```

        r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
        x best-external, a additional-path, c RIB-compressed,
        t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	172.20.0.3	0		0 65003	i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.3.0/24	172.20.0.3	0		0 65003	i
*> 192.168.10.0	0.0.0.0	0		32768	i

... 3 more lines

R3 show bgp ipv4 unicast neighbors 172.20.0.3 advertised-routes

```

BGP table version is 82, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	172.20.0.2	0		0 65002	i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	172.20.0.2	0		0 65002	i
*> 192.168.10.0	0.0.0.0	0		32768	i

... 3 more lines

R1 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```

BGP table version is 138, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 192.168.20.0	0.0.0.0	0		32768	i

Total number of prefixes 3

R1 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```

BGP table version is 170, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.3.1/32	0.0.0.0	0		32768	i
*> 10.30.3.0/24	0.0.0.0	0		32768	i
*> 192.168.30.0	0.0.0.0	0		32768	i

Total number of prefixes 3

PASS A Prefix Outside The Whitelist Is Not Advertised (10.5s, 20 commands)

The point of a whitelist. A spoke originates a prefix nobody authorised; it appears in that spoke's own BGP table, so the origination worked, and must still never reach the hub.

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback99

#{output} =

R2 ip address 10.77.77.77 255.255.255.255

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

```

R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.77.77.77 mask 255.255.255.255
  ${output} =
R2 end
  ${output} =
R2 show bgp ipv4 unicast 10.77.77.77
BGP routing table entry for 10.77.77.77/32, version 139
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  Local
    0.0.0.0 from 0.0.0.0 (172.16.2.1)
    Origin IGP, metric 0, localpref 100, weight 32768, valid, sourced, local, best
    rx pathid: 0, tx pathid: 0x0
    Updated on Sep 7 2026 23:28:10 UTC
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
BGP table version is 139, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.2.1/32       0.0.0.0              0           32768 i
*>  10.30.2.0/24       0.0.0.0              0           32768 i
*>  192.168.20.0      0.0.0.0              0           32768 i

Total number of prefixes 3
R1 show bgp ipv4 unicast 10.77.77.77
% Network not in table
R2 configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.77.77.77 mask 255.255.255.255
  ${output} =
R2 end
  ${output} =
R2 configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2 no interface Loopback99
  ${output} =
R2 end
  ${output} =

```

PASS Each Router Receives The Peer Prefix (0.1s, 2 commands)

```

R1 show bgp ipv4 unicast
BGP table version is 82, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop           Metric LocPrf Weight Path
*>  10.20.1.1/32       0.0.0.0              0           32768 i
*>  10.20.2.1/32       172.20.0.2           0             0 65002 i
*>  10.20.3.1/32       172.20.0.3           0             0 65003 i
*>  10.30.1.0/24       0.0.0.0              0           32768 i
*>  10.30.2.0/24       172.20.0.2           0             0 65002 i
... 4 more lines

```

R2 show bgp ipv4 unicast

```
BGP table version is 140, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	172.20.0.1	0		0	65001 i
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	172.20.0.3			0	65001 65003 i
*> 10.30.1.0/24	172.20.0.1	0		0	65001 i
*> 10.30.2.0/24	0.0.0.0	0		32768	i

... 4 more lines

PASS Peer Prefix Is Learned Via BGP And Not A Static Route (0.0s, 1 commands)

Distinguishes a genuinely BGP-installed route from one that merely happens to exist. eBGP is distance 20; a static is 1.

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:42 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:00:42 ago
    opaque_ptr 0x747B4EAD51F8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none
```

PASS BGP Route Resolves Out The Encrypted Tunnel (0.0s, 1 commands)

The next hop is a tunnel address, so the forwarding path must recurse to Tunnel0 rather than the cleartext link.

R1 show ip cef 10.20.2.1

```
10.20.2.1/32
  nexthop 172.20.0.2 Tunnel0
```

PASS Traffic Over The BGP-Learned Route Reaches The Peer (0.0s, 1 commands)

End to end: this only succeeds because BGP installed the route.

R1 ping 10.20.2.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

PASS BGP-Routed Traffic Is Encrypted (5.7s, 7 commands)

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 968, #pkts encrypt: 968, #pkts digest: 968
  #pkts decaps: 966, #pkts decrypt: 966, #pkts verify: 966
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2105, #pkts encrypt: 2105, #pkts digest: 2105
  #pkts decaps: 2100, #pkts decrypt: 2100, #pkts verify: 2100
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines

```

R1 ping 10.20.2.1 source Loopback1 repeat 20

```

Type escape sequence to abort.
Sending 20, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (20/20), round-trip min/avg/max = 1/1/2 ms

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 968, #pkts encrypt: 968, #pkts digest: 968
  #pkts decaps: 978, #pkts decrypt: 978, #pkts verify: 978
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2105, #pkts encrypt: 2105, #pkts digest: 2105
  #pkts decaps: 2100, #pkts decrypt: 2100, #pkts verify: 2100
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 33 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 1024, #pkts encrypt: 1024, #pkts digest: 1024
  #pkts decaps: 1027, #pkts decrypt: 1027, #pkts verify: 1027
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
current_peer 100.64.0.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2160, #pkts encrypt: 2160, #pkts digest: 2160

```



```

#pkts decaps: 2159, #pkts decrypt: 2159, #pkts verify: 2159
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 33 more lines

```

PASS BGP Control Plane Traffic Is Encrypted On The Wire (18.8s, 8 commands)

The reason for peering over the tunnel. Captures a keepalive window and asserts the session is invisible: no TCP, no peering addresses, no advertised prefixes -- only ESP.

R1 no monitor capture BGPCAP

Capture does not exist

R1 monitor capture BGPCAP interface GigabitEthernet2 both

#{output} =

R1 monitor capture BGPCAP match ipv4 any any

#{output} =

R1 monitor capture BGPCAP buffer size 5

#{output} =

R1 monitor capture BGPCAP start

Started capture point : BGPCAP

R1 monitor capture BGPCAP stop

Stopped capture point : BGPCAP

R1 show monitor capture BGPCAP buffer brief

```

-----
#  size  timestamp      source      destination  dscp  protocol
-----
0  138    0.000000    100.64.0.2   -> 100.64.0.1   48 CS6  ESP
1  138    0.000000    100.64.0.1   -> 100.64.0.2   48 CS6  ESP
2  138    0.063992    100.64.0.1   -> 100.64.0.2   48 CS6  ESP
3  138    0.224994    100.64.0.1   -> 100.64.0.2   48 CS6  ESP
4  138    0.224994    100.64.0.2   -> 100.64.0.1   48 CS6  ESP
5  138    0.248980    100.64.0.1   -> 100.64.0.3   48 CS6  ESP
6  138    0.248980    100.64.0.3   -> 100.64.0.1   48 CS6  ESP
7  138    0.350040    100.64.0.2   -> 100.64.0.1   48 CS6  ESP
8  138    0.426025    100.64.0.3   -> 100.64.0.1   48 CS6  ESP
9  138    0.426025    100.64.0.1   -> 100.64.0.3   48 CS6  ESP
10 138    0.445036    100.64.0.2   -> 100.64.0.1   48 CS6  ESP
... 319 more lines

```

R1 no monitor capture BGPCAP

#{output} =

PASS Withdrawing A Prefix Removes It From The Peer (60.5s, 23 commands)

Shuts the advertised interface on R2 and requires R1 to lose both the BGP path and the route, then restores it.

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback1

#{output} =

R2 shutdown

#{output} =

R2 end

#{output} =

R1 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:07 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:07 ago
    opaque_ptr 0x747B4EAD51F8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none

```

R1 show ip route 10.20.2.1

```

Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0

```

```
Tag 65002, type external
Last update from 172.20.0.2 00:01:12 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.2, 00:01:12 ago
  opaque_ptr 0x747B4EAD51F8
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 172.20.0.2 00:01:17 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.2, 00:01:17 ago
  opaque_ptr 0x747B4EAD51F8
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 172.20.0.2 00:01:22 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.2, 00:01:22 ago
  opaque_ptr 0x747B4EAD51F8
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 172.20.0.2 00:01:27 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.2, 00:01:27 ago
  opaque_ptr 0x747B4EAD51F8
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
Known via "bgp 65001", distance 20, metric 0
Tag 65002, type external
Last update from 172.20.0.2 00:01:32 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.2, 00:01:32 ago
  opaque_ptr 0x747B4EAD51F8
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65002
  MPLS label: none
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R2 configure terminal

```
Enter configuration commands, one per line. End with CNTL/Z.
```

R2 interface Loopback1

```
${output} =
```

R2 no shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

```

R1 show ip route 10.20.2.1
% Subnet not in table

R1 show ip route 10.20.2.1
% Subnet not in table

R1 show ip route 10.20.2.1
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:04 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:04 ago
      opaque_ptr 0x747B4EAD51F8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none

```

```

R2 show ip route 10.20.1.1
Routing entry for 10.20.1.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:02:07 ago
  Routing Descriptor Blocks:
    * 172.20.0.1, from 172.20.0.1, 00:02:07 ago
      opaque_ptr 0x7E32CED11C60
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none

```

PASS Session Recovers After A Hard Clear (10.2s, 5 commands)

```

R1 clear ip bgp *
${output} =

```

```

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 165/162 prefixes, 171/168 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:55:44.780 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

```

```

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 165/162 prefixes, 171/168 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:55:49.814 ago)

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines

```

```

R1 show bgp ipv4 unicast summary
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 10, main routing table version 10
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory

```

BGP activity 171/162 prefixes, 177/168 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:55:54.853 ago)

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:01 ago
  Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:01 ago
      opaque_ptr 0x747B4EAD51F8
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

PASS Mismatched MD5 Password Prevents The Session Coming Up (50.4s, 19 commands)

Negative control on BGP authentication, mirroring the IPsec wrong-PSK test: without it, a session that accepted any peer would still pass everything above.

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 neighbor 172.20.0.1 password WrongBgpSecret999

#{output} =

R2 end

#{output} =

R1 clear ip bgp *

#{output} =

R2 clear ip bgp *

#{output} =

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5000 total bytes of memory
BGP activity 177/171 prefixes, 183/177 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:56:40.023 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002

#{output} =

R2 neighbor 172.20.0.1 password LabBgpSecret

#{output} =

R2 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback1

#{output} =

R2 no shutdown

#{output} =

R2 end

#{output} =

R1 clear ip bgp *

#{output} =

R2 clear ip bgp *

`${output} =`

R1 show bgp ipv4 unicast summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3848 total bytes of memory
BGP activity 180/177 prefixes, 186/183 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:56:40.214 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
... 2 more lines									

R1 show bgp ipv4 unicast summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3848 total bytes of memory
BGP activity 180/177 prefixes, 186/183 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:56:45.249 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
... 2 more lines									

10 Host To Host

PASS Every Host Holds Its LAN Address (0.1s, 3 commands)

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
... 70 more lines
```

PASS Every Host Routes The Other Two LANs Via Its Own Router (0.0s, 3 commands)

H1 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
```

H2 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
```

H3 ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
```

```
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10
```

PASS Every Host Can Reach Its Default Gateway (12.3s, 3 commands)

Isolates the host-to-router leg so a failure further along is not misread as a broken LAN.

H1 ping -c 5 192.168.10.1

```
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=0.442 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=0.385 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=0.359 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=0.436 ms
64 bytes from 192.168.10.1: icmp_seq=5 ttl=255 time=0.377 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4103ms
rtt min/avg/max/mdev = 0.359/0.399/0.442/0.033 ms
```

H2 ping -c 5 192.168.20.1

```
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=0.543 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=0.411 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=0.396 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=0.389 ms
64 bytes from 192.168.20.1: icmp_seq=5 ttl=255 time=0.375 ms

--- 192.168.20.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4103ms
rtt min/avg/max/mdev = 0.375/0.422/0.543/0.061 ms
```

H3 ping -c 5 192.168.30.1

```
PING 192.168.30.1 (192.168.30.1) 56(84) bytes of data.
64 bytes from 192.168.30.1: icmp_seq=1 ttl=255 time=0.393 ms
64 bytes from 192.168.30.1: icmp_seq=2 ttl=255 time=0.446 ms
64 bytes from 192.168.30.1: icmp_seq=3 ttl=255 time=0.381 ms
64 bytes from 192.168.30.1: icmp_seq=4 ttl=255 time=0.333 ms
64 bytes from 192.168.30.1: icmp_seq=5 ttl=255 time=0.325 ms

--- 192.168.30.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4100ms
rtt min/avg/max/mdev = 0.325/0.375/0.446/0.043 ms
```

PASS Every Host Can Reach Every Other Host (24.1s, 6 commands)

The full mesh, including h2 <-> h3 which never touches a direct link and can only work if the hub re-advertises between spokes.

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.09 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.21 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.15 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.16 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.22 ms

--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.094/1.165/1.221/0.045 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.18 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.24 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.19 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.26 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.35 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.179/1.242/1.346/0.059 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.32 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.05 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=1.25 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.18 ms
```

```
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.18 ms

--- 192.168.10.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.053/1.195/1.318/0.088 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.15 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.15 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.13 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.20 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.09 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.089/1.143/1.198/0.035 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.27 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.45 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=1.23 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.12 ms
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.13 ms

--- 192.168.10.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.121/1.239/1.445/0.117 ms
```

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.30 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.15 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.11 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.20 ms

--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.111/1.201/1.296/0.065 ms
```

PASS Spoke To Spoke Traffic Is Routed Through The Hub (0.1s, 3 commands)

R2 must reach R3's LAN via the hub's tunnel address, not by any direct path -- there is no spoke-to-spoke link.

R2 show ip route 192.168.30.0

```
Routing entry for 192.168.30.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:01:29 ago
  Routing Descriptor Blocks:
    * 172.20.0.3, from 172.20.0.1, 00:01:29 ago
      opaque_ptr 0x7E32CED11C60
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 show ip cef 192.168.30.10

```
192.168.30.0/24
  nexthop 172.20.0.3 Tunnel0
```

R3 show ip cef 192.168.20.10

```
192.168.20.0/24
  nexthop 172.20.0.2 Tunnel0
```

PASS Hub Reaches Both Spoke LANs On Separate Tunnels (0.0s, 2 commands)

R1 show ip cef 192.168.20.10

```
192.168.20.0/24
  nexthop 172.20.0.2 Tunnel0
```

R1 show ip cef 192.168.30.10

```
192.168.30.0/24
  nexthop 172.20.0.3 Tunnel0
```


PASS Host Traffic Is Encrypted In Transit (19.6s, 5 commands)

No host runs crypto: the routers encrypt for them, so the ESP counters must move by the traffic the hosts generated.

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2544, #pkts encrypt: 2544, #pkts digest: 2544
  #pkts decaps: 2546, #pkts decrypt: 2546, #pkts verify: 2546
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 260, #pkts encrypt: 260, #pkts digest: 260
  #pkts decaps: 264, #pkts decrypt: 264, #pkts verify: 264
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

H1 ping -c 20 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.70 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.28 ms
64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=1.14 ms
64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.22 ms
64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=1.25 ms
64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=1.11 ms
64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.05 ms
64 bytes from 192.168.20.10: icmp_seq=11 ttl=62 time=1.14 ms
64 bytes from 192.168.20.10: icmp_seq=12 ttl=62 time=1.28 ms
64 bytes from 192.168.20.10: icmp_seq=13 ttl=62 time=1.21 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2673, #pkts encrypt: 2673, #pkts digest: 2673
  #pkts decaps: 2683, #pkts decrypt: 2683, #pkts verify: 2683
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 369, #pkts encrypt: 369, #pkts digest: 369
  #pkts decaps: 373, #pkts decrypt: 373, #pkts verify: 373
```

```
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

PASS Host Addresses Never Appear In Cleartext On The Spoke Link (11.7s, 9 commands)

Capture the R1<->R2 link while the hosts talk: only ESP between the link addresses should be visible.

R1 no monitor capture HOSTCAP

Capture does not exist

R1 monitor capture HOSTCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture HOSTCAP match ipv4 any any

\${output} =

R1 monitor capture HOSTCAP buffer size 5

\${output} =

R1 monitor capture HOSTCAP start

Started capture point : HOSTCAP

H1 ping -c 10 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=2.04 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.15 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.28 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.12 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.18 ms
64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=1.44 ms
64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=1.14 ms
64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.06 ms
```

--- 192.168.20.10 ping statistics ---

10 packets transmitted, 10 received, 0% packet loss, time 9012ms
... 1 more lines

R1 monitor capture HOSTCAP stop

Stopped capture point : HOSTCAP

R1 show monitor capture HOSTCAP buffer brief

#	size	timestamp	source	destination	dscp	protocol
0	138	0.000000	100.64.0.1	-> 100.64.0.3	48 CS6	ESP
1	138	0.000000	100.64.0.3	-> 100.64.0.1	48 CS6	ESP
2	170	0.002990	100.64.0.1	-> 100.64.0.2	0 BE	ESP
3	170	0.004989	100.64.0.2	-> 100.64.0.1	0 BE	ESP
4	138	0.037992	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
5	138	0.077983	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
6	138	0.077983	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
7	138	0.086986	100.64.0.3	-> 100.64.0.1	48 CS6	ESP
8	138	0.086986	100.64.0.1	-> 100.64.0.3	48 CS6	ESP
9	138	0.199986	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
10	138	0.199986	100.64.0.2	-> 100.64.0.1	48 CS6	ESP

... 230 more lines

R1 no monitor capture HOSTCAP

\${output} =

PASS Spoke To Spoke Traffic Is Encrypted On The Direct Path (11.5s, 11 commands)

The VTI lab this mirrors captured the hub's second link, because h2 to h3 crossed two tunnels and was decrypted and re-encrypted in between. Under DMVPN there is no second leg to capture: once the shortcut forms the hub is not on the path at all, so the capture moves to the spoke's own WAN interface -- the only place that traffic now appears. It should be ESP there, with neither host address visible.

R2 ping 10.20.3.1 source Loopback1 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:

Packet sent with a source address of 10.20.2.1

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
Tunnel0 created 00:01:15, expire 00:03:44
Type: dynamic, Flags: router nhop bfd
NBMA address: 100.64.0.3
```

R2 no monitor capture HOSTCAP

Capture does not exist

R2 monitor capture HOSTCAP interface GigabitEthernet2 both

\${output} =

R2 monitor capture HOSTCAP match ipv4 any any

\${output} =

R2 monitor capture HOSTCAP buffer size 5

\${output} =

R2 monitor capture HOSTCAP start

Started capture point : HOSTCAP

H2 ping -c 10 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.24 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.19 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.19 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.27 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.25 ms
64 bytes from 192.168.30.10: icmp_seq=6 ttl=62 time=1.23 ms
64 bytes from 192.168.30.10: icmp_seq=7 ttl=62 time=1.25 ms
64 bytes from 192.168.30.10: icmp_seq=8 ttl=62 time=1.27 ms
64 bytes from 192.168.30.10: icmp_seq=9 ttl=62 time=1.18 ms
64 bytes from 192.168.30.10: icmp_seq=10 ttl=62 time=1.32 ms
```

--- 192.168.30.10 ping statistics ---

10 packets transmitted, 10 received, 0% packet loss, time 9013ms

... 1 more lines

R2 monitor capture HOSTCAP stop

Stopped capture point : HOSTCAP

R2 show monitor capture HOSTCAP buffer brief

#	size	timestamp	source	destination	dscp	protocol
0	170	0.000000	100.64.0.2	-> 100.64.0.3	0 BE	ESP
1	170	0.001007	100.64.0.3	-> 100.64.0.2	0 BE	ESP
2	138	0.057996	100.64.0.3	-> 100.64.0.2	48 CS6	ESP
3	138	0.057996	100.64.0.2	-> 100.64.0.3	48 CS6	ESP
4	138	0.189993	100.64.0.2	-> 100.64.0.3	48 CS6	ESP
5	138	0.244982	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
6	138	0.244982	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
7	138	0.269990	100.64.0.2	-> 100.64.0.1	48 CS6	ESP
8	138	0.270982	100.64.0.1	-> 100.64.0.2	48 CS6	ESP
9	138	0.312987	100.64.0.2	-> 100.64.0.3	48 CS6	ESP
10	138	0.312987	100.64.0.3	-> 100.64.0.2	48 CS6	ESP

... 215 more lines

R2 no monitor capture HOSTCAP

\${output} =

PASS The Hub Is Not On The Spoke To Spoke Path (0.0s, 2 commands)

The routing consequence of the shortcut, from the hosts' point of view rather than the routers'.

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
Tunnel0 created 00:01:26, expire 00:03:33
Type: dynamic, Flags: router nhop bfd
NBMA address: 100.64.0.3
```

H2 traceroute -n -q 1 -w 1 192.168.30.10

```
traceroute to 192.168.30.10 (192.168.30.10), 30 hops max, 60 byte packets
 1 192.168.20.1 1.112 ms
 2 172.20.0.3 1.089 ms
 3 192.168.30.10 2.095 ms
```

11 Throughput

PASS Hub To Spoke Hosts Sustain The Target Throughput (32.0s, 2 commands)

h1 -> h2 crosses one tunnel.

```
H1 nc -l -p 5001 > /dev/null 2>&1 &
  ${output} =
H1 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=$(( $(date +%s) - S ))
  SECS=31
```

PASS Spoke To Spoke Hosts Sustain The Target Throughput (31.9s, 2 commands)

h2 -> h3 crosses two tunnels, being decrypted and re-encrypted at the hub, so it is the more demanding path.

```
H2 nc -l -p 5001 > /dev/null 2>&1 &
  ${output} =
H2 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.30.10 5001; echo SECS=$(( $(date +%s) - S ))
  SECS=31
```

PASS Bulk Throughput Traffic Is Encrypted (17.5s, 6 commands)

Rate alone says nothing about protection. A transfer of this size must move thousands of packets through ESP, so assert the counters rose in proportion to the bytes actually sent.

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 53004, #pkts encrypt: 53004, #pkts digest: 53004
#pkts decaps: 28767, #pkts decrypt: 28767, #pkts verify: 28767
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 50699, #pkts encrypt: 50699, #pkts digest: 50699
#pkts decaps: 26301, #pkts decrypt: 26301, #pkts verify: 26301
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

```
H1 nc -l -p 5001 > /dev/null 2>&1 &
[1]+  Stopped (tty input)          nc -l -p 5001 1>/dev/null 2>&1
```

```
H1 S=$(date +%s); dd if=/dev/zero bs=64k count=512 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=$(( $(date +%s) - S ))
  SECS=15
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
```

```
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 78000, #pkts encrypt: 78000, #pkts digest: 78000
#pkts decaps: 41206, #pkts decrypt: 41206, #pkts verify: 41206
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 50797, #pkts encrypt: 50797, #pkts digest: 50797
#pkts decaps: 26398, #pkts decrypt: 26398, #pkts verify: 26398
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines
```

12 Nat

PASS Every Router Has A Static Source NAT Mapping For Its Host (0.3s, 3 commands)

H1 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H2 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H3 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

PASS NAT Inside And Outside Interfaces Are Marked Correctly (0.3s, 7 commands)

The LAN faces inside, the tunnels face outside. Getting this backwards silently disables translation rather than erroring.

R1 show running-config interface GigabitEthernet3

Building configuration...

```
Current configuration : 145 bytes
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 negotiation auto
end
```

R1 show running-config interface Tunnel0

Building configuration...

```
Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface GigabitEthernet3

Building configuration...

Current configuration : 145 bytes

```

!
interface GigabitEthernet3
  description LAN toward the local host
  ip address 192.168.20.1 255.255.255.0
  ip nat inside
  negotiation auto
end

```

R2 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines

```

R3 show running-config interface GigabitEthernet3

Building configuration...

```

Current configuration : 145 bytes
!
interface GigabitEthernet3
  description LAN toward the local host
  ip address 192.168.30.1 255.255.255.0
  ip nat inside
  negotiation auto
end

```

R3 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 433 bytes
!
interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
... 6 more lines

```

R1 show running-config interface Tunnel0

Building configuration...

```

Current configuration : 399 bytes
!
interface Tunnel0
  description DMVPN hub
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines

```

PASS Mapped Prefixes Are Advertised Across The Fabric (0.1s, 3 commands)

Return traffic can only find its way home if each site's NAT range is reachable, so the mapped prefixes ride BGP like the real ones.

R2 show ip route 10.30.1.0

```

Routing entry for 10.30.1.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:03:47 ago
  Routing Descriptor Blocks:

```

```
* 172.20.0.1, from 172.20.0.1, 00:03:47 ago
  opaque_ptr 0x7E32CED118A0
  Route metric is 0, traffic share count is 1
  AS Hops 1
  Route tag 65001
  MPLS label: none
```

R2 show ip route 10.30.3.0

```
Routing entry for 10.30.3.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:03:47 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:03:47 ago
    opaque_ptr 0x7E32CED119E0
    Route metric is 0, traffic share count is 1
    AS Hops 2
    Route tag 65001
    MPLS label: none
```

R3 show ip route 10.30.1.0

```
Routing entry for 10.30.1.0/24
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:03:48 ago
  Routing Descriptor Blocks:
  * 172.20.0.1, from 172.20.0.1, 00:03:48 ago
    opaque_ptr 0x75671E6ABAD8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65001
    MPLS label: none
```

PASS Every Host Reaches Every Other Host At Its Mapped Address (24.1s, 6 commands)

H2 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
 64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.35 ms
 64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.17 ms
 64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.20 ms
 64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.11 ms
 64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.39 ms

--- 10.30.2.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4005ms
 rtt min/avg/max/mdev = 1.110/1.244/1.394/0.107 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
 64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.08 ms
 64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.21 ms
 64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.10 ms
 64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.20 ms
 64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.21 ms

--- 10.30.3.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4005ms
 rtt min/avg/max/mdev = 1.076/1.157/1.210/0.057 ms
```

H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
 64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.29 ms
 64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.58 ms
 64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.09 ms
 64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.10 ms
 64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.11 ms

--- 10.30.1.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4006ms
 rtt min/avg/max/mdev = 1.091/1.233/1.579/0.187 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
 64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.16 ms
 64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.27 ms
 64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.24 ms
 64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.17 ms
 64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.14 ms

--- 10.30.3.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4005ms
 rtt min/avg/max/mdev = 1.135/1.194/1.265/0.049 ms
```


H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
 64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.27 ms
 64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.23 ms
 64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.29 ms
 64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.16 ms
 64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.10 ms

--- 10.30.1.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4006ms
 rtt min/avg/max/mdev = 1.097/1.209/1.287/0.071 ms
```

H2 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
 64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.35 ms
 64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.26 ms
 64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.33 ms
 64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.02 ms
 64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.21 ms

--- 10.30.2.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4006ms
 rtt min/avg/max/mdev = 1.018/1.232/1.352/0.118 ms
```

PASS Translations Show The Host Rewritten To Its Mapped Address (4.2s, 2 commands)

The inside-local / inside-global pair is the direct evidence that the source address was rewritten.

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
 64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.17 ms
 64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.28 ms
 64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.15 ms
 64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.19 ms
 64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.16 ms

--- 10.30.2.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4006ms
 rtt min/avg/max/mdev = 1.147/1.188/1.284/0.049 ms
```

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.88	---	---
---	10.30.1.129	192.168.10.85	---	---
---	10.30.1.113	192.168.10.93	---	---
---	10.30.1.114	192.168.10.69	---	---
---	10.30.1.116	192.168.10.73	---	---
---	10.30.1.119	192.168.10.71	---	---
---	10.30.1.123	192.168.10.78	---	---
---	10.30.1.120	192.168.10.82	---	---
---	10.30.1.126	192.168.10.76	---	---
---	10.30.1.125	192.168.10.84	---	---
---	10.30.1.101	192.168.10.90	---	---
---	10.30.1.102	192.168.10.89	---	---
---	10.30.1.111	192.168.10.70	---	---
...	35 more lines			

PASS The Far Host Sees The Mapped Source Not The Real One (4.2s, 2 commands)

On the receiving router the peer appears as the sender's mapped address, which is the whole point of source NAT.

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
 64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.27 ms
 64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.11 ms
 64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.58 ms
 64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.25 ms
 64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.17 ms

--- 10.30.2.10 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4006ms
 rtt min/avg/max/mdev = 1.112/1.276/1.583/0.163 ms
```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.89	---	---
---	10.30.2.129	192.168.20.84	---	---
---	10.30.2.106	192.168.20.68	---	---

```

--- 10.30.2.105          192.168.20.88      ---
--- 10.30.2.111          192.168.20.92      ---
--- 10.30.2.108          192.168.20.80      ---
--- 10.30.2.101          192.168.20.64      ---
--- 10.30.2.102          192.168.20.65      ---
--- 10.30.2.126          192.168.20.77      ---
--- 10.30.2.125          192.168.20.86      ---
--- 10.30.2.123          192.168.20.79      ---
--- 10.30.2.120          192.168.20.74      ---
--- 10.30.2.116          192.168.20.82      ---
... 36 more lines

```

PASS Spoke To Spoke NAT Works Through The Hub (4.4s, 3 commands)

h2 -> h3 by mapped address crosses two tunnels and two translations, with no direct spoke-to-spoke path.

H2 ping -c 5 10.30.3.10

```

PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.19 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.23 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.24 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.21 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.18 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.179/1.208/1.241/0.023 ms

```

R2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.89	---	---
---	10.30.2.129	192.168.20.84	---	---
---	10.30.2.106	192.168.20.68	---	---
---	10.30.2.105	192.168.20.88	---	---
---	10.30.2.111	192.168.20.92	---	---
---	10.30.2.108	192.168.20.80	---	---
---	10.30.2.101	192.168.20.64	---	---
---	10.30.2.102	192.168.20.65	---	---
---	10.30.2.126	192.168.20.77	---	---
---	10.30.2.125	192.168.20.86	---	---
---	10.30.2.123	192.168.20.79	---	---
---	10.30.2.120	192.168.20.74	---	---
---	10.30.2.116	192.168.20.82	---	---
...	37 more lines			

R3 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.3.104	192.168.30.84	---	---
---	10.30.3.107	192.168.30.93	---	---
---	10.30.3.109	192.168.30.92	---	---
---	10.30.3.110	192.168.30.89	---	---
---	10.30.3.103	192.168.30.85	---	---
---	10.30.3.100	192.168.30.83	---	---
---	10.30.3.124	192.168.30.90	---	---
---	10.30.3.127	192.168.30.81	---	---
---	10.30.3.121	192.168.30.80	---	---
---	10.30.3.122	192.168.30.77	---	---
---	10.30.3.118	192.168.30.86	---	---
---	10.30.3.117	192.168.30.75	---	---
---	10.30.3.115	192.168.30.82	---	---
...	38 more lines			

PASS NATted Traffic Is Still Encrypted (19.6s, 5 commands)

Translation happens before encryption, so NATted traffic must still raise the ESP counters.

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 78330, #pkts encrypt: 78330, #pkts digest: 78330
  #pkts decaps: 41533, #pkts decrypt: 41533, #pkts verify: 41533
  #pkts compressed: 0, #pkts decompressed: 0

```

```

#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 51108, #pkts encrypt: 51108, #pkts digest: 51108
#pkts decaps: 26710, #pkts decrypt: 26710, #pkts verify: 26710
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

H1 ping -c 20 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.29 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.08 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.36 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.07 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.10 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.13 ms
... 11 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 78460, #pkts encrypt: 78460, #pkts digest: 78460
#pkts decaps: 41665, #pkts decrypt: 41665, #pkts verify: 41665
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 51217, #pkts encrypt: 51217, #pkts digest: 51217
#pkts decaps: 26818, #pkts decrypt: 26818, #pkts verify: 26818
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

PASS Mapped Addresses Never Appear In Cleartext On The WAN (14.0s, 18 commands)

R1 no monitor capture NATCAP

```
Capture does not exist
```

R1 monitor capture NATCAP interface GigabitEthernet2 both

```
${output} =
```

R1 monitor capture NATCAP match ipv4 any any

```

${output} =
R1 monitor capture NATCAP buffer size 5
${output} =
R1 monitor capture NATCAP start
Started capture point : NATCAP
R1 ping 100.64.0.2 repeat 2
Type escape sequence to abort.
Sending 2, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!
Success rate is 100 percent (2/2), round-trip min/avg/max = 1/1/2 ms
R1 monitor capture NATCAP stop
Stopped capture point : NATCAP
R1 show monitor capture NATCAP buffer brief
-----
#    size    timestamp      source          destination      dscp    protocol
-----
 0   114     0.000000    100.64.0.1      -> 100.64.0.2      0 BE    ICMP
 1   114     0.000000    100.64.0.2      -> 100.64.0.1      0 BE    ICMP
 2   114     0.001007    100.64.0.1      -> 100.64.0.2      0 BE    ICMP
 3   114     0.001998    100.64.0.2      -> 100.64.0.1      0 BE    ICMP
 4   138     0.049008    100.64.0.1      -> 100.64.0.2      48 CS6   ESP
 5   138     0.050000    100.64.0.2      -> 100.64.0.1      48 CS6   ESP
 6   138     0.133995    100.64.0.1      -> 100.64.0.3      48 CS6   ESP
 7   138     0.135002    100.64.0.3      -> 100.64.0.1      48 CS6   ESP
 8   138     0.161994    100.64.0.3      -> 100.64.0.1      48 CS6   ESP
 9   138     0.161994    100.64.0.1      -> 100.64.0.3      48 CS6   ESP
10   138     0.387040    100.64.0.2      -> 100.64.0.1      48 CS6   ESP
... 39 more lines
R1 no monitor capture NATCAP
${output} =
R1 no monitor capture NATCAP
Capture does not exist
R1 monitor capture NATCAP interface GigabitEthernet2 both
${output} =
R1 monitor capture NATCAP match ipv4 any any
${output} =
R1 monitor capture NATCAP buffer size 5
${output} =
R1 monitor capture NATCAP start
Started capture point : NATCAP
H1 ping -c 10 10.30.2.10
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.19 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.21 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.19 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.65 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.12 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.34 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.25 ms

--- 10.30.2.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
... 1 more lines
R1 monitor capture NATCAP stop
Stopped capture point : NATCAP
R1 show monitor capture NATCAP buffer brief
-----
#    size    timestamp      source          destination      dscp    protocol
-----
 0   170     0.000000    100.64.0.1      -> 100.64.0.2      0 BE    ESP
 1   170     0.000992    100.64.0.2      -> 100.64.0.1      0 BE    ESP
 2   138     0.005997    100.64.0.2      -> 100.64.0.1      48 CS6   ESP
 3   138     0.005997    100.64.0.1      -> 100.64.0.2      48 CS6   ESP
 4   138     0.035994    100.64.0.3      -> 100.64.0.1      48 CS6   ESP
 5   138     0.035994    100.64.0.1      -> 100.64.0.3      48 CS6   ESP
 6   138     0.130990    100.64.0.1      -> 100.64.0.3      48 CS6   ESP
 7   138     0.148995    100.64.0.3      -> 100.64.0.1      48 CS6   ESP
 8   138     0.248980    100.64.0.2      -> 100.64.0.1      48 CS6   ESP
 9   138     0.248980    100.64.0.1      -> 100.64.0.2      48 CS6   ESP
10   138     0.403971    100.64.0.1      -> 100.64.0.2      48 CS6   ESP
... 226 more lines

```

R1 no monitor capture NATCAP

```
${output} =
```

PASS The Untranslated LAN Path Still Works (8.0s, 2 commands)

Guards the route-map scoping. If translation ever widened to all traffic, replies on this path would arrive from a mapped address the sender never contacted, and this would fail.

H1 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.  
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.19 ms  
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.34 ms  
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.34 ms  
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.10 ms  
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.09 ms  
  
--- 192.168.20.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4006ms  
rtt min/avg/max/mdev = 1.093/1.210/1.337/0.107 ms
```

H2 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.  
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.20 ms  
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.19 ms  
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.18 ms  
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.28 ms  
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.22 ms  
  
--- 192.168.30.10 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4005ms  
rtt min/avg/max/mdev = 1.180/1.213/1.275/0.033 ms
```

13 Nat Pool

PASS Every Host Carries Its Full Block Of Addresses (0.1s, 6 commands)

10 statically mapped plus 30 pool-eligible, on one interface.

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H1 ip addr show eth1 | grep -c 'inet 192.168.10.'
40
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H2 ip addr show eth1 | grep -c 'inet 192.168.20.'
40
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    ... 70 more lines
```

```
H3 ip addr show eth1 | grep -c 'inet 192.168.30.'
40
```

PASS Every Router Configures Ten Static Mappings And A Thirty Address Pool (0.4s, 6 commands)

```
H1 show running-config | count ^ip nat inside source static
Number of lines which match regexp = 10
```

H1 show ip nat statistics

```
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
    Tunnel0
Inside interfaces:
    GigabitEthernet3
Hits: 0 Misses: 0
```

```

Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.1.100 end 10.30.1.129
... 9 more lines

H2 show running-config | count ^ip nat inside source static
Number of lines which match regexp = 10

H2 show ip nat statistics
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
    Tunnel0
Inside interfaces:
    GigabitEthernet3
Hits: 0 Misses: 0
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.2.100 end 10.30.2.129
... 9 more lines

H3 show running-config | count ^ip nat inside source static
Number of lines which match regexp = 10

H3 show ip nat statistics
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
    Tunnel0
Inside interfaces:
    GigabitEthernet3
Hits: 0 Misses: 0
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.3.100 end 10.30.3.129
... 9 more lines

```

PASS The Pool Access List Is One Range And Excludes The Static Block (0.1s, 3 commands)

A single /27 range rather than thirty host entries, and it must not overlap the statically mapped addresses -- otherwise which rule wins depends on evaluation order.

```

H1 show ip access-lists NAT-POOL-SRC
Extended IP access list NAT-POOL-SRC
    10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255

H2 show ip access-lists NAT-POOL-SRC
Extended IP access list NAT-POOL-SRC
    10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255

H3 show ip access-lists NAT-POOL-SRC
Extended IP access list NAT-POOL-SRC
    10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255

```

PASS All Ten Static Addresses Translate To Their Own Fixed Mapping (9.5s, 2 commands)

Static NAT is one-to-one and deterministic: inside .10-.19 must appear outside as .10-.19, each to its own.

```

H1 for i in $(seq 10 19) $(seq 64 93); do ping -c 2 -W 2 -I 192.168.10.$i 10.30.2.10 >/dev/null 2>&1 & done; wait;
echo SENT
SENT
[40]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[39]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[38]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[37]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[36]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1

```

```
[35]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[34]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[33]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[32]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[31]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[30]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[29]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[28]+ Done ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
... 27 more lines
```

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.82	---	---
---	10.30.1.113	192.168.10.77	---	---
---	10.30.1.114	192.168.10.66	---	---
---	10.30.1.116	192.168.10.69	---	---
---	10.30.1.119	192.168.10.86	---	---
---	10.30.1.123	192.168.10.93	---	---
---	10.30.1.120	192.168.10.76	---	---
---	10.30.1.126	192.168.10.71	---	---
---	10.30.1.125	192.168.10.72	---	---
---	10.30.1.101	192.168.10.84	---	---
---	10.30.1.102	192.168.10.83	---	---
---	10.30.1.111	192.168.10.67	---	---

... 68 more lines

PASS All Thirty Pool Addresses Translate Into The Pool And Are Distinct (0.5s, 1 commands)

Without overload each client takes its own pool address, so 30 clients must yield 30 different outside addresses.

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.82	---	---
---	10.30.1.113	192.168.10.77	---	---
---	10.30.1.114	192.168.10.66	---	---
---	10.30.1.116	192.168.10.69	---	---
---	10.30.1.119	192.168.10.86	---	---
---	10.30.1.123	192.168.10.93	---	---
---	10.30.1.120	192.168.10.76	---	---
---	10.30.1.126	192.168.10.71	---	---
---	10.30.1.125	192.168.10.72	---	---
---	10.30.1.101	192.168.10.84	---	---
---	10.30.1.102	192.168.10.83	---	---
---	10.30.1.111	192.168.10.67	---	---

... 68 more lines

PASS The Pool Is Exactly Consumed With No Misses (0.0s, 1 commands)

30 clients against a 30 address pool: full allocation and no misses is the evidence that none of them shared or failed.

R1 show ip nat statistics

```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 120 Misses: 40
Reserved port setting disabled provisioned no
Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
  pool SITE-POOL: id 1, netmask 255.255.255.0
    start 10.30.1.100 end 10.30.1.129
... 9 more lines
```


PASS A Thirty-first Client Gets No Translation Once The Pool Is Exhausted (7.3s, 8 commands)

The boundary the pool sizing never otherwise reaches. The pool holds exactly as many addresses as there are clients, which means the exhaustion path is never exercised by a normal run -- and an off-by-one in pool bounds would look identical to a correct configuration. So bring up one more client inside the pool ACL and require the router to refuse it: no translation, no borrowing of another client's address, and the miss counted. Without "overload" there is nothing to share, so silently reusing an address would be a correctness bug, not graceful degradation. Run against R1/H1, because it is R1's pool that the tests above fill: the suite drives traffic from H1 first and only reaches the other routers later. Asserting exhaustion on a router whose pool is still empty would test nothing.

```
R1 show ip nat statistics | include allocated
```

```
type generic, total addresses 30, allocated 30 (100%), misses 0
```

```
R1 show ip nat statistics
```

```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
```

```
Outside interfaces:
```

```
Tunnel0
```

```
Inside interfaces:
```

```
GigabitEthernet3
```

```
Hits: 120 Misses: 40
```

```
Reserved port setting disabled provisioned no
```

```
Dynamic overload mapping configured: 0
```

```
Expired translations: 0
```

```
Dynamic mappings:
```

```
-- Inside Source
```

```
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
```

```
pool SITE-POOL: id 1, netmask 255.255.255.0
```

```
start 10.30.1.100 end 10.30.1.129
```

```
... 9 more lines
```

```
H1 sudo ip addr add 192.168.10.94/24 dev eth1
```

```
${output} =
```

```
H1 ping -c 3 -W 2 -I 192.168.10.94 10.30.2.10
```

```
PING 10.30.2.10 (10.30.2.10) from 192.168.10.94 : 56(84) bytes of data.
```

```
From 192.168.10.1 icmp_seq=1 Destination Host Unreachable
```

```
From 192.168.10.1 icmp_seq=2 Destination Host Unreachable
```

```
From 192.168.10.1 icmp_seq=3 Destination Host Unreachable
```

```
--- 10.30.2.10 ping statistics ---
```

```
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2049ms
```

```
R1 show ip nat translations | include 192.168.10.94
```

```
${output} =
```

```
R1 show ip nat statistics
```

```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
```

```
Outside interfaces:
```

```
Tunnel0
```

```
Inside interfaces:
```

```
GigabitEthernet3
```

```
Hits: 120 Misses: 40
```

```
Reserved port setting disabled provisioned no
```

```
Dynamic overload mapping configured: 0
```

```
Expired translations: 0
```

```
Dynamic mappings:
```

```
-- Inside Source
```

```
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
```

```
pool SITE-POOL: id 1, netmask 255.255.255.0
```

```
start 10.30.1.100 end 10.30.1.129
```

```
... 9 more lines
```

```
R1 show ip nat statistics
```

```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
```

```
Outside interfaces:
```

```
Tunnel0
```

```
Inside interfaces:
```

```
GigabitEthernet3
```

```
Hits: 120 Misses: 40
```

```
Reserved port setting disabled provisioned no
```

```
Dynamic overload mapping configured: 0
```

```
Expired translations: 0
```

```
Dynamic mappings:
```

```
-- Inside Source
```

```
[Id: 1] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
```

```
pool SITE-POOL: id 1, netmask 255.255.255.0
```

```
start 10.30.1.100 end 10.30.1.129
```

```
... 9 more lines
```

```
H1 sudo ip addr del 192.168.10.94/24 dev eth1
```

```
${output} =
```

PASS The Pool Is Still Intact After The Exhaustion Attempt (0.5s, 2 commands)

The refusal must cost nothing: the thirty legitimate clients keep the addresses they held, and none was evicted to make room for the one that was turned away.

R1 show ip nat statistics | include allocated

type generic, total addresses 30, allocated 30 (100%), misses 3

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.82	---	---
---	10.30.1.113	192.168.10.77	---	---
---	10.30.1.114	192.168.10.66	---	---
---	10.30.1.116	192.168.10.69	---	---
---	10.30.1.119	192.168.10.86	---	---
---	10.30.1.123	192.168.10.93	---	---
---	10.30.1.120	192.168.10.76	---	---
---	10.30.1.126	192.168.10.71	---	---
---	10.30.1.125	192.168.10.72	---	---
---	10.30.1.101	192.168.10.84	---	---
---	10.30.1.102	192.168.10.83	---	---
---	10.30.1.111	192.168.10.67	---	---
...	68 more lines			

PASS Static And Pool Translations Coexist Across The Whole Block (0.5s, 1 commands)

All 40 of a host's addresses in flight, translated by two different mechanisms, with no outside address reused.

R1 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.1.128	192.168.10.89	---	---
---	10.30.1.129	192.168.10.82	---	---
---	10.30.1.113	192.168.10.77	---	---
---	10.30.1.114	192.168.10.66	---	---
---	10.30.1.116	192.168.10.69	---	---
---	10.30.1.119	192.168.10.86	---	---
---	10.30.1.123	192.168.10.93	---	---
---	10.30.1.120	192.168.10.76	---	---
---	10.30.1.126	192.168.10.71	---	---
---	10.30.1.125	192.168.10.72	---	---
---	10.30.1.101	192.168.10.84	---	---
---	10.30.1.102	192.168.10.83	---	---
---	10.30.1.111	192.168.10.67	---	---
...	68 more lines			

PASS Both Mechanisms Work On Every Router (19.5s, 4 commands)

The spokes carry the same scheme, not just the hub.

H2 for i in \$(seq 10 19) \$(seq 64 93); do ping -c 2 -W 2 -I 192.168.20.\$i 10.30.1.10 >/dev/null 2>&1 & done; wait; echo SENT

SENT

[40]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[39]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[38]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[37]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[36]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[35]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[34]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[33]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[32]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[31]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[30]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[29]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
[28]+	Done	ping -c 2 -W 2 -I 192.168.20.\${i} 10.30.1.10 1>/dev/null 2>&1
...	27 more lines	

H2 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.2.128	192.168.20.87	---	---
---	10.30.2.129	192.168.20.82	---	---
---	10.30.2.106	192.168.20.66	---	---
---	10.30.2.105	192.168.20.84	---	---

```

--- 10.30.2.111      192.168.20.77      ---      ---
--- 10.30.2.108      192.168.20.75      ---      ---
--- 10.30.2.101      192.168.20.91      ---      ---
--- 10.30.2.102      192.168.20.64      ---      ---
--- 10.30.2.126      192.168.20.72      ---      ---
--- 10.30.2.125      192.168.20.74      ---      ---
--- 10.30.2.123      192.168.20.93      ---      ---
--- 10.30.2.120      192.168.20.83      ---      ---
--- 10.30.2.116      192.168.20.79      ---      ---
... 108 more lines

```

H3 for i in \$(seq 10 19) \$(seq 64 93); do ping -c 2 -W 2 -I 192.168.30.\$i 10.30.1.10 >/dev/null 2>&1 & done; wait;
echo SENT

```

SENT
[40]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[39]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[38]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[37]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[36]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[35]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[34]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[33]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[32]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[31]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[30]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[29]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[28]+ Done          ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
... 27 more lines

```

H3 show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	10.30.3.104	192.168.30.75	---	---
---	10.30.3.107	192.168.30.92	---	---
---	10.30.3.109	192.168.30.93	---	---
---	10.30.3.110	192.168.30.91	---	---
---	10.30.3.103	192.168.30.88	---	---
---	10.30.3.100	192.168.30.87	---	---
---	10.30.3.124	192.168.30.90	---	---
---	10.30.3.127	192.168.30.83	---	---
---	10.30.3.121	192.168.30.86	---	---
---	10.30.3.122	192.168.30.82	---	---
---	10.30.3.118	192.168.30.89	---	---
---	10.30.3.117	192.168.30.84	---	---
---	10.30.3.115	192.168.30.77	---	---
---	...	68 more lines	---	---

PASS Only The Static Mappings Live In The Configuration (0.1s, 1 commands)

Static mappings are configuration and survive a reload; pool allocations are runtime state and do not. Confusing the two is how NAT surprises people after a reboot.

R1 show running-config | include ip nat inside source

```

ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL

```

PASS Traffic From Both Mechanisms Is Encrypted (38.6s, 6 commands)

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 79005, #pkts encrypt: 79005, #pkts digest: 79005
  #pkts decaps: 42201, #pkts decrypt: 42201, #pkts verify: 42201

```

```

#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 51572, #pkts encrypt: 51572, #pkts digest: 51572
#pkts decaps: 27172, #pkts decrypt: 27172, #pkts verify: 27172
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

H1 ping -c 20 -I 192.168.10.10 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) from 192.168.10.10 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.31 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.19 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.12 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.18 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.21 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.04 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.08 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.07 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.30 ms
... 11 more lines

```

H1 ping -c 20 -I 192.168.10.64 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.18 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.30 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.40 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.12 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.12 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.10 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.16 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.20 ms
... 11 more lines

```

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 79269, #pkts encrypt: 79269, #pkts digest: 79269
#pkts decaps: 42468, #pkts decrypt: 42468, #pkts verify: 42468
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

R2 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 51793, #pkts encrypt: 51793, #pkts digest: 51793

```

```

#pkts decaps: 27393, #pkts decrypt: 27393, #pkts verify: 27393
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
... 78 more lines

```

PASS Neither Real Nor Translated Addresses Appear In Cleartext (11.5s, 9 commands)

R1 no monitor capture POOLCAP

Capture does not exist

R1 monitor capture POOLCAP interface GigabitEthernet2 both

\${output} =

R1 monitor capture POOLCAP match ipv4 any any

\${output} =

R1 monitor capture POOLCAP buffer size 5

\${output} =

R1 monitor capture POOLCAP start

Started capture point : POOLCAP

H1 ping -c 10 -I 192.168.10.64 10.30.2.10

```

PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.36 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.06 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.45 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.21 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.49 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.55 ms

```

--- 10.30.2.10 ping statistics ---

10 packets transmitted, 10 received, 0% packet loss, time 9013ms

... 1 more lines

R1 monitor capture POOLCAP stop

Stopped capture point : POOLCAP

R1 show monitor capture POOLCAP buffer brief

```

-----
#   size  timestamp      source          destination    dscp  protocol
-----
0  170    0.000000    100.64.0.1      -> 100.64.0.2     0 BE   ESP
1  170    0.000000    100.64.0.2      -> 100.64.0.1     0 BE   ESP
2  138    0.148003    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
3  138    0.148003    100.64.0.1      -> 100.64.0.3    48 CS6  ESP
4  138    0.264986    100.64.0.2      -> 100.64.0.1    48 CS6  ESP
5  138    0.264986    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
6  138    0.309997    100.64.0.1      -> 100.64.0.3    48 CS6  ESP
7  138    0.310989    100.64.0.3      -> 100.64.0.1    48 CS6  ESP
8  138    0.325987    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
9  138    0.326995    100.64.0.2      -> 100.64.0.1    48 CS6  ESP
10 138    0.370983    100.64.0.1      -> 100.64.0.2    48 CS6  ESP
... 218 more lines

```

R1 no monitor capture POOLCAP

\${output} =

14 Ntp

PASS Every Router Is An NTP Client Of The NMS Over The Management VRF (0.3s, 3 commands)

R1 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

R2 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

R3 show running-config | include ^ntp

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
```

PASS The NMS Itself Synchronises To The Public Pool (2.4s, 2 commands)

The precondition the rest of this suite rests on. chrony is configured "local stratum 10", so it serves time whether or not it has an upstream -- meaning the routers can agree perfectly on time that came from nowhere. This requires a real upstream: a reference that is not the local clock, and a stratum that proves it came from somewhere else.

NMS chronyc tracking

```
Reference ID      : 44B793F0 (68.183.147.240)
Stratum          : 3
Ref time (UTC)   : Mon Sep 07 23:28:02 2026
System time      : 0.000295948 seconds fast of NTP time
Last offset      : -0.000096958 seconds
RMS offset       : 0.000707182 seconds
Frequency        : 8.692 ppm fast
Residual freq    : -0.005 ppm
Skew             : 0.229 ppm
Root delay       : 0.047848970 seconds
Root dispersion  : 0.003390881 seconds
Update interval  : 520.1 seconds
Leap status      : Normal
```

NMS chronyc -n sources

MS Name/IP address	Stratum	Poll	Reach	LastRx	Last sample
^+ 162.254.225.151	2	10	377	392	-3230us[-3230us] +/- 53ms
^* 68.183.147.240	2	10	377	582	+1658us[+1561us] +/- 28ms
^- 51.81.226.229	2	10	377	514	+862us[+862us] +/- 105ms
^+ 172.234.44.141	2	10	377	393	-401us[-401us] +/- 70ms
^+ 69.164.213.136	2	10	377	461	+239us[+239us] +/- 52ms
^+ 23.168.24.210	2	10	377	641	-2175us[-2271us] +/- 37ms

PASS Every Router Appears As A Client Of The NMS (0.1s, 1 commands)

Seen from the server, which is where a missing client shows up as an absence rather than as a router quietly using something else.

NMS sudo chronyc clients

Hostname	NTP	Drop	Int	IntL	Last	Cmd	Drop	Int	Last
192.168.99.3	74	0	7	-	88	0	0	-	-
192.168.99.2	82	0	7	-	37	0	0	-	-
192.168.99.1	75	0	7	-	22	0	0	-	-
localhost	0	0	-	-	-	24	0	7	0

PASS Every Router Forms Associations With Pool Servers (0.1s, 3 commands)

R1 show ntp associations

```
address      ref clock      st  when  poll reach  delay  offset  disp
*~192.168.99.10  68.183.147.240  3    22   128   377  0.075  0.224  4.458
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach  delay  offset  disp
*~192.168.99.10  68.183.147.240  3    37   128   377  0.066 -1.229  5.509
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach  delay  offset  disp
*~192.168.99.10  68.183.147.240  3    88   128   377  0.072  0.066  4.738
```

* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

PASS Associations Become Reachable (0.1s, 3 commands)

A non-zero reach register means packets are actually coming back, which distinguishes a working client from one that has merely been configured.

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    22   128  377 0.075 0.224 4.458
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    37   128  377 0.066 -1.229 5.509
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    88   128  377 0.072 0.066 4.738
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

PASS Every Router Selects A System Peer (0.1s, 3 commands)

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    22   128  377 0.075 0.224 4.458
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    38   128  377 0.066 -1.229 5.509
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    89   128  377 0.072 0.066 4.738
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

PASS Every Router Clock Is Synchronised (0.1s, 3 commands)

The end state: IOS declares the clock synchronised. This lags real accuracy by a long way. IOS holds the flag back until its loopfilter leaves the 'FREQ' drift-measurement phase, measured on this platform at about 17 minutes from a cold NTP start -- while the offset is already within milliseconds well before that. The wait below is bounded to cover a freshly provisioned lab; on a lab that has been up it returns on the first poll and costs nothing.

R1 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
ntp uptime is 499000 (1/100 of seconds), resolution is 4000
reference time is EE49C932.33B64630 (23:37:22.202 UTC Mon Sep 7 2026)
clock offset is 0.2243 msec, root delay is 47.92 msec
root dispersion is 9.37 msec, peer dispersion is 4.45 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008782 s/s
system poll interval is 128, last update was 23 sec ago.
```

R2 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 499000 (1/100 of seconds), resolution is 4000
reference time is EE49C7CF.28B439C8 (23:31:27.159 UTC Mon Sep 7 2026)
clock offset is -1.2297 msec, root delay is 47.91 msec
root dispersion is 13.09 msec, peer dispersion is 5.50 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007039 s/s
system poll interval is 128, last update was 378 sec ago.
```

R3 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0021 Hz, precision is 2**10
ntp uptime is 499000 (1/100 of seconds), resolution is 4000
reference time is EE49C6D8.2AC08388 (23:27:20.167 UTC Mon Sep 7 2026)
clock offset is 0.0665 msec, root delay is 50.63 msec
root dispersion is 14.92 msec, peer dispersion is 4.73 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008462 s/s
system poll interval is 128, last update was 625 sec ago.
```


PASS Stratum Is Sane On Every Router (0.1s, 3 commands)

Stratum 16 means unsynchronised. A client of a stratum 1-2 pool server should land well below the limit.

R1 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
ntp uptime is 499100 (1/100 of seconds), resolution is 4000
reference time is EE49C932.33B64630 (23:37:22.202 UTC Mon Sep 7 2026)
clock offset is 0.2243 msec, root delay is 47.92 msec
root dispersion is 9.37 msec, peer dispersion is 4.45 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008782 s/s
system poll interval is 128, last update was 23 sec ago.
```

R2 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 499000 (1/100 of seconds), resolution is 4000
reference time is EE49C7CF.28B439C8 (23:31:27.159 UTC Mon Sep 7 2026)
clock offset is -1.2297 msec, root delay is 47.91 msec
root dispersion is 13.09 msec, peer dispersion is 5.50 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007039 s/s
system poll interval is 128, last update was 378 sec ago.
```

R3 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0021 Hz, precision is 2**10
ntp uptime is 499000 (1/100 of seconds), resolution is 4000
reference time is EE49C6D8.2AC08388 (23:27:20.167 UTC Mon Sep 7 2026)
clock offset is 0.0665 msec, root delay is 50.63 msec
root dispersion is 14.92 msec, peer dispersion is 4.73 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008462 s/s
system poll interval is 128, last update was 625 sec ago.
```

PASS The Reference Is One Of The Peers The Router Actually Polls (0.1s, 6 commands)

Guards against a router calling itself synchronised to something that is not in its association list.

R1 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
ntp uptime is 499100 (1/100 of seconds), resolution is 4000
reference time is EE49C932.33B64630 (23:37:22.202 UTC Mon Sep 7 2026)
clock offset is 0.2243 msec, root delay is 47.92 msec
root dispersion is 9.37 msec, peer dispersion is 4.45 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008782 s/s
system poll interval is 128, last update was 23 sec ago.
```

R1 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    23    128   377  0.075  0.224  4.458
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0017 Hz, precision is 2**10
ntp uptime is 499000 (1/100 of seconds), resolution is 4000
reference time is EE49C7CF.28B439C8 (23:31:27.159 UTC Mon Sep 7 2026)
clock offset is -1.2297 msec, root delay is 47.91 msec
root dispersion is 13.09 msec, peer dispersion is 5.50 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000007039 s/s
system poll interval is 128, last update was 378 sec ago.
```

R2 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    38    128   377  0.066 -1.229  5.509
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```
Clock is synchronized, stratum 4, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0021 Hz, precision is 2**10
ntp uptime is 499000 (1/100 of seconds), resolution is 4000
reference time is EE49C6D8.2AC08388 (23:27:20.167 UTC Mon Sep 7 2026)
clock offset is 0.0665 msec, root delay is 50.63 msec
root dispersion is 14.92 msec, peer dispersion is 4.73 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008462 s/s
system poll interval is 128, last update was 625 sec ago.
```

R3 show ntp associations

```
address      ref clock      st  when  poll reach delay offset disp
*~192.168.99.10 68.183.147.240 3    89    128   377  0.072  0.066  4.738
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```


PASS Clock Offset Is Small On Every Router (0.1s, 3 commands)

R1 show ntp associations

```
address          ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  68.183.147.240  3    23   128   377  0.075  0.224  4.458
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address          ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  68.183.147.240  3    38   128   377  0.066 -1.229  5.509
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address          ref clock      st  when  poll reach delay offset disp
*~192.168.99.10  68.183.147.240  3    89   128   377  0.072  0.066  4.738
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

PASS All Routers Agree On The Time (0.0s, 3 commands)

The practical payoff: three independently synchronised clocks should read the same, which is what makes logs across the fabric correlatable.

R1 show clock

```
23:37:45.512 UTC Mon Sep 7 2026
```

R2 show clock

```
23:37:45.525 UTC Mon Sep 7 2026
```

R3 show clock

```
23:37:45.535 UTC Mon Sep 7 2026
```

PASS The Clock Is No Longer At The Boot Epoch (0.1s, 6 commands)

An unsynchronised IOS device reports a 1993 or 1900 clock; a synchronised one reports now. This is the crude check that catches a clock that never moved at all.

R1 show clock

```
23:37:45.548 UTC Mon Sep 7 2026
```

R1 show clock

```
23:37:45.560 UTC Mon Sep 7 2026
```

R2 show clock

```
23:37:45.575 UTC Mon Sep 7 2026
```

R2 show clock

```
23:37:45.588 UTC Mon Sep 7 2026
```

R3 show clock

```
23:37:45.601 UTC Mon Sep 7 2026
```

R3 show clock

```
23:37:45.614 UTC Mon Sep 7 2026
```

15 Syslog

PASS The Collector Is A Service On The NMS Serving Every Router On One Port (0.1s, 2 commands)

What the move bought: one rsyslog listener for all three routers instead of one busybox listener per router, and a system service rather than something a suite starts -- so no run can leave a stale listener swallowing another's messages.

```
NMS systemctl is-active rsyslog
active
NMS sudo ss -ltnp | grep ':514 ' || echo NONE
UNCONN 0      0      0.0.0.0:514      0.0.0.0:*      users:(("rsyslogd",pid=2925,fd=6))
UNCONN 0      0      [::]:514        [::]:*        users:(("rsyslogd",pid=2925,fd=7))
```

PASS Every Router Is Configured To Log To The Collector (0.3s, 3 commands)

```
R1 show running-config | include ^logging|^service (timestamps|sequence)
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
R2 show running-config | include ^logging|^service (timestamps|sequence)
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
R3 show running-config | include ^logging|^service (timestamps|sequence)
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
```

PASS Every Router Reports The Collector As An Active Logging Host (0.2s, 3 commands)

The router's own view: it should name the collector and show a non-zero count of messages sent to it.

```
R1 show logging | include Trap logging|Logging to 192.168.99.10
Trap logging: level informational, 1238 message lines logged
Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
R2 show logging | include Trap logging|Logging to 192.168.99.10
Trap logging: level informational, 1616 message lines logged
Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
R3 show logging | include Trap logging|Logging to 192.168.99.10
Trap logging: level informational, 974 message lines logged
Logging to 192.168.99.10 (MGMT) (udp port 514, audit disabled,
```

PASS The Hub Message Reaches The Collector (4.0s, 1 commands)

End to end over the LAN: emit a uniquely marked message and find it at the collector.

```
NMS sudo grep -c -- 'LABSYSLOG-HUB-193745' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

PASS Each Spoke Message Reaches The Collector Across The Tunnel (8.1s, 2 commands)

The spokes have no direct path to h1's LAN; their messages are carried by the tunnels and the hub's routing.

```
NMS sudo grep -c -- 'LABSYSLOG-R2-193745' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LABSYSLOG-R3-193745' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
1
```

PASS Received Messages Carry Facility Severity And Mnemonic (4.1s, 2 commands)

```
NMS sudo grep -c -- 'LABSYSLOG-FIELDS-193745' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
```

```
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
```

```
<189>207: 000203: *Sep 7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep 7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep 7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep 7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep 7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep 7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep 7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep 7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep 7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep 7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep 7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep 7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep 7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep 7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 1020 more lines
```

PASS Received Messages Carry Sequence Numbers (0.1s, 1 commands)

'service sequence-numbers' is configured, so every message should be numbered -- which is how a collector detects loss.

```
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
```

```
<189>207: 000203: *Sep 7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep 7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep 7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep 7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep 7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep 7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep 7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep 7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep 7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep 7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep 7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep 7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep 7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep 7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 1020 more lines
```

PASS Message Timestamps Agree With The Router Clock (4.1s, 3 commands)

Ties syslog to NTP: a message's timestamp should match the sending router's clock. Skewed timestamps make a collector's correlation across devices worthless.

```
NMS sudo grep -c -- 'LABSYSLOG-TIME-193745' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
R1 show clock
23:38:08.968 UTC Mon Sep 7 2026
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
<189>207: 000203: *Sep 7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep 7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep 7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep 7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep 7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep 7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep 7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep 7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep 7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep 7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep 7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep 7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep 7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep 7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 1021 more lines
```

PASS Messages Below The Trap Level Are Not Forwarded (14.0s, 1 commands)

Negative control. Trap level is informational (6), so a debug-severity (7) message must not reach the collector. If this ever passes traffic, the level filter is not working and the positive tests above prove much less than they appear to.

```
NMS sudo grep -c -- 'LABSYSLOG-DEBUG-193745' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
0
0
```

PASS All Three Routers Are Represented At The Collector (12.1s, 3 commands)

Messages from every router, distinguishable at the collector, which is the point of a central log host.

```
NMS sudo grep -c -- 'LABSYSLOG-ALL-R1-193745' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R2-193745' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R3-193745' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
1
```

16 Snmpv3

PASS Both Trap Receivers Are Services On The NMS (0.1s, 2 commands)

snmptrapd on the standard port and the raw capture on a second one, both system services rather than listeners a suite starts -- so no run can leave one behind, which is how a negative test once passed while nothing was arriving.

```
NMS systemctl is-active snmptrapd lab-trapcap | tr '\n' ' '
active active

NMS sudo ss -ltnp | grep -E ':(162|1162) ' || echo NONE
UNCONN 0      0      0.0.0.0:1162  0.0.0.0:*    users:(("python3",pid=3082,fd=3))
UNCONN 0      0      0.0.0.0:162  0.0.0.0:*
users:(("snmptrapd",pid=3043,fd=3),("systemd",pid=1,fd=124))
UNCONN 0      0      [::]:162     [::]:*
users:(("snmptrapd",pid=3043,fd=4),("systemd",pid=1,fd=125))
```

PASS Every Router Has An SNMPv3 User With Authentication And Privacy (0.1s, 3 commands)

v3 without auth and priv is barely better than v2c, so the protocols in use are asserted, not just the user's existence.

```
R1 show snmp user
User name: labmon
Engine ID: 800000000300525400C80101
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

```
R2 show snmp user
User name: labmon
Engine ID: 800000000300525400C80201
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

```
R3 show snmp user
User name: labmon
Engine ID: 800000000300525400C80301
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: LABGRP
```

PASS Every Router Has A v3 Group Requiring Privacy (0.1s, 3 commands)

```
R1 show snmp group
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                    security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
... 3 more lines
```

```
R2 show snmp group
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                    writeview: *ilmi
notifyview: <no notifyview specified>
row status: active
```

```

groupname: LABGRP                security model:v3 priv
contextname: <no context specified> storage-type: nonvolatile
... 3 more lines

R3 show snmp group
groupname: ILMI                  security model:v1
contextname: <no context specified> storage-type: permanent
readview : *ilmi                 writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                  security model:v2c
contextname: <no context specified> storage-type: permanent
readview : *ilmi                 writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                security model:v3 priv
contextname: <no context specified> storage-type: nonvolatile
... 3 more lines

```

PASS No Router Accepts SNMPv1 Or v2c Communities (0.3s, 3 commands)

Security posture: configuring v3 is pointless if a community string still gets you in.

```

R1 show running-config | include ^snmp-server community
${output} =

R2 show running-config | include ^snmp-server community
${output} =

R3 show running-config | include ^snmp-server community
${output} =

```

PASS Every Router Targets The Collector With v3 Privacy (0.1s, 3 commands)

```

R1 show snmp host
Notification host: 192.168.99.10    udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10    udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

R2 show snmp host
Notification host: 192.168.99.10    udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10    udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

R3 show snmp host
Notification host: 192.168.99.10    udp-port: 1162  VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

Notification host: 192.168.99.10    udp-port: 162   VRFName: MGMT  type: trap
user: labmon      security model: v3 priv

```

PASS Global Trap Sending Is Enabled (0.1s, 3 commands)

Enabling individual trap types leaves the global switch off: the router builds trap PDUs and never sends them, which looks exactly like a broken collector.

```

R1 show snmp | include global trap
SNMP global trap: enabled

R2 show snmp | include global trap
SNMP global trap: enabled

R3 show snmp | include global trap
SNMP global trap: enabled

```

PASS Traps From Every Router Reach The Collector (13.0s, 51 commands)

```

R1 configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.

```

```

R1 interface Loopback270
  ${output} =
R1 ip address 10.99.97.1 255.255.255.255
  ${output} =
R1 snmp trap link-status
  ${output} =
R1 end
  ${output} =
R1 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Loopback270
  ${output} =
R1 shutdown
  ${output} =
R1 end
  ${output} =
R1 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R1 interface Loopback270
  ${output} =
R1 no shutdown
  ${output} =
R1 end
  ${output} =
R1 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R1 no interface Loopback270
  ${output} =
R1 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
  ${output} =
R2 ip address 10.99.97.1 255.255.255.255
  ${output} =
R2 snmp trap link-status
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
  ${output} =
R2 shutdown
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 interface Loopback270
  ${output} =
R2 no shutdown
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R2 no interface Loopback270
  ${output} =
R2 end
  ${output} =
R3 configure terminal
  Enter configuration commands, one per line. End with CNTL/Z.
R3 interface Loopback270

```

```
{output} =
R3 ip address 10.99.97.1 255.255.255.255
{output} =
R3 snmp trap link-status
{output} =
R3 end
{output} =
R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3 interface Loopback270
{output} =
R3 shutdown
{output} =
R3 end
{output} =
R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3 interface Loopback270
{output} =
R3 no shutdown
{output} =
R3 end
{output} =
R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3 no interface Loopback270
{output} =
R3 end
{output} =
NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
d7c3450c8b69b1810963c1c351af94e662bb62948440836307965026c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
b66e773f6cb948e6c84fbc1c22baffabc9a7c820adbbccc0607f9fe235937749231d4e94bb8fdcc059dcffbbba830d801e015c85863d7
158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
ef6ddaf4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdfc96
8ee7b7c9e197db97da0102a62640bcd82abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
0102020200a104066c61626d6f6e040c1a06b8620cf2bdfae082242d0408a7c7be9c6efe40570481a3f13563f7a9d2b23f9d6aed6db
b3ee8cb9bfc314d080afa0b0fb48addca5f604a787508df6374629f81bf9c5e8ed322c4fbc9fd5daf2e665dcbc312cc01027c101b15d
95d74b6c51ef9faf06c0670f81620af82d566719918aab054f908d5415c32688a26757db8041390a1fd440c938835bf8a171ee3321ef
59f0d1a550a2e55083266bba89add3ddb3176843bbfc92d3de40fb901ba83f7d8c7124f27c7fe6db1dce
1788819441.982 192.168.99.1 3081f1020103300d020138020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c54e0b089dfb6d3286639954d04089b184efb1b39c21a0481a38399f66019ba392443b719d26a
fd64cb17b4f7e47e6efc18540d7a161bb4df34a5d637d8baf88d7959b126d7f786e1521239899f10d3d27949f04880309d9a8e456a78
4b512b0558a7215ca4ba9844e8fac890bae9434f944d06cc273d3d6bb9f4842b28c44aa4b6f7723acfc5a8d378b7f6d6019f1c750dbd
2b44a96e3c05c10c5df1deb3e7c8f3b1022054494304e4456fad68c0f554b6016aa90535c790e85d921b
1788819442.019 192.168.99.2 3081f1020103300d020140020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cc5e940312135bbb74315f07f0408bdf4dc46c97074950481a3d122cd69208ea63a33c15acfa8
7a45102a3d13ff4011a6fe0077d1190d97e78309f972e8bd391feaf8f439f142fa831bc51566d9eb8017d4cb97e7436f0ff3c8ef79ef
ea5d0c4c1aab31cc9b7f8cff78355da4f4fcfa92bdd9281133bec7fae933da59e755762b226e899cb3c6998c3fb887f3397554d5152
dcbea7abdb77818b788f02536a47b0893019f437a968d65cf16414aaec7588c7acf31617d6f2d26c240f
1788819442.056 192.168.99.3 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c97f59d812586c8cbc82aeec80408a7c7be9c6efe40590481a3944f05c35a695e8ab698a77fc0
679d4817a25d2867b284b3553e5b9560d1abeb0194ef14399db882e9fa2c2b7a2e5a3a8776478141012d370690d28f11c26bfc8aa1fd
53215e87e900d9dda1de7d41080b83bf95c9c0743949bd5c4af9ea11abc6a118939fad85eb0735991d6b9d8e891ccf0bd9cb99abf172
805a8c9b5e5d778a279e1a0384713bb102e8ba0b0be1d56a9527722ef00de7e6baabfe3aae5f497a44f4
1788819442.233 192.168.99.1 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040ce97d7d93c11dac8ea0caaafe04089b184efb1b39c21c0481a3ee52a2e7caf58003f3aaef04a
dcaabcf43bd1c13a51d4087e88c6a9a9cfb270aec98cf2cbdffc6f4803608b8bc6591cde59384e91da399d6f1080102f98fe25152158b
139ebf2f59507f8016a328a4c7a9d1b80a4219982fe2f74978cda2b2ef1bf7d7faf9d835a5cf3f9042d5c744ae898d39833cc1dbe052
02b505f2d02dbbe48944635bb890fd08e79bbaef5523f4dbd8e7aa0f86d71427b43736a5d9d7bef4403
1788819442.267 192.168.99.2 3081f1020103300d020142020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cd6f8a07f793b069d069759170408bdf4dc46c97074970481a33116ac4348d5c39ada28f96d78
e74d155c5cc9bffa42fe9fdadacbc9f298c97a79d1335ef72ad9631dc12834db86976c08bc0949678ce7926afec969d51ff59beb6fd1
7b912bab7415c3068dc730665587dd829e838b378fbcd81aeea7be721b39cc8ec406c7f91f5bf3e02ef0f84d884b62698b64590b35f1
8921e3a8316c7d66a27f4300e3b40e92dcf8449782b483b7eb1c0a3da45fbc4c24622a7d600f36d7023
1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c3bb1f3582527a19034ec5d610408a7c7be9c6efe405b0481a3fa3381e2249898d509d85e1d9b
3389b3128c7d34aaa0145cd77854357e6d69cd388db0d86c165a4fdf8a387f9bb961854ba187a1984c45dbbf58adca5541a5485c344
583a562913d9b44704b325f28a9b0c3c74d18b4c3bd08fc9130c877165dc4c6d22a6e4226cd8546bac4ed78bd313ac2d76bfad8165bf
```



```
2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aecfc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
0201020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcde825ab5152edf654545b7ab79bde2ffadf60703003bf05fd372b55c6598a7edececf536ad432095fb9260748bd46f1cc973fce
7d0c5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fde25afe53f39a555ecb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babb3db4f4ce0ebb07f01f9ffd45e6fb29ef8feac58621e676b92935f9ce739373e633a53e74e506575e6f46825bb060ff
e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecdde96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67078f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d69d85b798ee860408950a0e0a5c9d4a79f2784d07c6aca87f8a39
ea45e822985914ea2cb93b21d583ae4f38bbbbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be6e04256f22
03359df9f4b249e253bfc47820d97178378d92b51f24de8d8455b5342f276797f484d9471253faea500058a89ac76482836c4d83314
b637ad7fc13ded93b704de011795c6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27cd550b7799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020140020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfc1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbeb0d2965d0cedde29f5e6919fa5f14ad44ce1b2fffc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197adb64e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 3229 more lines
```

PASS Received Traps Are SNMP Version 3 (0.2s, 3 commands)

Reads the version field out of the datagram itself rather than trusting the configuration.

```
NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true
```

```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
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... 3230 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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... 3230 more lines
```

PASS Received Traps Carry The Configured User And Privacy Flags (0.2s, 3 commands)

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

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... 3231 more lines
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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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... 3231 more lines

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbeb0d2965d0cedde29f5e6919fa5f14ad44ce1b2ffcc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 3231 more lines
```

PASS Trap Payloads Are Encrypted (0.2s, 3 commands)

With privacy in force the scoped PDU arrives as an opaque OCTET STRING. If it were a readable SEQUENCE the varbinds would be on the wire in clear.

```
NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true
```

```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
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1788819441.768 192.168.99.2 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
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... 3232 more lines

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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0102020200a204066c61626d6f6e040ce97d7d93c11dac8ea0caaafe04089b184efb1b39c21c0481a3ee52a2e7caf58003f3aaef04a
dcaabcf43bd1c13a51d4087e88c6ba9a9cfcfb270aec98cf2cbdfcf6f8d03608b8bc6591cde59384e91da399d6f1080102f98fe25152158b
139ebf2f59507f8016a328a4c7a9d1b80a4219982fe2f74978cda2b2ef1bf7d7faf9d835a5cf3f9042d5c74a4e898d39833cc1dbe052
02b505f5d02dbbe48944635b5bb890fd08e79bbaef5523fd4bd8e7aa0f86d71427b43736a5d9d7bef4403
1788819442.267 192.168.99.1 3081f1020103300d020142020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cd6f8a0f7f93b069d069759170408bdf4dc46c97074970481a33116ac4348d5c39ada28f96d78
e74d155c5cc9bffa42f9f9dadacbc9f298c97a79d1335ef72ad9631dc12834db86976c08bc0949678ce7926afec969d51ff59beb6fd1
7b912bab7415c3068dc730665587db829e838b378fbc81aeeaf72b1b39cc8ec406c7f91f5bf3e02ef0f84d884b62698b64590b35f1
8921e3a8316c7d66a27f4300e3b40e92dc8f449782b483b7eb1b1c0a34a5f4c24622a7d600f36d7023
1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c3bb1f3582527a19034ac5d610408a7c7be9c6efe405b0481a3fa3381e2249898d509d85e1d9b
3389b3128c7d34aa0145cd77854357e6d69cd388db0d86c165a4fdcf8a3b7f9bb961854ba187a1984c45dbbf58adac5541a5485c344
583a562913d9b44704b325f28a9b0c3c74d18b4c3bd08fc9130c877165dc4c6d22a6e4226cd8546bac4ed78bd313ac2d76bfad8165bf
d95c8ff1fe7981815276c5c81bab3f027acbecb251449958991c430e2e4f57b98530410546f515e98124
1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c73a76bc49e552aab0d08f64d04089b184efb1b39c21e0481a3b1a6978d7875aefa7484d41426
fcac7bd01a45e729fb041e4e0053bcf6e1e7b4453849a0950b51754d5a8abb1e683b91e18dedd70c8eeb400fd5392669c3139475a2e
f9cea2e5f556ab286affc24b299f7257a7f9aaa3362d18525a4037ae861fb964fcd2e3f759405bf4911bb9c8e340b9c3c32a39f9b
caf86df01c6728b43aa5493736b1193e7dccaec3f3f0abd1c6f25b53985eb169cb05adfaa725d29e505e5
1788819442.518 192.168.99.2 3081f1020103300d020144020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040ce22c3e1cccc3a0816b2de3090408bdf4dc46c97074990481a362ccc6cfc3fcb1f5df6a36ba9
061e6bd5f8e4cea22d2f8283e04ff2f72e21962da8a88d91b1c4fbdcc6a5d339e9dd070feed2193e628e3bdc3aeaea01f49bc99930a11e
69a68b57bc34e6500c2a923e4766fa1263512636869017b82cec5982625Feded62ba54a93598fd3604d189683a16346138220d45f3a5
2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aefc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
020102020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcde825ab5152edf65455b7ab79bde2ffad6f0703003bf05fd372b55c6598a7edececf536ad432095fb9260748bd46f1cc973fce
7d0cd5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fe25afe53f39a555eb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babdb3db4f4ce0ebbb07f01f9ff445e6fb29ef8faec58621e67b92935f9ce739373e633a53e74e506575e6f46825bb060ff
e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecdcd96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67078f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d698485b8798ee860408950a0e0a5c9d4a79f2784d07c6aca87f839a
ea45e822985914ea2cb93b21d583ae4f38bbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be0e04256f622
03359df9f4b249e253bfc47820d97178378d9b251f24de8d8455b5342f2d76797f484d9471253faea500058a89ac76482836c4d83314
b637ad7fc13ded93b704de011795c6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27dc5500b799d7f6f31678bbff135288e19a3de00a02ab05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfcfe88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183bc3c2e4abf6a37517d3ba7fbed02965d0cedde29cf56919fa5f14ad44c1eb2ffcc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 3232 more lines
```

PASS snmptrapd Decrypts Traps From Every Router (0.2s, 3 commands)

The strongest single statement this suite makes: a real SNMP manager, holding only the configured user and each router's engine ID, authenticated and decrypted the traps. That cannot happen unless the keys, the engine IDs and the privacy protocol all match on both sides.

```
NMS sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true
```

```
NET-SNMP version 5.9.4.pre2
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.21 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.95 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:34.92 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.35 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:36.09 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.06 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
... 6479 more lines
```

```
NMS sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true
```

```
NET-SNMP version 5.9.4.pre2
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.21 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.95 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:34.92 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.35 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:36.09 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.06 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
... 6479 more lines
```

```
NMS sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true
```

```
NET-SNMP version 5.9.4.pre2
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.21 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.95 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:34.92 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.26 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.26 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.26 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.26 "lab"
2026-09-07 22:17:21 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.35 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
```



```
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:36.09      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:03:35.06      iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.27 1      iso.3.6.1.4.1.9.9.43.1.1.6.1.4.27 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.27 2      iso.3.6.1.4.1.9.9.43.1.1.6.1.8.27 "lab"
2026-09-07 22:17:22 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
... 6481 more lines
```

PASS One Raw Listener Receives Every Router (0.1s, 1 commands)

The property the per-router-port workaround existed to dodge: a single socket now serves all three senders.

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`

```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14007f04089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
d7c3450c8b69b1810963c1c351af94e662bb62948440836307965026c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
b66e773f6cb948e6c84fbc122baffabc9a7c820adbcbccc0607f9fe235937749231d4e94bb8fdcc059dcffbbba830d801e015c85863d7
158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
ef6ddaf4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdfc96
8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
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... 3233 more lines
```

PASS Each Router Identifies Itself By Its Own Engine ID (0.3s, 6 commands)

The engine ID in the received trap must match the one the router reports, and must differ between routers -- that is what lets a collector attribute a trap to a device.

R1 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80101
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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... 3233 more lines
```

R2 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80201
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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... 3233 more lines
```

R3 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80301
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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0102020200a204066c61626d6f6e040ce22c3e1cccc3a0816b2de3090408bdf4dc46c97074990481a362ccc6cfc3fcb1f5dfe6a36ba9
061eb6d5f8e4cea2b22f8283e04ff127fe21962da888d91b1c2f6ba6a5d339e9dd070feed2193e628e3bdc3aeaea01f49bc99930a11e
69a68b57bc34e6500c2a923e4766fa1263512636869017b2ccec5982625feded62ba54a93598fd3604d189683a16346138220d45f3a5
2b61fddeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8ab2de5e8564fe10aecfc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d02014020205dc04010302010304373035040c800000090300525400c80301
020102020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcdce825ab5152edf6f54545b7ab79bde2ffad6f0703003bf05fd372b55c6598a7edececfd536ad432095fb9260748bd46f1cc973fce
7d0c5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fde25afe53f39a555ecb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babb3db4f4ce0ebb07f01f9fffd456fb29ef8feac58621e676b92935f9ce739373e633a53e74e506575e6f46825bb060ff
```


e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecdcd96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67078f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d69d85b8798ee860408950a0e0a5c9d4a79f2784d07c6aca87f8a39
ea45e822985914ea2cb93b21d583ae4f38bbbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be6e04256f622
03359df9f4b249e253bfc47820d97178378d92b51f24de8d8455b5342f2d76797f484d9471253faea500058a89ac76482836c4d83314
b637ad7fc13ded93b704de011795c6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27cd5500b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbe0d2965d0cedde29f5e6919fa5f14ad44ce1b2ffc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 3233 more lines

17 Snmp Polling

PASS The NMS Runs Real Net-SNMP Tooling (0.1s, 2 commands)

The point of this VM: unlike the cirros hosts it has a genuine SNMP client, so the polling below is the real thing rather than something hand rolled for the test.

```
NMS snmpget --version 2>&1 | head -2; snmpwalk --version 2>&1 | head -1
NET-SNMP version: 5.9.4.pre2
NET-SNMP version: 5.9.4.pre2

NMS command -v snmpget snmpwalk snmptrapd
/usr/bin/snmpget
/usr/bin/snmpwalk
/usr/sbin/snmptrapd
```

PASS Every Router Answers An SNMPv3 authPriv Poll (0.2s, 3 commands)

The agent must be reachable and must accept the credentials the provisioner configured.

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

PASS Polled sysName Matches The Router's Own Hostname (0.4s, 6 commands)

Cross-check: the value SNMP returns is compared against what the device says on its CLI, so a plausible-looking answer from the wrong box cannot pass.

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

R1 show running-config | include ^hostname
hostname c8000v-r1

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

R2 show running-config | include ^hostname
hostname c8000v-r2

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"

R3 show running-config | include ^hostname
hostname c8000v-r3
```

PASS Polled sysDescr Identifies The Platform (0.2s, 6 commands)

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"

R1 show version | include Version
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.1.0
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R2 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.1.0
```

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```

R3 show version | include Version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0. For more details, see the
```

PASS sysUpTime Advances Between Two Polls (5.1s, 2 commands)

Proves the answers come from a live agent rather than a cached or replayed response.

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
```

```
.1.3.6.1.2.1.1.3.0 = Timeticks: (512734) 1:25:27.34
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
```

```
.1.3.6.1.2.1.1.3.0 = Timeticks: (513236) 1:25:32.36
```

PASS Walking The Interface Table Returns Every Interface The CLI Shows (0.2s, 6 commands)

A walk, not a get: the agent must return the whole subtree. Every physical interface named by the CLI must appear in it.

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

R1 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	100.64.0.1	YES	TFTP	up	up
GigabitEthernet3	192.168.10.1	YES	TFTP	up	up
GigabitEthernet4	192.168.99.1	YES	TFTP	up	up
Loopback0	172.16.1.1	YES	TFTP	up	up
Loopback1	10.20.1.1	YES	manual	up	up
Tunnel0	172.20.0.1	YES	manual	up	up

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
```

```
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

R2 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	100.64.0.2	YES	TFTP	up	up
GigabitEthernet3	192.168.20.1	YES	TFTP	up	up
GigabitEthernet4	192.168.99.2	YES	TFTP	up	up
Loopback0	172.16.2.1	YES	TFTP	up	up
Loopback1	10.20.2.1	YES	manual	up	up
Tunnel0	172.20.0.2	YES	manual	up	up

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3

1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

R3 show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1	10.0.2.15	YES	DHCP	up	up
GigabitEthernet2	100.64.0.3	YES	TFTP	up	up
GigabitEthernet3	192.168.30.1	YES	TFTP	up	up
GigabitEthernet4	192.168.99.3	YES	TFTP	up	up
Loopback0	172.16.3.1	YES	TFTP	up	up
Loopback1	10.20.3.1	YES	manual	up	up
Tunnel0	172.20.0.3	YES	manual	up	up

PASS Polled Interface Count Agrees With The Interface Table (0.3s, 6 commands)

ifNumber is a scalar the agent maintains separately from the table, so the two disagreeing would mean a broken agent.

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1

1.3.6.1.2.1.2.1.0

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1

1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.2.1.0

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3

1.3.6.1.2.1.2.1.0

```
.1.3.6.1.2.1.2.1.0 = INTEGER: 9
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3

1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
```

```
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

PASS The LAN Interface Is Reported Up By Both SNMP And The CLI (0.4s, 9 commands)

ifOperStatus 1 is up(1). Checked against the CLI line protocol state for the same interface.

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

```
R1 show interfaces GigabitEthernet3 | include line protocol
```

GigabitEthernet3 is up, line protocol is up

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

```
R2 show interfaces GigabitEthernet3 | include line protocol
```

GigabitEthernet3 is up, line protocol is up

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.2
```

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"
```

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.8
```

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
```

```
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

R3 show interfaces GigabitEthernet3 | include line protocol
 GigabitEthernet3 is up, line protocol is up

PASS Spokes Are Polled Across The Encrypted Tunnel (1.2s, 16 commands)

The NMS shares a LAN with the hub only. Traffic to R2 and R3 crosses the IPsec tunnels, so a burst of polling must show up on the ESP counters at both ends.

R1 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 79927, #pkts encrypt: 79927, #pkts digest: 79927
  #pkts decaps: 43120, #pkts decrypt: 43120, #pkts verify: 43120
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
current_peer 100.64.0.3 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 52413, #pkts encrypt: 52413, #pkts digest: 52413
  #pkts decaps: 28013, #pkts decrypt: 28013, #pkts verify: 28013
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0
... 78 more lines
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2

1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 79935, #pkts encrypt: 79935, #pkts digest: 79935
    #pkts decaps: 43126, #pkts decrypt: 43126, #pkts verify: 43126
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  ... 78 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
  current_peer 100.64.0.3 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 52418, #pkts encrypt: 52418, #pkts digest: 52418
    #pkts decaps: 28018, #pkts decrypt: 28018, #pkts verify: 28018
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  ... 78 more lines

```

PASS A Wrong Authentication Password Is Refused (0.1s, 2 commands)

Negative control. The agent answers the correct credentials first, so a refusal here cannot be a dead router.

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A NotTheRightPassword -x AES -X LabPrivPass123 -t 2 -r 0
192.168.99.1 1.3.6.1.2.1.1.5.0
snmpget: Authentication failure (incorrect password, community or key)

```

PASS A Wrong Privacy Password Is Refused (2.2s, 3 commands)

Authentication succeeds but the payload will not decrypt, so the router drops the request silently and the client times out. Bracketed by a good poll to prove it is still alive.

```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X NotTheRightPassword -t 2 -r 0
192.168.99.1 1.3.6.1.2.1.1.5.0
Timeout: No Response from 192.168.99.1.

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

```

PASS Polling Without Privacy Is Refused By Every Router (0.3s, 6 commands)

The group is configured "v3 priv". Correct credentials at a lower security level must still be denied, otherwise the privacy requirement is decorative.

```
NMS snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
Error in packet
Reason: authorizationError (access denied to that object)

NMS snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
Error in packet
Reason: authorizationError (access denied to that object)

NMS snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0
Error in packet
Reason: authorizationError (access denied to that object)

NMS snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0
Error in packet
Reason: authorizationError (access denied to that object)

NMS snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0
Error in packet
Reason: authorizationError (access denied to that object)

NMS snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0
Error in packet
Reason: authorizationError (access denied to that object)
```

PASS An Unknown User Is Refused (0.1s, 1 commands)

Only the provisioned user exists in the USM table.

```
NMS snmpget -On -v3 -l authPriv -u intruder -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 2 -r 0 192.168.99.1
1.3.6.1.2.1.1.5.0
snmpget: Unknown user name
```

PASS SNMPv2c Gets No Answer At The Wire (6.3s, 6 commands)

Suite 16 asserts no community is configured. This asserts the consequence on the network: a v2c poll gets nothing back, while a v3 poll to the same address succeeds.

```
NMS snmpget -On -v2c -c public -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0
Timeout: No Response from 192.168.99.1.

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"

NMS snmpget -On -v2c -c public -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0
Timeout: No Response from 192.168.99.2.

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v2c -c public -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0
Timeout: No Response from 192.168.99.3.

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```


18 Mtu

PASS Every Tunnel Reports The Expected Transport MTU (0.3s, 9 commands)

The tunnel MTU must be below the physical one by the ESP overhead. Equal would mean encryption costs nothing, which would mean it is not happening.

```
R1 show ip interface brief | include ^Tunnel
Tunnel0          172.20.0.1      YES manual up
R1 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1472 bytes
R1 show ip interface Tunnel0 | include MTU
MTU is 1472 bytes
R2 show ip interface brief | include ^Tunnel
Tunnel0          172.20.0.2      YES manual up
R2 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1472 bytes
R2 show ip interface Tunnel0 | include MTU
MTU is 1472 bytes
R3 show ip interface brief | include ^Tunnel
Tunnel0          172.20.0.3      YES manual up
R3 show interfaces Tunnel0 | include Tunnel transport MTU
Tunnel transport MTU 1472 bytes
R3 show ip interface Tunnel0 | include MTU
MTU is 1472 bytes
```

PASS The Overhead Is What GRE With A Tunnel Key Costs (0.0s, 0 commands)

Records the number rather than assuming it: 20 bytes of outer IP, 4 of GRE and 4 for the tunnel key. If the encapsulation changes -- dropping the key, or moving IPsec from transport to tunnel mode -- this is the test that should fail first.

No device commands (assertion used earlier output).

PASS A Packet Exactly At The Tunnel MTU Crosses With DF Set (0.1s, 2 commands)

The permissive half of the boundary. Sourced from Loopback1 so the traffic is inside the protected selector. Warmed up first: the first packet between two spokes is consumed establishing the shortcut and its security association, so a cold measurement here reports 80 percent and says nothing about the MTU. That first-packet loss is DMVPN working, not failing.

```
R2 ping 10.20.3.1 source Loopback1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
R2 ping 10.20.3.1 source Loopback1 size 1472 df-bit repeat 5
Type escape sequence to abort.
Sending 5, 1472-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

PASS One Byte Over The Tunnel MTU Does Not Cross With DF Set (10.0s, 1 commands)

The restrictive half. Without this the test above would pass just as well against a tunnel with no MTU limit at all.

```
R2 ping 10.20.3.1 source Loopback1 size 1473 df-bit repeat 5
Type escape sequence to abort.
Sending 5, 1473-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
.....
Success rate is 0 percent (0/5)
```

PASS Oversize Traffic Still Crosses When Fragmentation Is Allowed (0.1s, 2 commands)

Distinguishes an MTU limit from a blackhole: the same sizes that fail with DF set must succeed without it.

```
R2 ping 10.20.3.1 source Loopback1 size 1500 repeat 5
Type escape sequence to abort.
Sending 5, 1500-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms

R2 ping 10.20.3.1 source Loopback1 size 2000 repeat 5
Type escape sequence to abort.
Sending 5, 2000-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

PASS Hosts Pass Sub-MTU Traffic With DF Set (4.0s, 1 commands)

End to end across two encrypt/decrypt hops, from machines that know nothing about the tunnel.

```
H2 ping -c 5 -M do -s 1444 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 1444(1472) bytes of data.
1452 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.48 ms
1452 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.43 ms
1452 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.36 ms
1452 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.33 ms
1452 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.50 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.329/1.421/1.503/0.066 ms
```

PASS An End Host Learns The Tunnel Path MTU (2.0s, 1 commands)

The test that matters most in practice. The host is two hops and an encryption boundary away from the constraint, yet it must end up knowing the real limit -- otherwise a client retransmits into a black hole instead of backing off. The host reporting the exact tunnel MTU is proof that path MTU discovery survived the tunnel.

```
H2 ping -c 2 -M do -s 1472 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
From 192.168.20.1 icmp_seq=1 Frag needed and DF set (mtu = 1472)
ping: local error: message too long, mtu=1472

--- 192.168.30.10 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1005ms
```

PASS Hosts Pass Oversize Traffic Without DF Set (4.0s, 1 commands)

The counterpart: the same payload that was refused with DF set gets through when it may be fragmented.

```
H2 ping -c 5 -s 1472 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
1480 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.47 ms
1480 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.46 ms
1480 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=2.01 ms
1480 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.57 ms
1480 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.41 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.414/1.582/2.008/0.218 ms
```

19 Oob

PASS Every Router Has A Management VRF (0.0s, 3 commands)

R1 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R2 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R3 show vrf

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

PASS Only The OOB Interface Is In The Management VRF (0.1s, 3 commands)

A VRF containing a data interface would defeat the point, so this asserts membership is exactly one interface.

R1 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R2 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

R3 show vrf MGMT

Name	Default RD	Protocols	Interfaces
MGMT	<not set>	ipv4	Gi4

PASS Every Router Holds Its OOB Address On The Flat Network (0.1s, 3 commands)

R1 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.1	YES	TFTP	up	up
------------------	--------------	-----	------	----	----

R2 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.2	YES	TFTP	up	up
------------------	--------------	-----	------	----	----

R3 show ip interface brief | include GigabitEthernet4

GigabitEthernet4	192.168.99.3	YES	TFTP	up	up
------------------	--------------	-----	------	----	----

PASS The Management VRF Routing Table Holds Only The Flat Network (0.1s, 3 commands)

The isolation, stated as routing: the VRF must know how to reach the management network and nothing else. A default route or a data prefix here would be a leak.

R1 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

... 5 more lines

R2 show ip route vrf MGMT

Routing Table: MGMT

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary

```

o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

```

... 5 more lines

R3 show ip route vrf MGMT

```

Routing Table: MGMT
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

```

... 5 more lines

PASS The Global Table Cannot Reach The Management Network (0.1s, 3 commands)

The other direction. Traffic in the global table must have no path to the OOB network, otherwise the VRF is decorative.

R1 show ip route 192.168.99.0

```
% Network not in table
```

R2 show ip route 192.168.99.0

```
% Network not in table
```

R3 show ip route 192.168.99.0

```
% Network not in table
```

PASS Every Router Answers SNMP On Its OOB Address (0.3s, 6 commands)

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

R1 show running-config | include ^hostname

```
hostname c8000v-r1
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

R2 show running-config | include ^hostname

```
hostname c8000v-r2
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

R3 show running-config | include ^hostname

```
hostname c8000v-r3
```

PASS Management Traffic Never Touches The Data Path (3.2s, 28 commands)

The negative property that makes this out-of-band. Polling a spoke used to cross an IPsec tunnel and was asserted to advance the ESP counters; now it must do the opposite. The comparison is against an idle window of the same length, because BGP and BFD tick those counters continuously regardless -- so an absolute delta proves nothing.

R1 show crypto ipsec sa

```

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
current_peer 100.64.0.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 80145, #pkts encrypt: 80145, #pkts digest: 80145
  #pkts decaps: 43337, #pkts decrypt: 43337, #pkts verify: 43337

```



```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 80151, #pkts encrypt: 80151, #pkts digest: 80151
    #pkts decaps: 43343, #pkts decrypt: 43343, #pkts verify: 43343
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  ... 78 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 80151, #pkts encrypt: 80151, #pkts digest: 80151
    #pkts decaps: 43343, #pkts decrypt: 43343, #pkts verify: 43343
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  ... 78 more lines

R1 show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
  current_peer 100.64.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 80165, #pkts encrypt: 80165, #pkts digest: 80165
    #pkts decaps: 43356, #pkts decrypt: 43356, #pkts verify: 43356
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
  ... 78 more lines

```

PASS Syslog Reaches The Collector From The OOB Address (12.2s, 4 commands)

rsyslog files are named for the sender. A file named after a LAN address would mean the router is logging from the data path despite the configuration. Each router is made to emit a marked message rather than waiting on ambient traffic: whether a router happens to have logged recently is not what this test is about, and depending on it makes the result a matter of timing.

```

NMS sudo grep -c -- 'LAB00B-SRC-R1-193959' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LAB00B-SRC-R2-193959' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
NMS sudo grep -c -- 'LAB00B-SRC-R3-193959' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
1
NMS sudo ls -l /var/log/lab
total 3412
-rw-r--r-- 1 syslog syslog 162978 Sep  7 23:40 192.168.99.1.log

```

```
-rw-r--r-- 1 syslog syslog 204503 Sep 7 23:40 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 127160 Sep 7 23:40 192.168.99.3.log
-rw-r--r-- 1 root root 1136638 Sep 7 23:39 snmptrapd.log
-rw-r--r-- 1 root root 1832337 Sep 7 23:39 traps-raw.log
```

PASS A Spoke Stays Manageable With Its Tunnel Down (14.4s, 15 commands)

The reason out-of-band management exists. Shutting R2's tunnel removes every data path between the hub and the spoke -- BGP drops, the LAN becomes unreachable -- and the spoke must still answer SNMP and keep logging, because none of that ever used the tunnel.

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c800v-r2.lab.local"
```

R2 configure terminal

```
Enter configuration commands, one per line. End with CNTL/Z.
```

R2 interface Tunnel0

```
${output} =
```

R2 shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show ip route 192.168.20.0

```
Routing entry for 192.168.20.0/24
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:08:27 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:08:27 ago
    opaque_ptr 0x747B4EAD55B8
    Route metric is 0, traffic share count is 1
    AS Hops 1
    Route tag 65002
    MPLS label: none
```

R1 show ip route 192.168.20.0

```
% Network not in table
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
```

```
.1.3.6.1.2.1.1.5.0 = STRING: "c800v-r2.lab.local"
```

```
NMS sudo grep -c -- 'LAB00B-194017' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
1
```

R2 configure terminal

```
Enter configuration commands, one per line. End with CNTL/Z.
```

R2 interface Tunnel0

```
${output} =
```

R2 no shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 13, main routing table version 13
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 189/183 prefixes, 195/189 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (01:06:22.165 ago)
```

```
Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 16, main routing table version 16
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
```

2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 192/183 prefixes, 198/189 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (01:06:27.205 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
... 2 more lines									

20 Tacacs

PASS The TACACS Server Runs On The NMS (0.1s, 2 commands)

```
NMS systemctl is-active tacacs-lab
active
```

```
NMS sudo ss -ltnp | grep ':49 ' || echo NONE
LISTEN 0      128          0.0.0.0:49      0.0.0.0:*      users:(("tac_plus",pid=3254,fd=4))
LISTEN 0      128          [::]:49         [::]:*         users:(("tac_plus",pid=3254,fd=5))
```

PASS The Server Holds Both Remote Users And The Automation Account (0.1s, 1 commands)

The automation account has to exist on the server as well as locally on each router. Without it the first router to get AAA becomes unmanageable by this very test suite, while the server sits there perfectly healthy.

```
NMS sudo grep -oE '^user = [A-Za-z0-9_-]+' /etc/tacacs+/tac_plus.conf
user = netadmin
user = netops
user = lab
```

PASS Every Router Points At The Server Through The Management VRF (0.4s, 3 commands)

```
R1 show running-config | include ^aaa|^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source
```

```
aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10
```

```
R2 show running-config | include ^aaa|^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source
```

```
aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10
```

```
R3 show running-config | include ^aaa|^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source
```

```
aaa new-model
aaa group server tacacs+ LABTACGRP
ip vrf forwarding MGMT
ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
address ipv4 192.168.99.10
```

PASS Every Method List Falls Back To The Local Database (0.3s, 3 commands)

A method list with no local fallback locks the lab out the moment the server is unreachable -- including from the console, which is the one way back in.

```
R1 show running-config | include ^aaa authentication login|^aaa authorization exec
```

```
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

```
R2 show running-config | include ^aaa authentication login|^aaa authorization exec
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated

R3 show running-config | include ^aaa authentication login|^aaa authorization exec
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

PASS Every Router Reports The Server Alive (0.1s, 3 commands)

R1 show tacacs

```
Tacacs+ Server - public :
  Server name: LAB-TAC
  Server address: 192.168.99.10
  Server port: 49
  Socket opens:      1279
  Socket closes:     1279
  Socket aborts:      0
  Socket errors:      0
  Socket Timeouts:    0
  Failed Connect Attempts: 0
  Total Packets Sent: 1348
  Total Packets Recv: 1348
  Server Status: Alive
Continous Authc fail count: 3
... 1 more lines
```

R2 show tacacs

```
Tacacs+ Server - public :
  Server name: LAB-TAC
  Server address: 192.168.99.10
  Server port: 49
  Socket opens:      1974
  Socket closes:     1974
  Socket aborts:      0
  Socket errors:      0
  Socket Timeouts:    0
  Failed Connect Attempts: 0
  Total Packets Sent: 2010
  Total Packets Recv: 2010
  Server Status: Alive
Continous Authc fail count: 2
... 1 more lines
```

R3 show tacacs

```
Tacacs+ Server - public :
  Server name: LAB-TAC
  Server address: 192.168.99.10
  Server port: 49
  Socket opens:      733
  Socket closes:     733
  Socket aborts:      0
  Socket errors:      0
  Socket Timeouts:    0
  Failed Connect Attempts: 0
  Total Packets Sent: 799
  Total Packets Recv: 799
  Server Status: Alive
Continous Authc fail count: 2
... 1 more lines
```

PASS A Remote Administrator Authenticates And Is Given Privilege 15 (0.2s, 0 commands)

No device commands (assertion used earlier output).

PASS A Remote Operator Authenticates And Is Held At Privilege 1 (0.3s, 0 commands)

The pair that proves the level comes from the server. Two users, same router, same method list, different levels -- which cannot happen if the router is applying a default.

No device commands (assertion used earlier output).

PASS A Wrong Password Is Refused (2.1s, 0 commands)

No device commands (assertion used earlier output).

PASS An Unknown User Is Refused (2.1s, 0 commands)

The server rejects the user and the router honours that rather than quietly trying the local database.

No device commands (assertion used earlier output).

PASS The Server Records Authentication In Its Accounting Log (0.1s, 1 commands)

Accounting is what makes remote authentication auditable; without it the router knows who logged in and nobody else ever finds out.

```
NMS sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true
Sep  7 23:37:39 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4959 timezone=UTC
service=shell start_time=1788824259 priv-lvl=15 cmd=show monitor capture POOLCAP buffer brief <cr>
Sep  7 23:37:41 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4960 timezone=UTC
service=shell start_time=1788824261 priv-lvl=15 cmd=no monitor capture POOLCAP <cr>
Sep  7 23:37:41 192.168.99.2 lab tty434 10.0.2.2 stop task_id=5649 timezone=UTC
service=shell start_time=1788824172 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=89 stop_time=1788824261
Sep  7 23:37:42 192.168.99.3 lab tty434 10.0.2.2 stop task_id=4434 timezone=UTC
service=shell start_time=1788824172 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=90 stop_time=1788824262
Sep  7 23:37:42 192.168.99.1 lab tty435 10.0.2.2 start task_id=4962 timezone=UTC
service=shell start_time=1788824262
Sep  7 23:37:42 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4941 timezone=UTC
service=shell start_time=1788824172 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=90 stop_time=1788824262
Sep  7 23:37:42 192.168.99.2 lab tty435 10.0.2.2 start task_id=5657 timezone=UTC
service=shell start_time=1788824262
Sep  7 23:37:42 192.168.99.3 lab tty434 10.0.2.2 start task_id=4440 timezone=UTC
service=shell start_time=1788824262
Sep  7 23:37:42 192.168.99.1 lab tty435 10.0.2.2 stop task_id=4962 timezone=UTC
service=shell start_time=1788824262 priv-lvl=15 cmd=show running-config <cr>
elapsed_time=3 stop_time=1788824265 priv-lvl=15 cmd=show running-config <cr>
Sep  7 23:37:45 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4440 timezone=UTC
service=shell start_time=1788824262 priv-lvl=15 cmd=show running-config <cr>
elapsed_time=3 stop_time=1788824265
Sep  7 23:37:45 192.168.99.2 lab tty435 10.0.2.2 stop task_id=4962 timezone=UTC
service=shell start_time=1788824262 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788824265
Sep  7 23:37:45 192.168.99.1 lab tty435 10.0.2.2 stop task_id=5657 timezone=UTC
service=shell start_time=1788824262 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788824265
Sep  7 23:37:45 192.168.99.2 lab tty434 10.0.2.2 start task_id=4964 timezone=UTC
service=shell start_time=1788824265
... 186 more lines
```

PASS Accounting Records Carry The Privilege Level And Come From The OOB Address (0.1s, 1 commands)

```
NMS sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true
Sep  7 23:37:39 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4959 timezone=UTC
service=shell start_time=1788824259 priv-lvl=15 cmd=show monitor capture POOLCAP buffer brief <cr>
Sep  7 23:37:41 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4960 timezone=UTC
service=shell start_time=1788824261 priv-lvl=15 cmd=no monitor capture POOLCAP <cr>
Sep  7 23:37:41 192.168.99.2 lab tty434 10.0.2.2 stop task_id=5649 timezone=UTC
service=shell start_time=1788824172 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=89 stop_time=1788824261
Sep  7 23:37:42 192.168.99.3 lab tty434 10.0.2.2 stop task_id=4434 timezone=UTC
service=shell start_time=1788824172 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=90 stop_time=1788824262
Sep  7 23:37:42 192.168.99.1 lab tty435 10.0.2.2 start task_id=4962 timezone=UTC
service=shell start_time=1788824262
Sep  7 23:37:42 192.168.99.1 lab tty434 10.0.2.2 stop task_id=4941 timezone=UTC
service=shell start_time=1788824172 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=90 stop_time=1788824262
Sep  7 23:37:42 192.168.99.2 lab tty435 10.0.2.2 start task_id=5657 timezone=UTC
service=shell start_time=1788824262
Sep  7 23:37:42 192.168.99.3 lab tty434 10.0.2.2 start task_id=4440 timezone=UTC
service=shell start_time=1788824262
Sep  7 23:37:42 192.168.99.1 lab tty435 10.0.2.2 stop task_id=4962 timezone=UTC
service=shell start_time=1788824262 priv-lvl=15 cmd=show running-config <cr>
Sep  7 23:37:42 192.168.99.2 lab tty435 10.0.2.2 stop task_id=5657 timezone=UTC
```

```

service=shell start_time=1788824262 priv-lvl=15 cmd=show running-config <cr>
Sep 7 23:37:42 192.168.99.3 lab tty434 10.0.2.2 stop task_id=4440 timezone=UTC
service=shell start_time=1788824262 priv-lvl=15 cmd=show running-config <cr>
Sep 7 23:37:45 192.168.99.1 lab tty435 10.0.2.2 stop task_id=4962 timezone=UTC
service=shell start_time=1788824262 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788824265
Sep 7 23:37:45 192.168.99.2 lab tty435 10.0.2.2 stop task_id=5657 timezone=UTC
service=shell start_time=1788824262 disc-cause=1 disc-cause-ext=9 pre-session-time=0
elapsed_time=3 stop_time=1788824265
Sep 7 23:37:45 192.168.99.1 lab tty434 10.0.2.2 start task_id=4964 timezone=UTC
service=shell start_time=1788824265
... 186 more lines

```

PASS The Local Account Still Works When The Server Is Unreachable (19.6s, 21 commands)

The resilience property. R2's out-of-band interface is shut, which is the only path to the server, so TACACS becomes unreachable rather than merely unwilling. The automation account must still get in through the local database, and a remote-only user must not -- it exists nowhere else. The precondition is a ping inside the VRF, not "show tacacs". That field reports cached state -- it keeps saying Alive until a transaction actually fails -- so waiting on it would either hang or, worse, let this test run with the server still reachable and prove nothing.

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface GigabitEthernet4

#{output} =

R2 shutdown

#{output} =

R2 end

#{output} =

R2 ping vrf MGMT 192.168.99.10 repeat 2

% VRF MGMT does not have a usable source address

R1 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1 interface GigabitEthernet4

#{output} =

R1 no shutdown

#{output} =

R1 end

#{output} =

R2 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2 interface GigabitEthernet4

#{output} =

R2 no shutdown

#{output} =

R2 end

#{output} =

R3 configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3 interface GigabitEthernet4

#{output} =

R3 no shutdown

#{output} =

R3 end

#{output} =

R2 ping vrf MGMT 192.168.99.10 repeat 5

% VRF MGMT does not have a usable source address

R2 ping vrf MGMT 192.168.99.10 repeat 5

% VRF MGMT does not have a usable source address

R2 ping vrf MGMT 192.168.99.10 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:

.!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms

R2 ping vrf MGMT 192.168.99.10 repeat 5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:

!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

21 Vty Acl

PASS Every Router Applies The ACL Inbound On All VTY Lines (0.3s, 3 commands)

R1 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R2 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R3 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

PASS Only SSH Is Accepted On The VTY Lines (0.3s, 3 commands)

An ACL restricting who may connect is worth less if telnet is still an option for those who may.

R1 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R2 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R3 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

PASS The ACL Permits The Management Network And Denies The Rest (0.1s, 3 commands)

R1 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (4 matches)
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (186 matches)
 30 deny any log (6 matches)
```

R2 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (212 matches)
 30 deny any log (2 matches)
```

R3 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (184 matches)
 30 deny any log (2 matches)
```

PASS The Documented Exception Is Present And Is Only The Hypervisor Path (0.1s, 3 commands)

Pinned deliberately. This permit is the one concession in the policy, so it should be visible in a test rather than only in a comment -- and widening it should break something.

R1 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (4 matches)
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (186 matches)
 30 deny any log (6 matches)
```

R2 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (212 matches)
 30 deny any log (2 matches)
```

R3 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (184 matches)
 30 deny any log (2 matches)
```

PASS SSH From The Management Network Is Allowed (0.0s, 1 commands)

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.99.10',0))
try:
    s.connect(('192.168.99.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
SSH-2.0-Cisco-1.25
```

PASS SSH From Outside The Management Network Is Refused (0.1s, 1 commands)

The same machine, a different source address, and a target it is directly connected to -- so nothing but the ACL can account for the difference. Before the access-class was applied this path returned a banner just like the one above.

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

PASS Every Router Refuses A Login From Outside The Management Network (0.2s, 3 commands)

```
NMS python3 -c "import socket,sys
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

```
NMS python3 -c "import socket,sys
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.20.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

```
NMS python3 -c "import socket,sys
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.30.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

PASS The Refusal Is Logged And Reaches The Collector (0.1s, 2 commands)

"deny any log" makes a rejected connection auditable rather than merely blocked, and the message travels the OOB network to the collector like every other log.

```
NMS python3 -c "import socket,sys
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused

NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
<189>207: 000203: *Sep 7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep 7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep 7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep 7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep 7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user '' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep 7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep 7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source: 10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep 7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep 7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0 (10.0.2.2)
<190>216: 000212: *Sep 7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep 7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user '' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep 7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
```



```

authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep 7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep 7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR: Tunnel0: Recieved wrong
authentication string for Registration Request , Reason: authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 1098 more lines

```

PASS The Harness Path Still Works (0.1s, 3 commands)

The reason the second permit exists. If this fails the lab is unreachable by every other suite, so it is asserted here rather than discovered in the next run. Asserted from the ACL's own hit counter rather than from "show users". That column prints a resolved name when reverse DNS answers -- inside the packaged appliance the same session shows as "_gateway" instead of 10.0.2.2 -- so matching on the rendered address made this depend on name resolution rather than on the thing being tested. A hit on that ACE means a session was admitted by exactly the permit under discussion.

R1 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (6 matches)
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (186 matches)
 30 deny any log (9 matches)

```

R2 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (212 matches)
 30 deny any log (3 matches)

```

R3 show ip access-lists VTY-MGMT

```

Standard IP access list VTY-MGMT
 10 permit 192.168.99.0, wildcard bits 0.0.0.255
 20 permit 10.0.2.0, wildcard bits 0.0.0.255 (184 matches)
 30 deny any log (3 matches)

```

22 Banner

PASS Every Router Has A Login Banner Configured (0.3s, 3 commands)

```
R1 show running-config | include ^banner
  banner login ^C
R2 show running-config | include ^banner
  banner login ^C
R3 show running-config | include ^banner
  banner login ^C
```

PASS The Banner Is Presented Before Authentication (6.2s, 0 commands)

Captured from a login that is refused, so the banner demonstrably reaches an unauthenticated client.

No device commands (assertion used earlier output).

PASS The Presented Banner Matches The Source File Exactly (6.3s, 0 commands)

Line for line, not a substring match: a device carrying a stale or truncated copy should fail rather than pass on one memorable phrase still being present.

No device commands (assertion used earlier output).

PASS The Banner States The Warning It Exists To Give (2.1s, 0 commands)

A banner that does not actually warn is decoration. These are the elements that make it one: restricted access, a statement of monitoring, and consent following from use.

No device commands (assertion used earlier output).

PASS The Banner Discloses Nothing About The Platform (6.2s, 0 commands)

An unauthenticated visitor should learn that they are not welcome, and nothing else. A banner naming the vendor, model, software or hostname helps whoever should not be there more than it helps anyone who should.

No device commands (assertion used earlier output).

PASS The Banner Survives A Configuration Save (0.1s, 3 commands)

Guards the case where a banner is applied to the running configuration and never written -- present until the next reload, and absent exactly when it would be needed.

```
R1 show startup-config | include ^banner
  banner login ^C
R2 show startup-config | include ^banner
  banner login ^C
R3 show startup-config | include ^banner
  banner login ^C
```

Appendix — captured device state

Written by suite *05 Capture State* during the run. Routing tables and protocol state first, then what the NMS sees for each router -- polled over SNMPv3, and received as syslog and traps -- then the full running configuration of each router.

R1 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```
S* 0.0.0.0/0 [254/0] via 10.0.2.2
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C 10.0.2.0/24 is directly connected, GigabitEthernet1
L 10.0.2.15/32 is directly connected, GigabitEthernet1
C 10.20.1.1/32 is directly connected, Loopback1
B 10.20.2.1/32 [20/0] via 172.20.0.2, 00:03:08
B 10.20.3.1/32 [20/0] via 172.20.0.3, 00:15:29
S 10.30.1.0/24 is directly connected, Null0
B 10.30.2.0/24 [20/0] via 172.20.0.2, 00:03:08
B 10.30.3.0/24 [20/0] via 172.20.0.3, 00:15:29
100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 100.64.0.0/24 is directly connected, GigabitEthernet2
L 100.64.0.1/32 is directly connected, GigabitEthernet2
172.16.0.0/32 is subnetted, 1 subnets
C 172.16.1.1 is directly connected, Loopback0
172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.20.0.0/24 is directly connected, Tunnel0
L 172.20.0.1/32 is directly connected, Tunnel0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet3
L 192.168.10.1/32 is directly connected, GigabitEthernet3
B 192.168.20.0/24 [20/0] via 172.20.0.2, 00:03:08
B 192.168.30.0/24 [20/0] via 172.20.0.3, 00:15:29
```

R2 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```
S* 0.0.0.0/0 [254/0] via 10.0.2.2
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C 10.0.2.0/24 is directly connected, GigabitEthernet1
L 10.0.2.15/32 is directly connected, GigabitEthernet1
B 10.20.1.1/32 [20/0] via 172.20.0.1, 00:03:09
C 10.20.2.1/32 is directly connected, Loopback1
B 10.20.3.1/32 [20/0] via 172.20.0.3, 00:03:09
B 10.30.1.0/24 [20/0] via 172.20.0.1, 00:03:09
S 10.30.2.0/24 is directly connected, Null0
B 10.30.3.0/24 [20/0] via 172.20.0.3, 00:03:09
100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 100.64.0.0/24 is directly connected, GigabitEthernet2
L 100.64.0.2/32 is directly connected, GigabitEthernet2
172.16.0.0/32 is subnetted, 1 subnets
```

```

C      172.16.2.1 is directly connected, Loopback0
      172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.20.0.0/24 is directly connected, Tunnel0
L      172.20.0.2/32 is directly connected, Tunnel0
B      192.168.10.0/24 [20/0] via 172.20.0.1, 00:03:09
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.20.0/24 is directly connected, GigabitEthernet3
L      192.168.20.1/32 is directly connected, GigabitEthernet3
B      192.168.30.0/24 [20/0] via 172.20.0.3, 00:03:09

```

R3 — ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr
& - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

```

S*    0.0.0.0/0 [254/0] via 10.0.2.2
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C      10.0.2.0/24 is directly connected, GigabitEthernet1
L      10.0.2.15/32 is directly connected, GigabitEthernet1
B      10.20.1.1/32 [20/0] via 172.20.0.1, 00:15:29
B      10.20.2.1/32 [20/0] via 172.20.0.2, 00:02:46
C      10.20.3.1/32 is directly connected, Loopback1
B      10.30.1.0/24 [20/0] via 172.20.0.1, 00:15:29
B      10.30.2.0/24 [20/0] via 172.20.0.2, 00:02:46
S      10.30.3.0/24 is directly connected, Null0
      100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      100.64.0.0/24 is directly connected, GigabitEthernet2
L      100.64.0.3/32 is directly connected, GigabitEthernet2
      172.16.0.0/32 is subnetted, 1 subnets
C      172.16.3.1 is directly connected, Loopback0
      172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.20.0.0/24 is directly connected, Tunnel0
L      172.20.0.3/32 is directly connected, Tunnel0
B      192.168.10.0/24 [20/0] via 172.20.0.1, 00:15:29
B      192.168.20.0/24 [20/0] via 172.20.0.2, 00:02:46
      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.30.0/24 is directly connected, GigabitEthernet3
L      192.168.30.1/32 is directly connected, GigabitEthernet3

```

R1 — bgp summary

BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 28, main routing table version 28
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 136/127 prefixes, 139/130 paths, scan interval 60 secs
22 networks peaked at 22:33:59 Sep 7 2026 UTC (00:48:55.594 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.2	4	65002	26	29	28	0	0	00:03:08	3
172.20.0.3	4	65003	106	115	28	0	0	00:15:40	3

R2 — bgp summary

BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 46, main routing table version 46
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory

2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 160/151 prefixes, 162/153 paths, scan interval 60 secs
31 networks peaked at 22:34:42 Sep 7 2026 UTC (00:48:12.674 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	29	26	46	0	0	00:03:08	6

R3 — bgp summary

BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 146, main routing table version 146
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 72/63 prefixes, 77/68 paths, scan interval 60 secs
9 networks peaked at 22:15:22 Sep 7 2026 UTC (01:07:32.746 ago)

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
172.20.0.1	4	65001	115	106	146	0	0	00:15:41	6

R1 — bgp table

BGP table version is 28, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	0.0.0.0	0		32768	i
*> 10.20.2.1/32	172.20.0.2	0		0	65002 i
*> 10.20.3.1/32	172.20.0.3	0		0	65003 i
*> 10.30.1.0/24	0.0.0.0	0		32768	i
*> 10.30.2.0/24	172.20.0.2	0		0	65002 i
*> 10.30.3.0/24	172.20.0.3	0		0	65003 i
*> 192.168.10.0	0.0.0.0	0		32768	i
*> 192.168.20.0	172.20.0.2	0		0	65002 i
*> 192.168.30.0	172.20.0.3	0		0	65003 i

R2 — bgp table

BGP table version is 46, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.20.1.1/32	172.20.0.1	0		0	65001 i
*> 10.20.2.1/32	0.0.0.0	0		32768	i
*> 10.20.3.1/32	172.20.0.3	0		0	65001 65003 i
*> 10.30.1.0/24	172.20.0.1	0		0	65001 i
*> 10.30.2.0/24	0.0.0.0	0		32768	i
*> 10.30.3.0/24	172.20.0.3	0		0	65001 65003 i
*> 192.168.10.0	172.20.0.1	0		0	65001 i
*> 192.168.20.0	0.0.0.0	0		32768	i
*> 192.168.30.0	172.20.0.3	0		0	65001 65003 i

R3 — bgp table

BGP table version is 146, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

```

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-state,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.20.1.1/32	172.20.0.1	0		0	65001 i
*>	10.20.2.1/32	172.20.0.2			0	65001 65002 i
*>	10.20.3.1/32	0.0.0.0	0		32768	i
*>	10.30.1.0/24	172.20.0.1	0		0	65001 i
*>	10.30.2.0/24	172.20.0.2			0	65001 65002 i
*>	10.30.3.0/24	0.0.0.0	0		32768	i
*>	192.168.10.0	172.20.0.1	0		0	65001 i
*>	192.168.20.0	172.20.0.2			0	65001 65002 i
*>	192.168.30.0	0.0.0.0	0		32768	i

R1 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
1	100.64.0.1/500	100.64.0.2/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/100 sec				
CE id: 1029, Session-id: 26				
Local spi: 674CC240C9A38803 Remote spi: 6B960B9A3952A45C				

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
2	100.64.0.1/500	100.64.0.3/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/100 sec				
CE id: 1030, Session-id: 27				
Local spi: C7ABC0CAC6503940 Remote spi: 54006597FF55CAD2				

IPv6 Crypto IKEv2 SA

R2 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
1	100.64.0.2/500	100.64.0.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/101 sec				
CE id: 1029, Session-id: 24				
Local spi: 6B960B9A3952A45C Remote spi: 674CC240C9A38803				

IPv6 Crypto IKEv2 SA

R3 — crypto ikev2 sa

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
1	100.64.0.3/500	100.64.0.1/500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK				
Life/Active Time: 86400/101 sec				
CE id: 1026, Session-id: 26				
Local spi: 54006597FF55CAD2 Remote spi: C7ABC0CAC6503940				

IPv6 Crypto IKEv2 SA

H1 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
   link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 82334sec preferred_lft 71534sec

```

```

inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
    valid_lft 86212sec preferred_lft 14212sec
inet6 fe80::5054:ff:feaa:101/64 scope link
    valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.15/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.16/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.17/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.18/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.19/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.64/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.65/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.66/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.67/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.68/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.69/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.70/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.71/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.72/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.73/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.74/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.75/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.76/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.77/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.78/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.79/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.80/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.81/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.82/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.83/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.84/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.85/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.86/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.87/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.88/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.89/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.10.90/24 scope global secondary eth1

```

```

    valid_lft forever preferred_lft forever
inet 192.168.10.91/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.92/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.10.93/24 scope global secondary eth1
    valid_lft forever preferred_lft forever

```

H1 — ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.229.55 metric 1003
192.168.10.0/24 dev eth1 scope link src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1

```

H2 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 82337sec preferred_lft 71537sec
    inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
        valid_lft 86083sec preferred_lft 14083sec
    inet6 fe80::5054:ff:feaa:201/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.15/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.16/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.17/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.18/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.19/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.64/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.65/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.66/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.67/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.68/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.69/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.70/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.71/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.72/24 scope global secondary eth1
        valid_lft forever preferred_lft forever
    inet 192.168.20.73/24 scope global secondary eth1
        valid_lft forever preferred_lft forever

```



```

inet 192.168.20.74/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.75/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.76/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.77/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.78/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.79/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.80/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.81/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.82/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.83/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.84/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.85/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.86/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.87/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.88/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.89/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.90/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.91/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.92/24 scope global secondary eth1
    valid_lft forever preferred_lft forever
inet 192.168.20.93/24 scope global secondary eth1
    valid_lft forever preferred_lft forever

```

H2 — ip route

```

default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.199.41 metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1

```

H3 — ip addr

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 82336sec preferred_lft 71536sec
    inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
        valid_lft 86292sec preferred_lft 14292sec
    inet6 fe80::5054:ff:feaa:301/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
        valid_lft forever preferred_lft forever

```

[illegible]

H3 — ip route

```
default via 10.0.2.2 dev eth0 src 10.0.2.15 metric 1002
```

```
10.0.2.0/24 dev eth0 scope link src 10.0.2.15 metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link src 169.254.207.242 metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link src 192.168.30.10
```

NMS — network

```
lo UNKNOWN 127.0.0.1/8 ::1/128
ens3 UP 10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4 UP 192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5 UP 192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS — snmp version

NET-SNMP version: 5.9.4.pre2

NMS — collectors

active
active
active

NMS — collector files

```
total 2660
-rw-r--r-- 1 syslog syslog 128123 Sep 7 23:22 192.168.99.1.log
-rw-r--r-- 1 syslog syslog 159924 Sep 7 23:22 192.168.99.2.log
-rw-r--r-- 1 syslog syslog 105045 Sep 7 23:22 192.168.99.3.log
-rw-r--r-- 1 root root 881546 Sep 7 23:22 snmptrapd.log
-rw-r--r-- 1 root root 1416130 Sep 7 23:22 traps-raw.log
```

NMS — R1 syslog received

tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory

NMS — R2 syslog received

tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory

NMS — R3 syslog received

tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory

NMS — snmptrapd

```
2026-09-07 23:22:26 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:39.57 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.29 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.29 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.29.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.29.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.29.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.29.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.29.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.29.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.29.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.29.1 0
2026-09-07 23:22:26 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:40.66 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.29 3600 iso.3.6.1.4.1.9.9.171.1.3.2.1.8.29 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.29.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.29.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.29.1 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.3.3.1.9.29.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.29.1 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.3.3.1.11.29.1 "64 40 00 03 "
```

iso.3.6.1.4.1.9.9.171.1.3.3.1.6.29.1 47 iso.3.6.1.4.1.9.9.171.1.3.3.1.7.29.1 0
2026-09-07 23:22:27 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:40.41 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708049 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708049 0
iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:27 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.50 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708035 0 iso.3.6.1.4.1.9.10.137.1.2.1.8.1708035 0
iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.71 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.29 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.29 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.1.6.29 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.29 4
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.1.10.29 213 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.28 213
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.28 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.28 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.28 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.28 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.28 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.28 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.28 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.28 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.28 0
2026-09-07 23:22:28 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.72 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.29 214 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.28 214
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.28 3600 iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.28 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.28 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.28 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.28 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.28 "64 40 00 03 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.28 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.28 "64 40 00 02 " iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.28 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.28 0
2026-09-07 23:22:29 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.80 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.2
iso.3.6.1.4.1.9.10.137.1.2.1.8.987139 3 iso.3.6.1.4.1.9.10.137.1.2.1.8.987139 3 iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:29 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.72 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-07 23:22:29 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.81 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.2.1.4.100.64.0.3.29 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.2.1.4.100.64.0.3.29 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.29 2 iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.29 6
2026-09-07 23:22:29 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:41.72 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.2
iso.3.6.1.4.1.9.10.137.1.2.1.8.987153 3 iso.3.6.1.4.1.9.10.137.1.2.1.8.987153 3 iso.3.6.1.2.1.2.2.1.1.8 8
2026-09-07 23:22:29 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:43.64 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.59 "DMVPN" iso.3.6.1.4.1.9.9.41.1.2.3.1.3.59 4
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.59 "DMVPN_NHRP_ERROR" iso.3.6.1.4.1.9.9.41.1.2.3.1.5.59 " Tunnel0: NHRP
Encap Error for Purge Reply , Reason: protocol generic error (7) on (Tunnel: 172.20.0.2 NBMA: 100.64.0.2)"
iso.3.6.1.4.1.9.9.41.1.2.3.1.6.59 0:1:08:43.64
2026-09-07 23:22:29 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:42.55 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.50 "DMVPN" iso.3.6.1.4.1.9.9.41.1.2.3.1.3.50 4
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.50 "DMVPN_NHRP_ERROR" iso.3.6.1.4.1.9.9.41.1.2.3.1.5.50 " Tunnel0: NHRP
Encap Error for Purge Request , Reason: protocol generic error (7) on (Tunnel: 172.20.0.3 NBMA:
100.64.0.3)" iso.3.6.1.4.1.9.9.41.1.2.3.1.6.50 0:1:08:42.55
2026-09-07 23:22:38 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:52.91 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.435.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.57102 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.57102 3555
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.57102 3504
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.57102 52339 iso.3.6.1.4.1.9.2.9.2.1.18.435 "lab"
2026-09-07 23:22:38 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:51.84 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.55618 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.55618 3547
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.55618 2784
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.55618 29731 iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-07 23:22:38 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:52.21 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.435.1 6 iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.36348 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.36348 3571
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.36348 2688
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.36348 28707 iso.3.6.1.4.1.9.2.9.2.1.18.435 "lab"
2026-09-07 23:22:41 <UNKNOWN> [UDP: [192.168.99.1]:54980->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:08:55.08 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1

```

iso.3.6.1.4.1.9.9.43.1.1.6.1.3.174 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.174 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.174 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.174 "lab"
2026-09-07 23:22:46 <UNKNOWN> [UDP: [192.168.99.2]:49602->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:00.46 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.387 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.387 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.387 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.387 "lab"
2026-09-07 23:22:50 <UNKNOWN> [UDP: [192.168.99.3]:50049->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:1:09:03.74 iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.127 1 iso.3.6.1.4.1.9.9.43.1.1.6.1.4.127 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.127 2 iso.3.6.1.4.1.9.9.43.1.1.6.1.8.127 "lab"

```

NMS — R1 polled over SNMPv3: snmp system

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (415000) 1:09:10.00
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

NMS — R1 polled over SNMPv3: snmp interfaces

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

```

NMS — R2 polled over SNMPv3: snmp system

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (415086) 1:09:10.86
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00

```

NMS — R2 polled over SNMPv3: snmp interfaces

```

.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Tunnel0"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Loopback1"

```

NMS — R3 polled over SNMPv3: snmp system

```

.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (414990) 1:09:09.90

```

NMS — R3 polled over SNMPv3: snmp interfaces

R1 — running config

3000V IPsec Testbed - test evidence

C8000V IPsec Testbed - test evidence

```

604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28
quit
!
!
!
!
!
!
!
!
!
license udi pid C8000V sn 9VRWIY571LW
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootstrap level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$uia9E1anibovxE$Qyklcz.SLvH8epz2w2TjV6d9TggTO3p9KSxEGgTTD2
username lab privilege 15 secret 9 $9$kSVY7W/rHjFg0.$AdIeocLbIuwaSOCYSowG0l/6LNagQ58URuynu0Dbtq6
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
 encryption aes-cbc-256
 integrity sha256
 group 14
!
crypto ikev2 policy LAB-POL
 proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
 peer ANY
 address 0.0.0.0 0.0.0.0
 pre-shared-key LabPreSharedKey123
!
!
!
crypto ikev2 profile LAB-PROF
 match identity remote address 0.0.0.0
 authentication remote pre-share
 authentication local pre-share
 keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
 mode transport
!
crypto ipsec profile LAB-IPSEC
 set transform-set LAB-TS
 set ikev2-profile LAB-PROF
!
!
!
!
!
```



```

!
!
!
interface Loopback0
 description Protected endpoint
 ip address 172.16.1.1 255.255.255.255
!
interface Loopback1
 description BGP-advertised prefix
 ip address 10.20.1.1 255.255.255.255
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
 bfd interval 500 min_rx 500 multiplier 3
 tunnel source GigabitEthernet2
 tunnel mode gre multipoint
 tunnel key 100
 tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
 description MGMT - QEMU user-mode net (DHCP)
 ip address dhcp
 negotiation auto
!
interface GigabitEthernet2
 description WAN - shared provider segment, DMVPN underlay
 ip address 100.64.0.1 255.255.255.0
 negotiation auto
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 negotiation auto
!
interface GigabitEthernet4
 description OOB management - NTP, syslog, SNMP
 vrf forwarding MGMT
 ip address 192.168.99.1 255.255.255.0
 negotiation auto
!
router bgp 65001
 bgp router-id 172.16.1.1
 bgp log-neighbor-changes
 neighbor 172.20.0.2 remote-as 65002
 neighbor 172.20.0.2 password LabBgpSecret
 neighbor 172.20.0.2 timers 10 30
 neighbor 172.20.0.2 fall-over bfd
 neighbor 172.20.0.3 remote-as 65003
 neighbor 172.20.0.3 password LabBgpSecret
 neighbor 172.20.0.3 timers 10 30
 neighbor 172.20.0.3 fall-over bfd
!
address-family ipv4
 network 10.20.1.1 mask 255.255.255.255
 network 10.30.1.0 mask 255.255.255.0
 network 192.168.10.0
 neighbor 172.20.0.2 activate
 neighbor 172.20.0.2 send-community both
 neighbor 172.20.0.2 route-map RM-IN-R2 in
 neighbor 172.20.0.2 route-map RM-OUT-R2 out
 neighbor 172.20.0.2 maximum-prefix 15 80
 neighbor 172.20.0.3 activate
 neighbor 172.20.0.3 send-community both
 neighbor 172.20.0.3 route-map RM-IN-R3 in
 neighbor 172.20.0.3 route-map RM-OUT-R3 out
 neighbor 172.20.0.3 maximum-prefix 15 80
 exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1

```

```

ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65002$
ip as-path access-list 20 permit ^65003$
no ip http server
ip http secure-server
ip route 10.30.1.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.1.100 10.30.1.129 prefix-length 24
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-FAR-R2 seq 40 permit 10.20.2.1/32
ip prefix-list PL-FAR-R2 seq 45 permit 10.30.2.0/24
ip prefix-list PL-FAR-R2 seq 50 permit 192.168.20.0/24
!
ip prefix-list PL-FAR-R3 seq 40 permit 10.20.3.1/32
ip prefix-list PL-FAR-R3 seq 45 permit 10.30.3.0/24
ip prefix-list PL-FAR-R3 seq 50 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.10.0/24
!
ip prefix-list PL-OWN-L0 seq 5 permit 10.20.1.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.1.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.10.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.10.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.10.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.10.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.10.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.10.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.10.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.10.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.10.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.10.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R2 permit 10
  description R1's own L0, tagged 65001:1
  match ip address prefix-list PL-OWN-L0
  set community 65001:1
!
route-map RM-OUT-R2 permit 20
  description R1's own NAT, tagged 65001:2
  match ip address prefix-list PL-OWN-NAT
  set community 65001:2
!
route-map RM-OUT-R2 permit 30
  description R1's own LAN, tagged 65001:3
  match ip address prefix-list PL-OWN-LAN
  set community 65001:3
!
route-map RM-OUT-R2 permit 55
  description re-advertise R3, preserving its communities

```

```

match ip address prefix-list PL-FAR-R3
!
route-map RM-OUT-R3 permit 10
description R1's own LO, tagged 65001:1
match ip address prefix-list PL-OWN-LO
set community 65001:1
!
route-map RM-OUT-R3 permit 20
description R1's own NAT, tagged 65001:2
match ip address prefix-list PL-OWN-NAT
set community 65001:2
!
route-map RM-OUT-R3 permit 30
description R1's own LAN, tagged 65001:3
match ip address prefix-list PL-OWN-LAN
set community 65001:3
!
route-map RM-OUT-R3 permit 55
description re-advertise R2, preserving its communities
match ip address prefix-list PL-FAR-R2
!
route-map NAT-TO-DOMAIN permit 10
match ip address NAT-DOMAIN
!
route-map RM-IN-R2 permit 10
description accept only known communities on an AS path R2 can produce
match as-path 10
match community CL-VALID
!
route-map RM-IN-R3 permit 10
description accept only known communities on an AS path R3 can produce
match as-path 20
match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold

```

```

snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch

```

```

snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
*****
                        AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0

```


C8000V IPsec Testbed - test evidence


```

!
interface Loopback0
description Protected endpoint
ip address 172.16.2.1 255.255.255.255
!
interface Loopback1
description BGP-advertised prefix
ip address 10.20.2.1 255.255.255.255
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.2 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode gre multipoint
tunnel key 100
tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
description MGMT - QEMU user-mode net (DHCP)
ip address dhcp
negotiation auto
!
interface GigabitEthernet2
description WAN - shared provider segment, DMVPN underlay
ip address 100.64.0.2 255.255.255.0
negotiation auto
!
interface GigabitEthernet3
description LAN toward the local host
ip address 192.168.20.1 255.255.255.0
ip nat inside
negotiation auto
!
interface GigabitEthernet4
description OOB management - NTP, syslog, SNMP
vrf forwarding MGMT
ip address 192.168.99.2 255.255.255.0
negotiation auto
!
router bgp 65002
bgp router-id 172.16.2.1
bgp log-neighbor-changes
neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 password LabBgpSecret
neighbor 172.20.0.1 timers 10 30
neighbor 172.20.0.1 fall-over bfd
!
address-family ipv4
network 10.20.2.1 mask 255.255.255.255
network 10.30.2.0 mask 255.255.255.0
network 192.168.20.0
neighbor 172.20.0.1 activate
neighbor 172.20.0.1 send-community both
neighbor 172.20.0.1 route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-OUT-R1 out
neighbor 172.20.0.1 maximum-prefix 15 80
exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$
ip as-path access-list 10 permit ^65001_65003$
no ip http server

```

```

ip http secure-server
ip route 10.30.2.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.2.100 10.30.2.129 prefix-length 24
ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.20.0/24
!
ip prefix-list PL-OWN-LO seq 5 permit 10.20.2.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.2.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.20.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.20.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.20.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.20.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.20.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.20.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.20.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.20.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.20.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.20.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
 description R2's own LO, tagged 65002:1
 match ip address prefix-list PL-OWN-LO
 set community 65002:1
!
route-map RM-OUT-R1 permit 20
 description R2's own NAT, tagged 65002:2
 match ip address prefix-list PL-OWN-NAT
 set community 65002:2
!
route-map RM-OUT-R1 permit 30
 description R2's own LAN, tagged 65002:3
 match ip address prefix-list PL-OWN-LAN
 set community 65002:3
!
route-map NAT-TO-DOMAIN permit 10
 match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
 description accept only known communities on an AS path R1 can produce
 match as-path 10
 match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change

```

```

snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp

```

```

snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!

```

```

!
!
!
!
banner login ^C
*****
                        AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end

```

R3 — running config

Building configuration...

```

Current configuration : 19832 bytes
!
! Last configuration change at 23:21:14 UTC Mon Sep 7 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
hostname c8000v-r3
!
boot-start-marker
boot-end-marker
!
!
vrf definition MGMT
  description out-of-band management
  !
  address-family ipv4
  exit-address-family
!
aaa new-model
!
!

```

[illegible]

```

D92847C4 DB87D6F8 C7D48A2 D1BB826A 822083B9 88DDDD430 2DFA57EA FD22715D
D73197D0 5A6C3582 5D12968B 56DE28B1 1D162D61 119BB282 E2239093 D96D95DD
D1122D35 0BF5780D 0FD3EB53 EF40D3D1 61693953 E97694CD C6F369F8 6B010182
D4978C4B 9316184D FDCDF993 82047F48 557B580A 8AFDB4D8 AB1CBD94 80E67D03
EB61EACC 075A9CDF F272A80F 02544535 24886FBF 500A61CC D2F56993 9457B80D
C3C63E80 359E03A4 1256F1C9 7F905BBA 20E341AD B99A8BCC 02C5B9FC 96C55D7B
66EE9989 ABC1C984 101DE474 631A5848 E41FB71A 9654FF7A 0302C615 EF5FE298B
A2C2F138 17AE6A7D 9B630B20 26159088 5AF665AF

quit
crypto pki certificate chain SLA-TrustPoint
certificate ca 01
30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7 D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C08D23CF 58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
C7479096 B4BC2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AEA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 D4565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDDEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEB7CF9F 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28

quit
!
!
!
!
!
!
!
!
!
!
license udi pid C8000V sn 98WSPSI3CQ2
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootstrap level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$Bw47UP1ye/.yVU$K5CSpZUsnABYfZykCttzeySotiNrBY/frbuYNtoXY2
!
username lab privilege 15 secret 9 $9$nTmPdPNMf5C6Nk$TZUyweyM6Pgquc3KoKrRbcfdLyeESXbtbt75p43gz7g
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
 encryption aes-cbc-256
 integrity sha256
 group 14
!
crypto ikev2 policy LAB-POL
 proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
 peer ANY
 address 0.0.0.0 0.0.0.0
 pre-shared-key LabPreSharedKey123
!
!
!
crypto ikev2 profile LAB-PROF
 match identity remote address 0.0.0.0

```

```

authentication remote pre-share
authentication local pre-share
keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
mode transport
!
crypto ipsec profile LAB-IPSEC
set transform-set LAB-TS
set ikev2-profile LAB-PROF
!
!
!
!
!
!
!
!
!
interface Loopback0
description Protected endpoint
ip address 172.16.3.1 255.255.255.255
!
interface Loopback1
description BGP-advertised prefix
ip address 10.20.3.1 255.255.255.255
!
interface Tunnel0
description DMVPN spoke
ip address 172.20.0.3 255.255.255.0
no ip redirects
ip nat outside
ip nhrp authentication LabNhrp
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
ip tcp adjust-mss 1360
bfd interval 500 min_rx 500 multiplier 3
tunnel source GigabitEthernet2
tunnel mode gre multipoint
tunnel key 100
tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
description MGMT - QEMU user-mode net (DHCP)
ip address dhcp
negotiation auto
!
interface GigabitEthernet2
description WAN - shared provider segment, DMVPN underlay
ip address 100.64.0.3 255.255.255.0
negotiation auto
!
interface GigabitEthernet3
description LAN toward the local host
ip address 192.168.30.1 255.255.255.0
ip nat inside
negotiation auto
!
interface GigabitEthernet4
description OOB management - NTP, syslog, SNMP
vrf forwarding MGMT
ip address 192.168.99.3 255.255.255.0
negotiation auto
!
router bgp 65003
bgp router-id 172.16.3.1
bgp log-neighbor-changes

```



```

neighbor 172.20.0.1 remote-as 65001
neighbor 172.20.0.1 password LabBgpSecret
neighbor 172.20.0.1 timers 10 30
neighbor 172.20.0.1 fall-over bfd
!
address-family ipv4
network 10.20.3.1 mask 255.255.255.255
network 10.30.3.0 mask 255.255.255.0
network 192.168.30.0
neighbor 172.20.0.1 activate
neighbor 172.20.0.1 send-community both
neighbor 172.20.0.1 route-map RM-IN-R1 in
neighbor 172.20.0.1 route-map RM-OUT-R1 out
neighbor 172.20.0.1 maximum-prefix 15 80
exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$
ip as-path access-list 10 permit ^65001_65002$
no ip http server
ip http secure-server
ip route 10.30.3.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.3.100 10.30.3.129 prefix-length 24
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-L0 seq 5 permit 10.20.3.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.3.0/24
!
ip access-list standard VTY-MGMT
10 remark out-of-band management network
10 permit 192.168.99.0 0.0.0.255
20 remark hypervisor user-mode NAT: how the harness reaches these devices
20 permit 10.0.2.0 0.0.0.255
30 deny any log
!
ip access-list extended NAT-DOMAIN
10 permit ip host 192.168.30.10 10.30.0.0 0.0.255.255
20 permit ip host 192.168.30.11 10.30.0.0 0.0.255.255
30 permit ip host 192.168.30.12 10.30.0.0 0.0.255.255
40 permit ip host 192.168.30.13 10.30.0.0 0.0.255.255
50 permit ip host 192.168.30.14 10.30.0.0 0.0.255.255
60 permit ip host 192.168.30.15 10.30.0.0 0.0.255.255
70 permit ip host 192.168.30.16 10.30.0.0 0.0.255.255
80 permit ip host 192.168.30.17 10.30.0.0 0.0.255.255
90 permit ip host 192.168.30.18 10.30.0.0 0.0.255.255
100 permit ip host 192.168.30.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
description R3's own L0, tagged 65003:1
match ip address prefix-list PL-OWN-L0
set community 65003:1
!

```

```

route-map RM-OUT-R1 permit 20
description R3's own NAT, tagged 65003:2
match ip address prefix-list PL-OWN-NAT
set community 65003:2
!
route-map RM-OUT-R1 permit 30
description R3's own LAN, tagged 65003:3
match ip address prefix-list PL-OWN-LAN
set community 65003:3
!
route-map NAT-T0-DOMAIN permit 10
match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
description accept only known communities on an AS path R1 can produce
match as-path 10
match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs

```

```

snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsr
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics

```

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snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
  address ipv4 192.168.99.10
  key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
*****
                        AUTHORISED ACCESS ONLY

Access to this device is restricted to authorised personnel. If you are not
an authorised user, disconnect now.

All activity is monitored, logged and audited. Continued use of this system
constitutes consent to that monitoring.

Unauthorised access or use may result in disciplinary action and civil or
criminal penalties.
*****
^C
!
line con 0
  exec-timeout 0 0
  activation-character 13
  logging synchronous
  stopbits 1
line aux 0
  activation-character 13
line vty 0 4
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
line vty 5 15
  access-class VTY-MGMT in vrf-also
  exec-timeout 0 0
  activation-character 13
  transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end

```